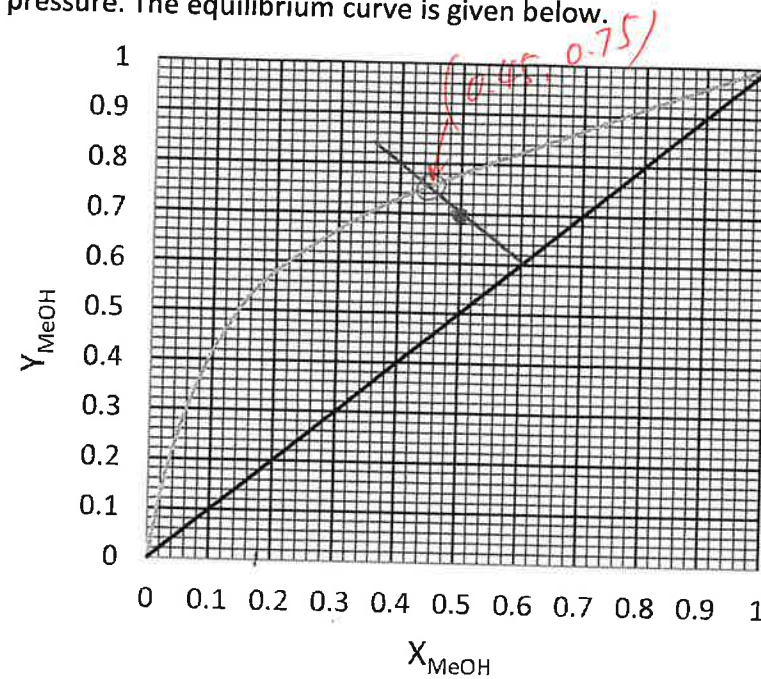


1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1.1) Overall Mass Balance: $F = L + V$

Component Mass Bal: $Fz = Vy - Lx$ ✓ 3/3

Operating Equation: $y = -\frac{L}{V}x + \frac{F}{V}z$

1.2) $F = 150 \text{ kmol/h}$ $z = 0.6$ $V/F = 0.5$ $V?$ $L?$

$$\frac{V}{F} = 0.5 \rightarrow \frac{V}{150} = 0.5 \rightarrow V = 75 \text{ kmol/h}$$

$$L = F - V = (150 - 75) \frac{\text{kmol}}{\text{h}} = 75 \frac{\text{kmol}}{\text{h}}$$

$$y = -\frac{L}{V}x + \frac{F}{V}z \rightarrow \text{Slope: } -\frac{L}{V} = -1 \quad y_{\text{int}} = 2(0.6) = 1.2 \text{ too high}$$

$$x = 0.5 \rightarrow y = -1(0.5) + 1.2 = 0.7 \quad (0.5, 0.7) \quad 4/5$$

1. Vapor and Liquid mole Fractions:

$$Fz = \frac{V(y(x)) + Lx}{150} = \frac{75(0.7) + 75(0.5)}{90} = \boxed{1}$$

From graph $\rightarrow (0.45, 0.75)$

1.3) $z = 0.3$ $Lx = 0.18$ $F = 150 \text{ kmol/h}$

$$F = L + V \quad Fz = Lx + Vy$$

$$y = -\frac{L}{V}x + \frac{F}{V}z \quad y_{\text{int}} = \frac{F}{V}z \rightarrow V =$$

2.) 1. Set a T_{drum} guess ✓

2. Find $K_i(T, P)$ ✓

3. Set up RR equations ✓

4. Set a V/F guess ✓

5. Solve RR eq 1 and 2 ✓

6. Plug into RR eq 3 for new V/F check if $\epsilon < 0.005$

7. If not solve again ✓

8. Once good solve for x_i, y_i, L, V ✓

9. Check T_{drum} $\rightarrow |T_{\text{drum}} - T| < 0.02$ ✓

10. If not repeat ✓

10/10

3.7)

$$\text{Comp: } F_2 Z_2 + V'y = L'x + Dx_0$$

$$y = \frac{L'}{V'}x + \frac{Dx_0 - F_2 Z_2}{V'}$$

$$x = y = \frac{Dx_0 - F_2 Z_2}{V' - L'}$$

$$F_1 + F_2 = D + D$$

$$0 = F_1 Z_1 + F_2 Z_2 - \frac{(F_1 + F_2)x_B}{x_D - x_B}$$

$$B = F_1 + F_2 - D$$

$$L = D$$

$$L' = F_2 q_2 + L$$

$$V' = F_2 + L' - D$$

$$\text{slope} = \frac{L'}{V'}$$

✓ 8/10

3.8) 5 stages without middle op

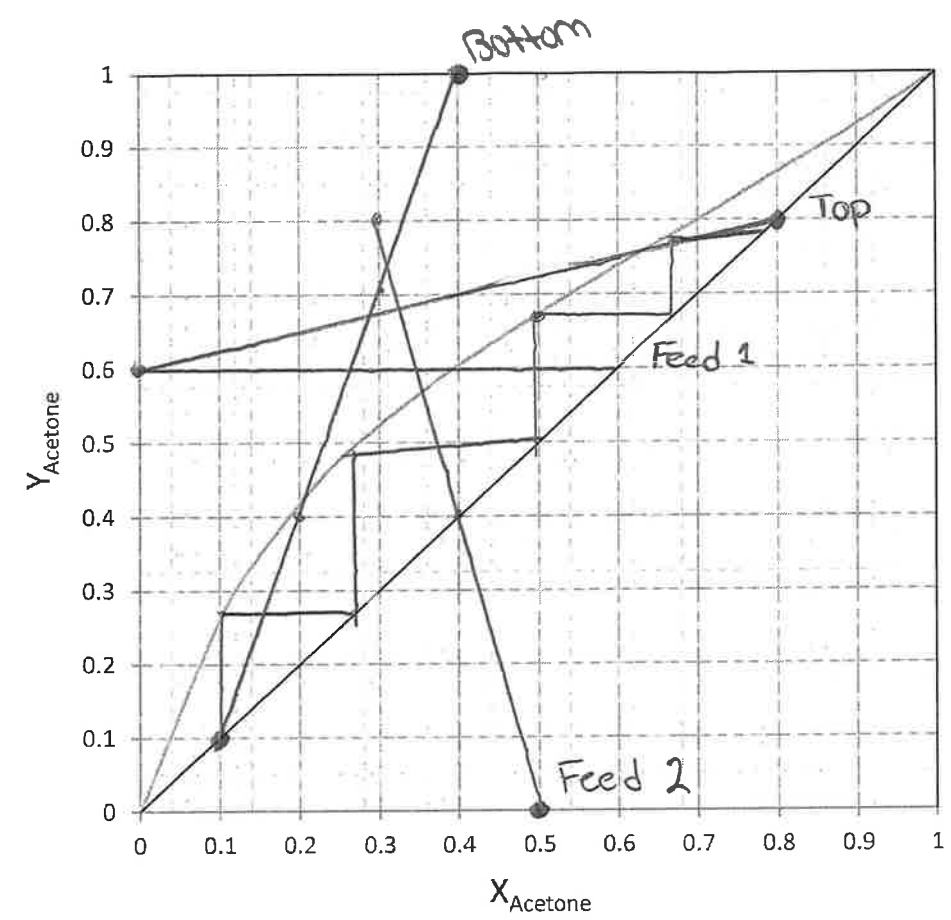
3/5

3.9) stage ~~4~~ and ~~5~~

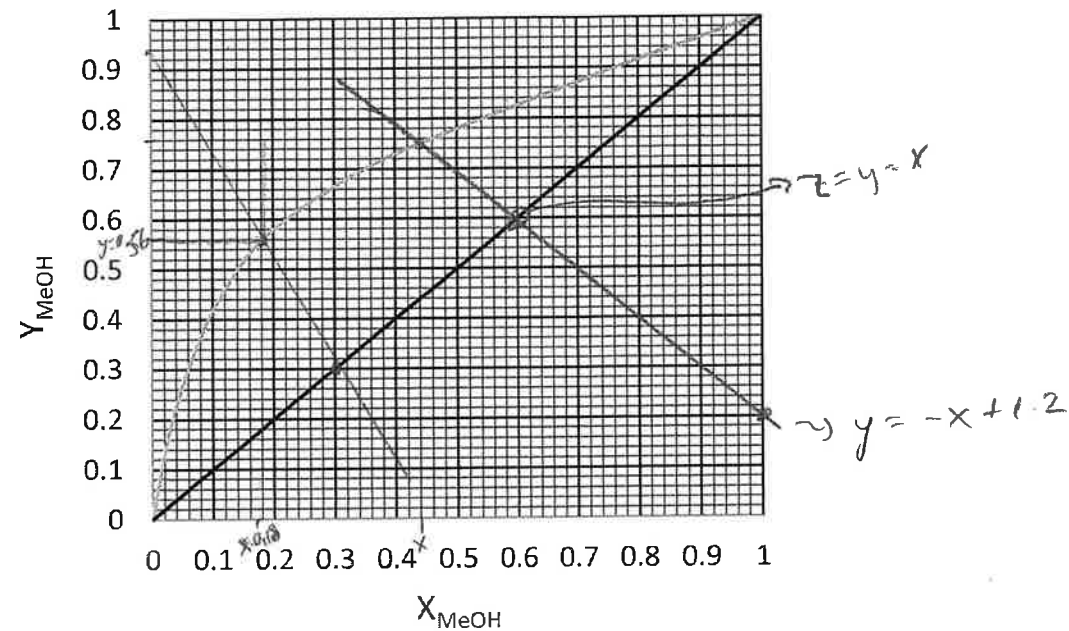
~~2/3~~ 2/3

3.10) $Z = 0.5$

0/2



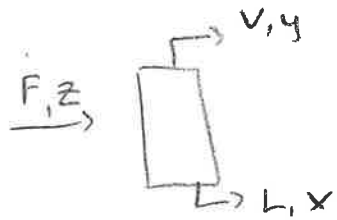
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①

1)



$$\text{OV MB: } F = V + L$$

$$\text{MVC MB: } Fz = Vy + Lx$$

solving for y:

$$yV = -Lx + Fz$$

$$y = -\frac{L}{V}x + \frac{F}{V}z \quad 3/3$$

$$y = -\frac{L}{V}x + \frac{(V+L)}{V}z$$

$$y = -\frac{L}{V}x + \left(1 + \frac{L}{V}\right)z$$

2) $F = 150 \text{ kmol/h}$

$z = 0.6$

$\frac{V}{F} = 0.5$

$V = 0.5F = (0.5)(150) = 75 \text{ kmol/h}$

$F = V + L \Rightarrow L = F - V = 75 \text{ kmol/h}$

slope: $\frac{L}{V} = -1$

$y_{\text{int}}: +\frac{F}{V}z = \frac{150(0.6)}{75} = 1.2$

$y = -x + 1.2$

1 pt (0.6, 0.6)

$x = 1$

$y = -1 + 1.2 = 0.2$

$$x = 0.42 \text{ or } 0.43 \text{ (interception)}$$

$$y \approx 0.76 \approx 0.77 \text{ at O.P. of Eq. (C)}$$

3) $z = 0.3$, $x = 0.18$, $F = 150 \text{ kmol/h}$

on Eq. curve $x = 0.18$, $y = 0.56$

$$y_{\text{in}}: \frac{F}{V}z \text{ op line points: } z = x = y = 0.3 \text{ and } (0.18, 0.56) \text{ to set for Eq. (C)}$$

$Fz = yV + Lx$, $L = F - V$

$Fz = yV + (F - V)x$

$$0.3(150) = 0.56(V) + (150(0.18)) - 0.18(V)$$

$$45 = V(0.56 - 0.18) + 27$$

$$\frac{45 - 27}{0.38} = V \Rightarrow 47.37 \text{ kmol/h}$$

$$\frac{V}{F} = \frac{47.37}{150} = 0.32 \approx \frac{1}{3}$$

$L = 102.63 \text{ kmol/h}$

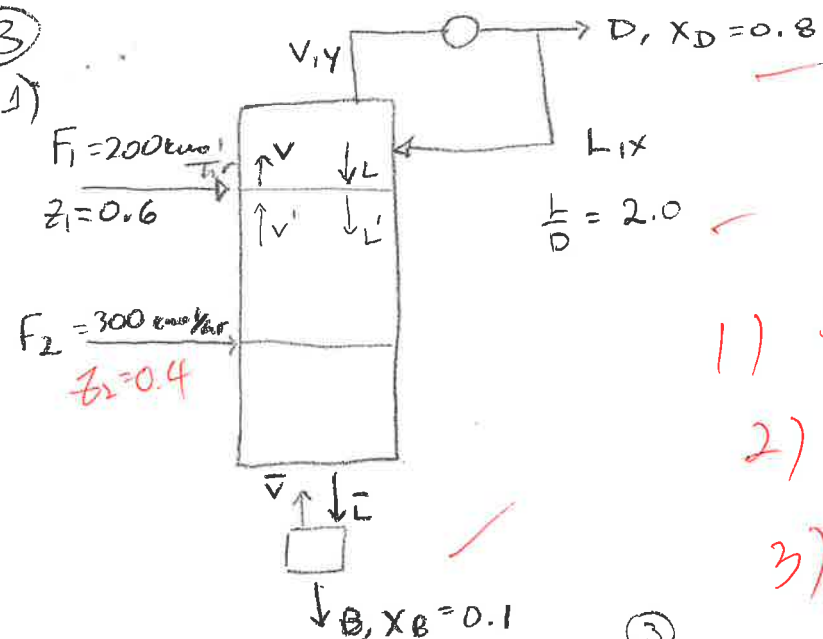
$y_{\text{int}} = \frac{F}{V}z = 0.95$

Katerina
Baupar

10/10

③

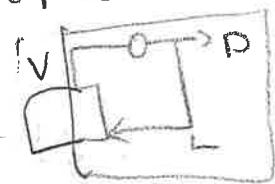
1)



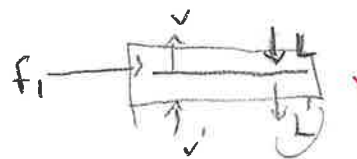
1) 3/3
2) 8/8
3) 2/15

② M.B Envelopes:

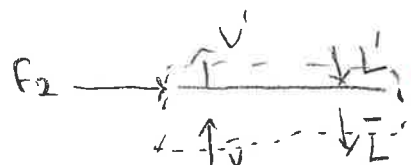
Enriching



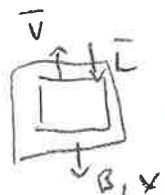
Feed 1:



Feed 2:



stripping section:



OV M.B %

$$V = D + L$$

CMB %

$$V y = D x_D + L x \quad ; \quad y = x_D \text{ or } x$$

$$V x_D = D x_D + L x_D$$

OV M.B %

$$F_1 + V' + L = V + L'$$

$$\text{CMB: } F_1 z_1 + V' y' + L x_D = V x_D + L' x'$$

OV M.B %

$$F_2 + L' + \bar{V} = V' + \bar{L}$$

$$\text{CMB: } F_2 z_2 + L' x' + \bar{V} y = V' y' + \bar{L} x$$

OV M.B %

$$\bar{L} = \bar{V} + B$$

$$\text{CMB: } \bar{L} x_B = \bar{V} y + B x_B$$

④

$$y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D \quad , \quad V = D + L$$

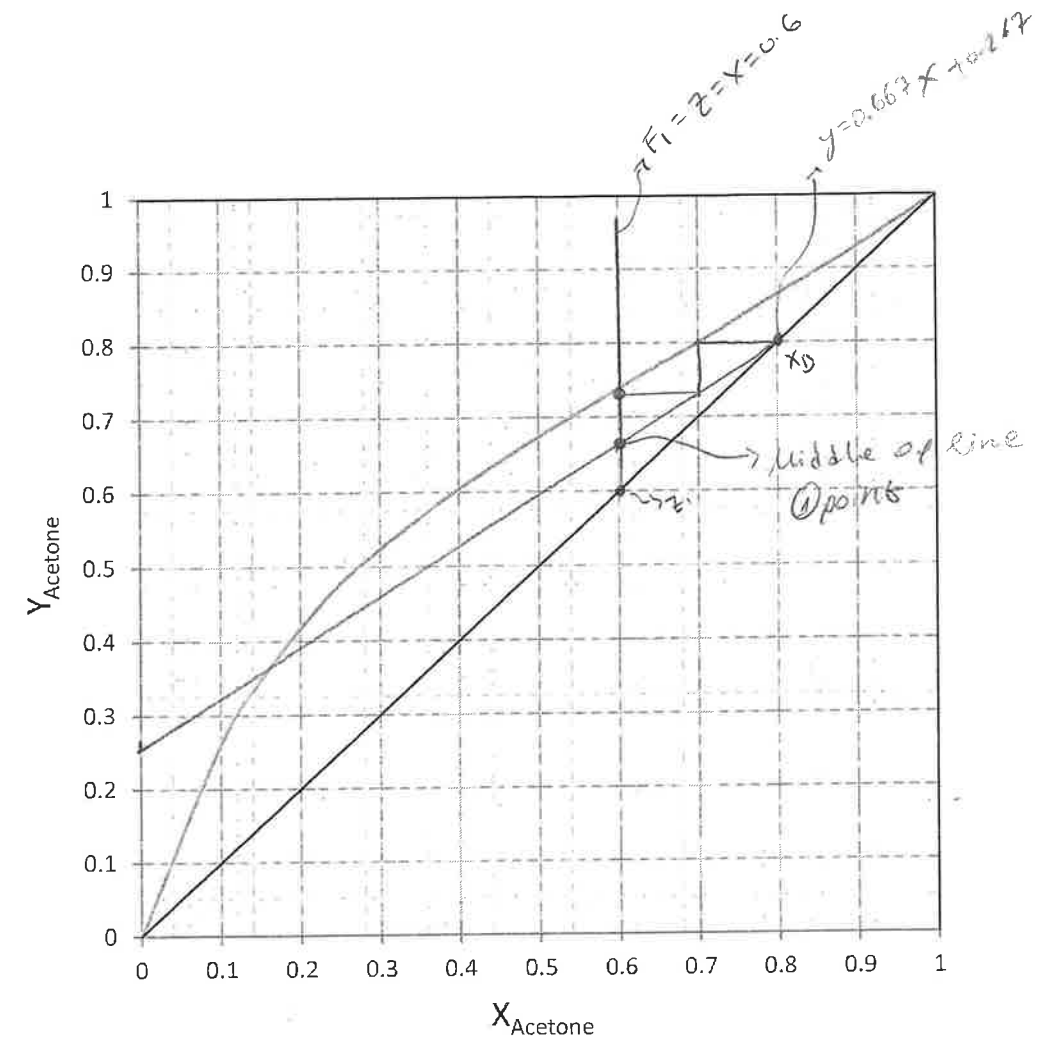
$$\frac{L}{V} = \frac{L}{D+L} = \frac{\frac{L}{D}}{1 + \frac{L}{D}} = \frac{2}{1+2} = \frac{2}{3} \quad \text{slope} \approx 0.667$$

$$y\text{-int: } \left(1 - \frac{L}{V}\right)x_D = \left(1 - \frac{2}{3}\right)0.8 = 0.267$$

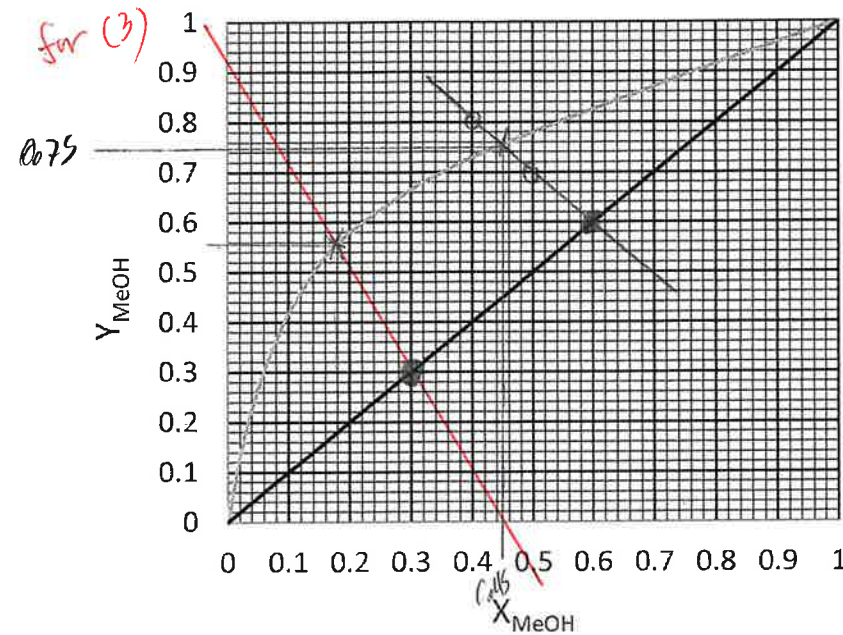
Top of line:

$$y = 0.667x + 0.267 \quad (0, 0.267) \text{ y-int}$$

$$(0.8, 0.8) (x_D, x_D)$$



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



$$y = mx + b$$

↑
slope $\frac{F}{V}$ x_{int}

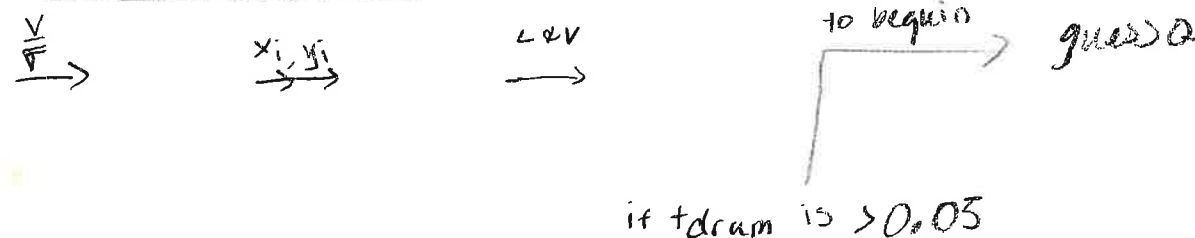
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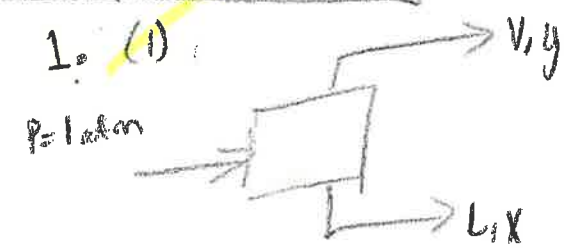
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- 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.

$$x = 0.18 \quad y = 0.55 \text{ int}$$

2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V





overall MB: $F = V + L$

Comp MB: $Fz = Vy + Lx$ 3/3

op eqn: $y = -\frac{L}{V}x + \frac{F}{V}z$

(2) $F = 150 \frac{\text{kmol}}{\text{hr}}$

$z = 0.6$

$\frac{V}{F} = 0.5$



Remember (z_F, z_B)
 $(0.6, 0.6)$ on x, y line

$V = 0.5(150) \frac{\text{kmol}}{\text{hr}}$

$V = 75 \frac{\text{kmol}}{\text{hr}}$ Vap ✓

$L = 75 \frac{\text{kmol}}{\text{hr}}$ Liq ✓

op eqn
to find
slope: $y = -\frac{L}{V}x + \frac{F}{V}z$

$= -1x + 2(0.6)$

$y = -1x + 1.2$

↑ slope from the slope & equilib line

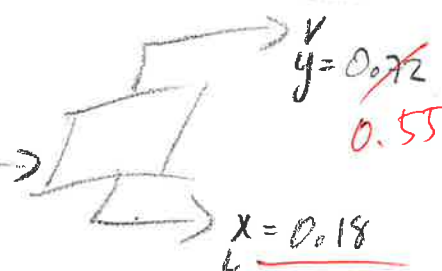
$y = 0.75$ ✓

$x = 0.45$

(3)

$F = 150 \frac{\text{kmol}}{\text{hr}}$

$z = 0.3$



$y = 0.72$

0.55

$x = 0.18$

$(z, z) = \text{feed point}$

$(0.3, 0.3)$

Know $F, z,$

unk: V, L

2 3/10

Comp MB: $Fz = Vy + Lx$

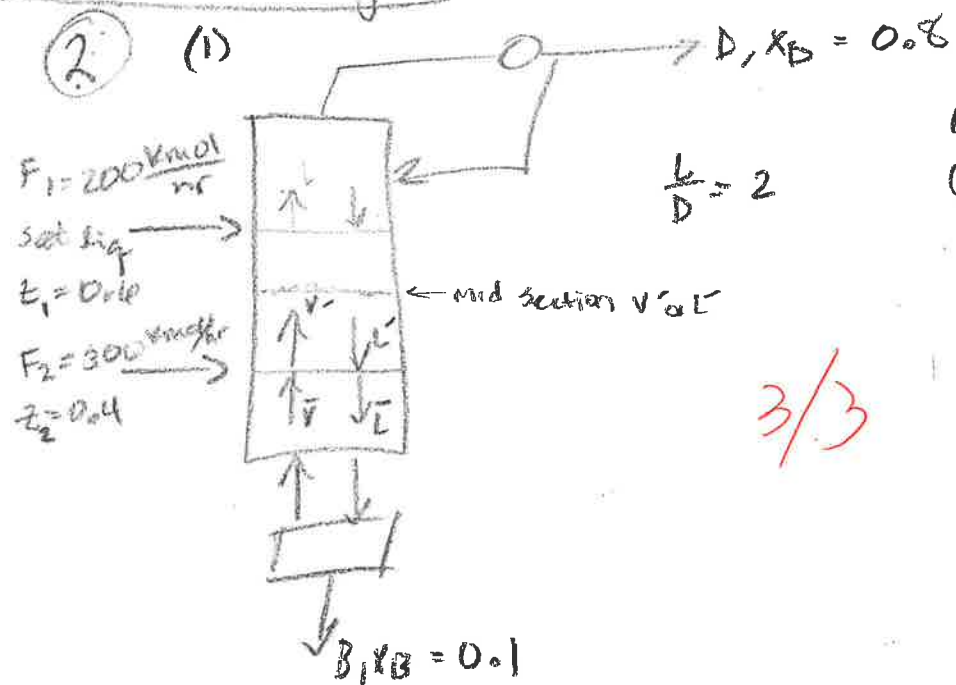
$45 = V$

$\frac{V}{F} = \frac{1}{3} \leftarrow \text{from graph}$

$V = 50 \text{ kmol/hr}$

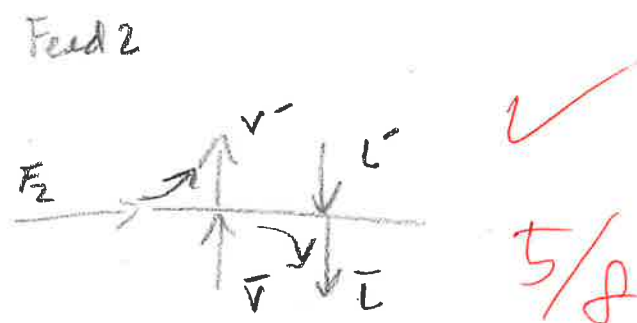
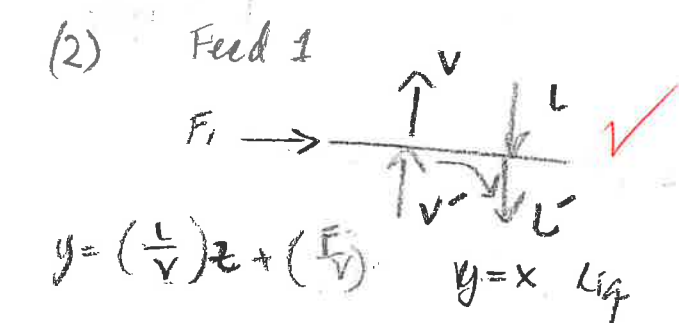
$L = 100$

overall MB: $F = V + L$
 $L = F - V$

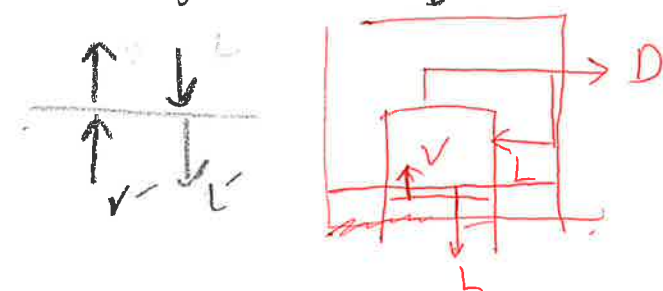


Overall MB: $F = D + B$
 Comp MB: $Fz = Dx_D + Bx_B$

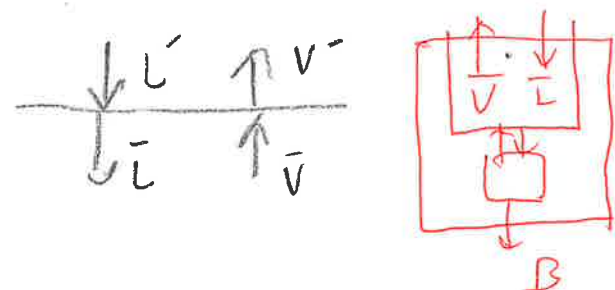
3/3



Enrich $yV = Lx + Dx_D$



strip: $y(V) = (L - \bar{L})x - Bx_B$



(4) $L/D = 2 \rightarrow L = 2D$

$\frac{L}{V} = \frac{L}{D+L} \rightarrow \frac{L/D}{1+L/D} = \frac{2}{1+2} = \frac{2}{3}$ $\approx$ slope top
 $\frac{\bar{L}}{V} = \frac{L}{D+L}$
 (0.8, 0.8) on $y = x$

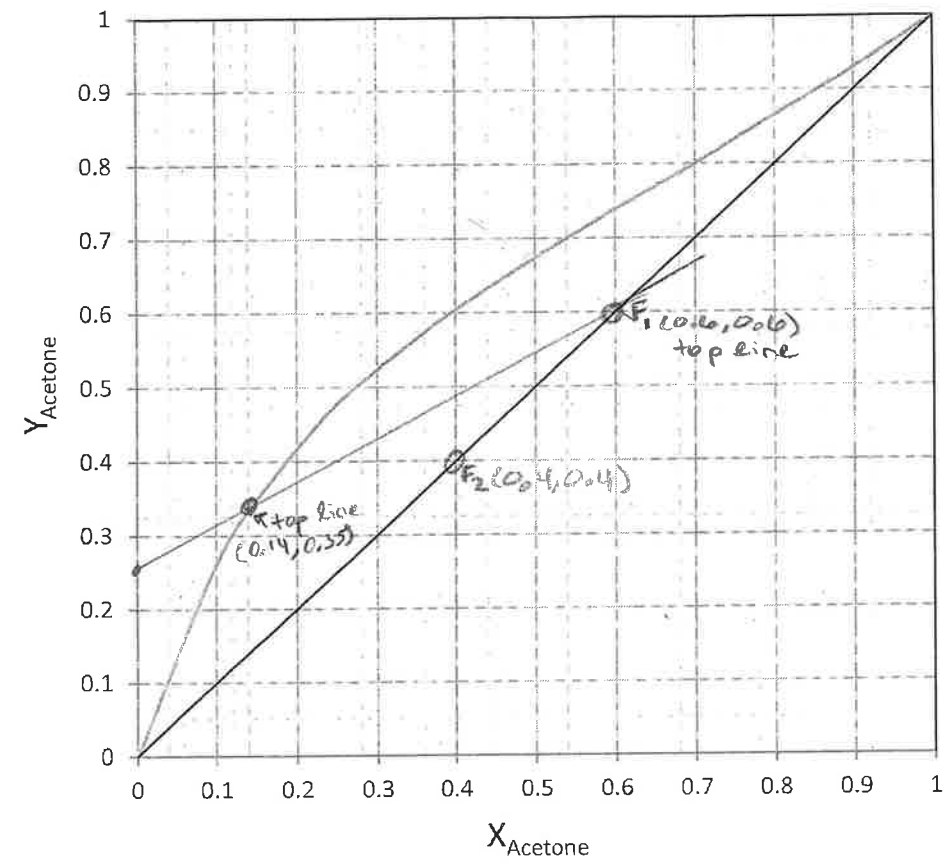
y-int top
 $\left(1 - \frac{2}{3}\right)(0.8) = 0.26$
 0.33

Page 3 Trina Murrell

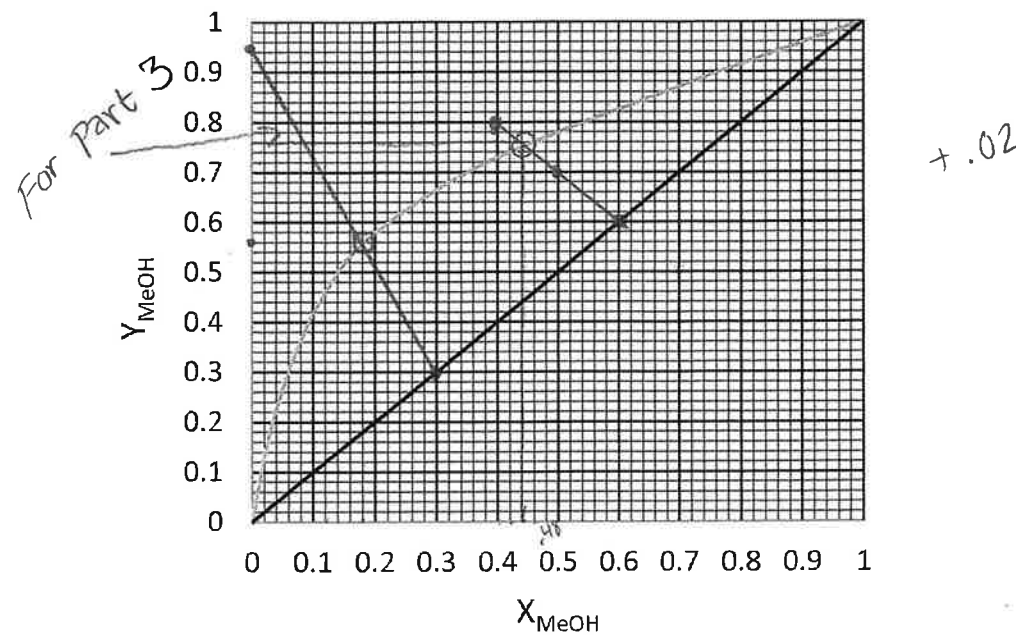
Top Equ Op $y = \frac{L}{V} x + \frac{L'}{V} z$

MB = $F_1 z_1 = D x_D +$

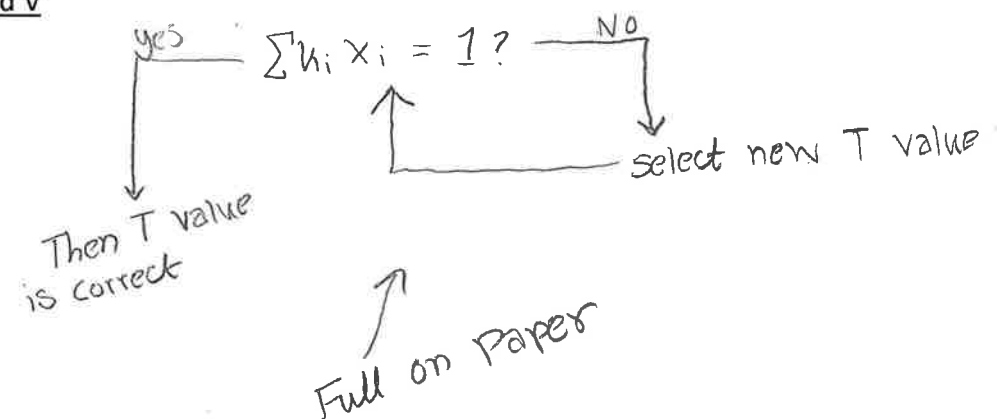
Trina Murrell

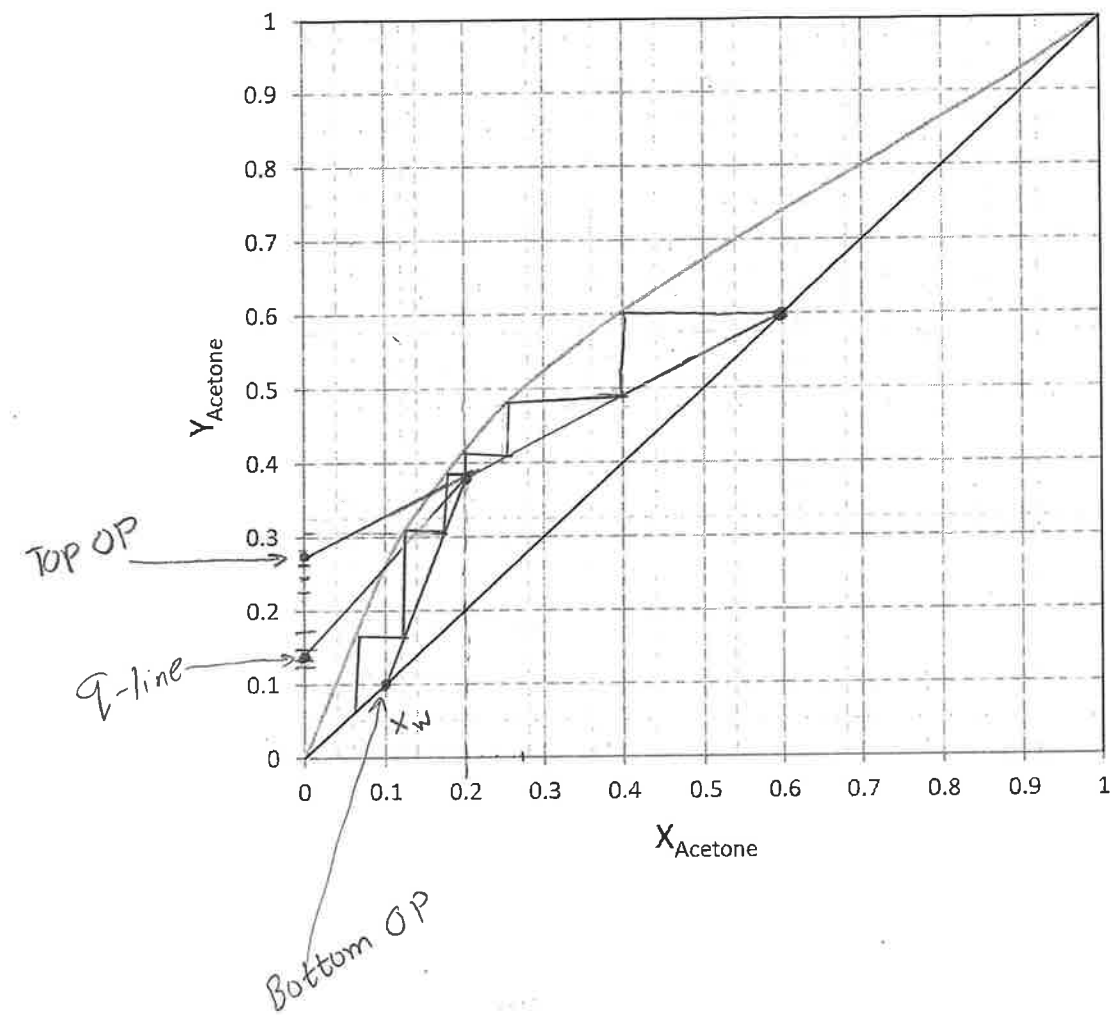


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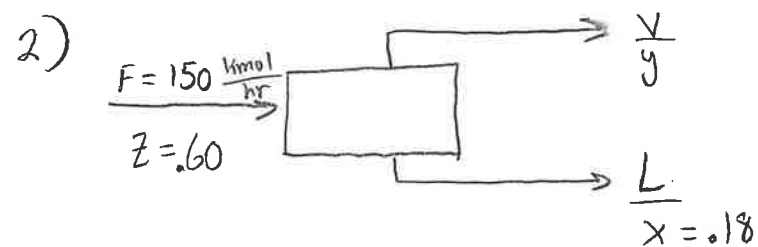


→ 6 stages

1) Component MB: $FZ_{\text{MEOH}} = Vy_{\text{MEOH}} + Lx_{\text{MEOH}}$

$$y_M = \frac{F}{V} z_M + \frac{L}{V} x_M$$

Overall: $F = V + L$



$$\frac{V}{F} = .50$$

$$V = .50(150) = 75 \frac{\text{kmol}}{\text{hr}}$$

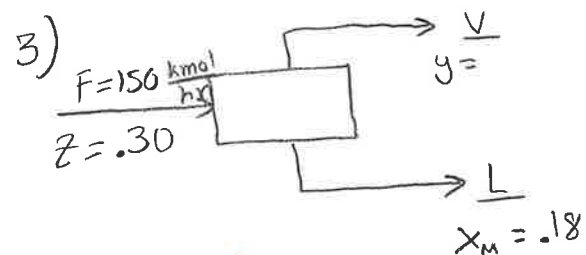
$$L = 75 \frac{\text{kmol}}{\text{hr}}$$

$$y = \frac{(.5-1)}{.5}x + \frac{1}{.5}$$

$$y = -1x + 2 \leftarrow \text{OP line start @ } (.60, .60)$$

$$(x_{\text{MEOH}}, y_{\text{MEOH}}) \approx (.44, .76)$$

Circled on graph



$$\frac{V}{F} = \frac{z - x}{y - x}$$

at $x = .18$, $y \approx .56 \rightarrow$ Squared on graph

$$\frac{V}{F} = \frac{.3 - .18}{.56 - .18} \approx .32$$

$$V = 47.37 \frac{\text{kmol}}{\text{hr}}$$

$$L = 102.63 \frac{\text{kmol}}{\text{hr}}$$

$$y = \frac{150}{47.37}(.3) - \frac{(102.63)}{47.37}x_M$$

$$y = .95 - 2.2x$$

y-int. slope

2) $y_i = x_i K_i \rightarrow$ check energy balance

① DePriester: $\sum K_i x_i = 1$

yes
Use T value & corresponding K values

no
select new guess for T

Sorry it's messy, was trying to fit it all

Can find $\frac{V}{F}$ from Rachford-Rice

$$Q = \frac{\sum z_i (K_i - 1)}{1 + (K_i - 1)(\frac{V}{F})}$$

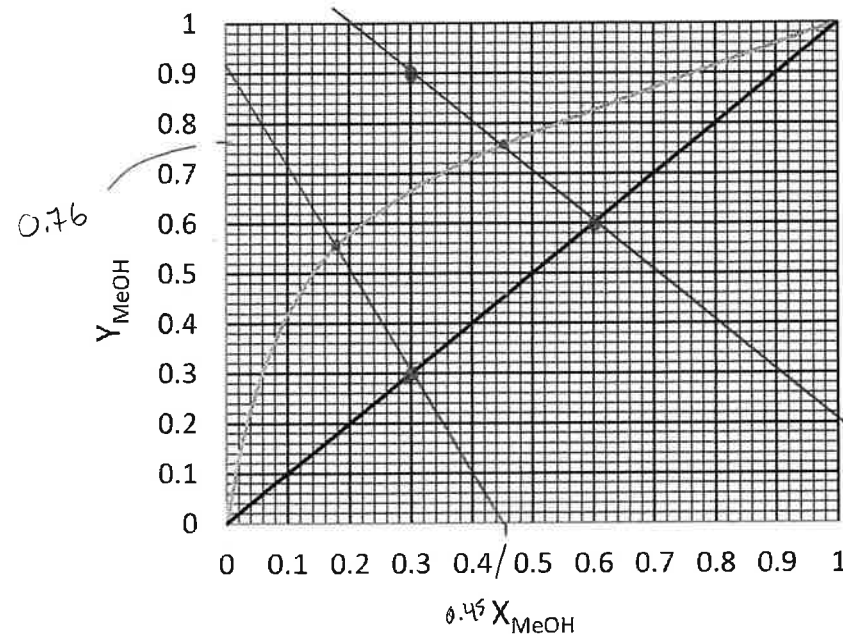
8/10

10/10

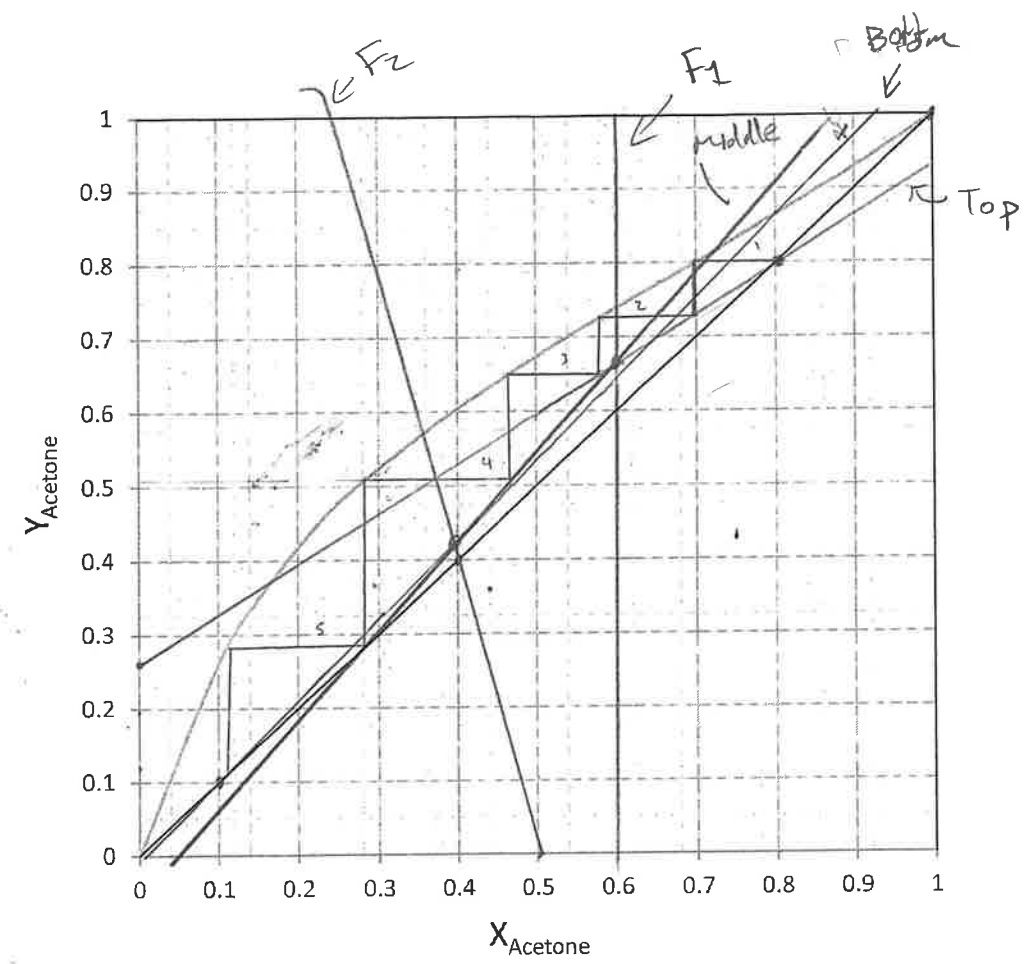
3/3

5/5

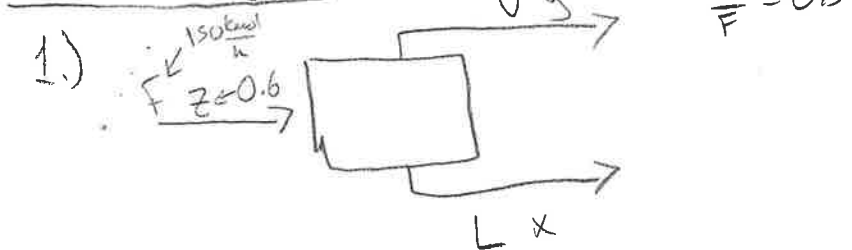
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- C
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Annam Chougale



1.1) Overall M.B. $F = V + L$
 Component M.B. $Fz = Vy + Lx$
 Operating Eqn $\Rightarrow Vy = Fz - Lx$
 $y = -\frac{L}{V}x + \frac{Fz}{V}$

3/3

1.2) $F = 150 \frac{\text{kmol}}{\text{h}}$ $\frac{V}{F} = 0.5 \Rightarrow V = 0.5F \Rightarrow V = 0.5(150 \frac{\text{kmol}}{\text{h}}) = 75 \frac{\text{kmol}}{\text{h}}$

$L = F - V = 150 - 75 = 75 \frac{\text{kmol}}{\text{h}}$ $L = 75 \frac{\text{kmol}}{\text{h}}$ $V = 75 \frac{\text{kmol}}{\text{h}}$

$y = -\frac{75}{75}x + \frac{(150)(0.6)}{75}$

$y = -x + 1.2$

@ $x = 0.3$ $(0.3, 0.9)$
 $y = -0.3 + 1.2 = 0.9$

@ $x = 0.6$ $(0.6, 0.6)$
 $y = -0.6 + 1.2 = 0.6$

from graph: $y = 0.76$
 $x = 0.45$ ← For meth

5/5

1.3) on $y=x$: $(0.3, 0.3)$ another at $(0.14, 0.156)$

$y\text{-intercept} = 0.92 = \frac{F}{V}z$ ✓

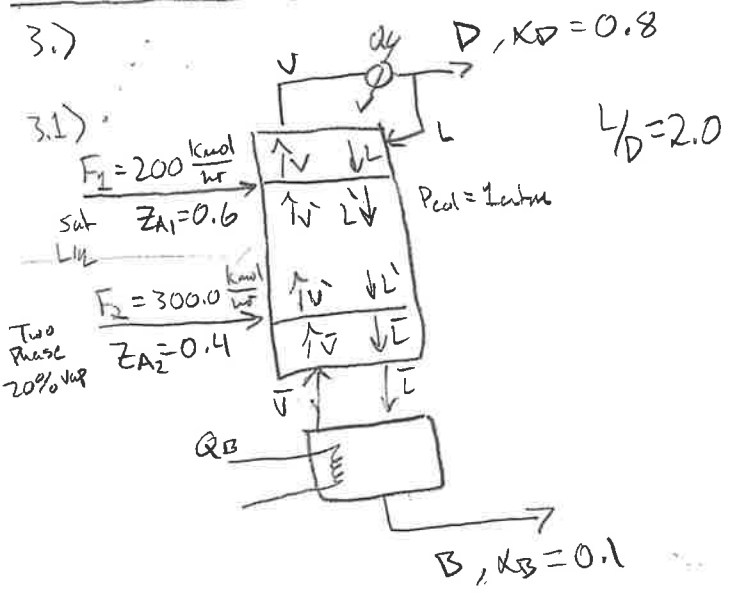
$\frac{F}{V} = \frac{0.92}{z} \Rightarrow \frac{V}{F} = \frac{0.3}{0.92} = 0.326$

$\frac{V}{F} = 0.326$ ✓

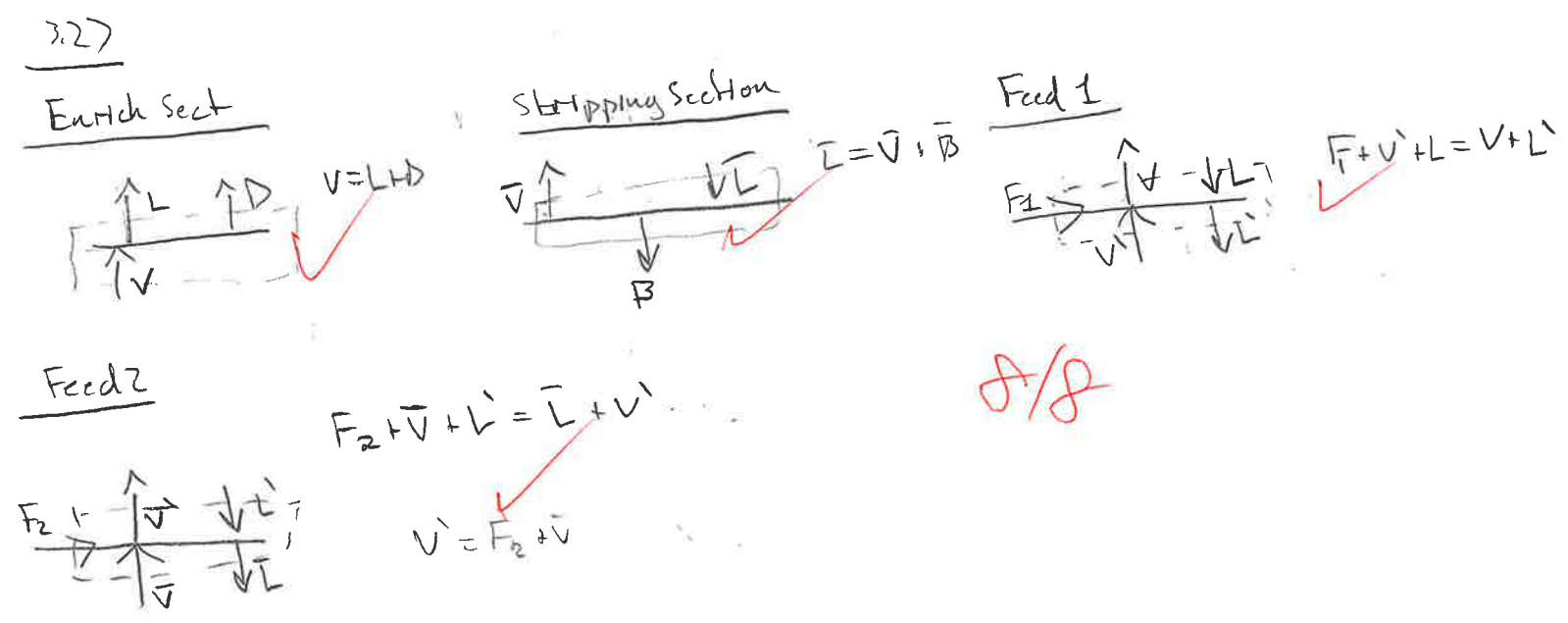
$V \approx 50 \text{ kmol/h}$
 $L = 100$

8/10

Amaan Chougale



3/3



8/8

3.3)

$$F_1 + F_2 = D + B \Rightarrow B = F_1 + F_2 - D$$

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B$$

$$F_1 z_1 + F_2 z_2 = D x_D + x_B (F_1 + F_2 - D)$$

$$200(0.6) + 300(0.4) = D(0.8) + 0.1(200 + 300 - D)$$

$$240 = 0.8D + 20 + 30 - 0.1D$$

$$190 = 0.7D$$

$$D = 271.43 \frac{\text{kmol}}{\text{hr}}$$

$$B = 200 + 300 - 271.43 = 228.57 \frac{\text{kmol}}{\text{hr}}$$

$\frac{L}{D} = 2.0 \Rightarrow L = 2(D) = 2(271.43)$

$$L = 542.86 \frac{\text{kmol}}{\text{hr}}$$

$$V = L + D$$

$$V = 542.86 + 271.43$$

$$V = 814.29 \frac{\text{kmol}}{\text{hr}}$$

$\bar{V} = V \leftarrow \text{No vap in } F_1$

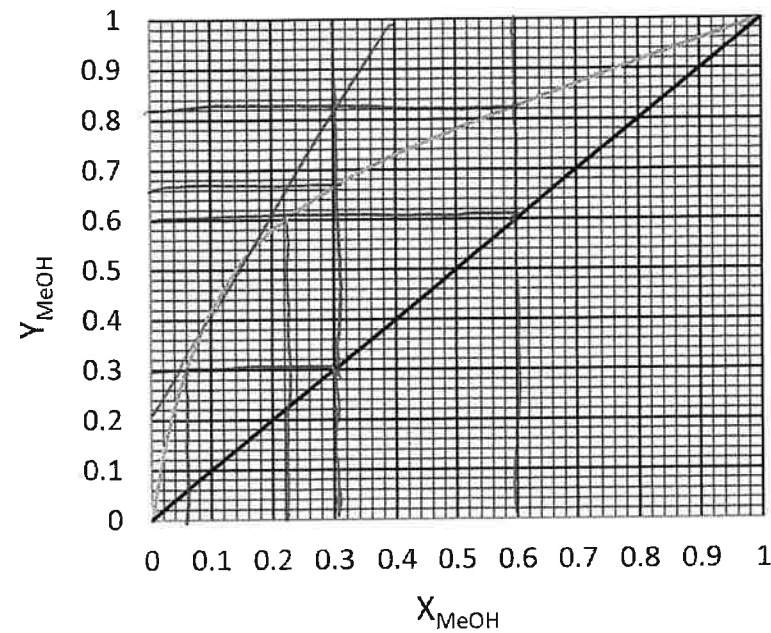
$$\bar{V} = 814.29 \frac{\text{kmol}}{\text{hr}}$$

American Choughale

8.) 5 stages 5/5
9.) F_1 stage 2 F_2 stage 4 3/3

10.) $x = 0.58$ $y = 0.65$
0.49 0.67 method is dc! 2/2

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1) Overall M.B

$$(1) F = B + D \Rightarrow (F = V + L)$$

Component M.B

$$Fz = B(x_B) + D(x_D) \Rightarrow (Fz = Vy + Lx)$$

Op equation

$$\text{Top} = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$$

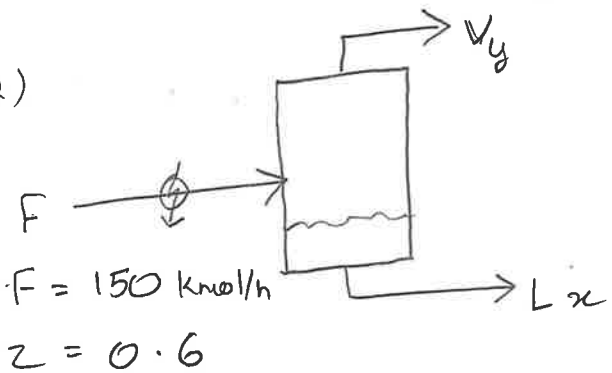
$$\text{Bottom} \left(\frac{L}{V}\right)x - \left(\frac{L}{V} - 1\right)x_B$$

$$\text{Feed line} \Rightarrow y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}$$

$$(y = -\frac{Lx}{V} + \frac{Fz}{V})$$

3/3

(2)



$$F = 150 \text{ kmol/h}$$

$$z = 0.6$$

$$f = \frac{V}{F} = 0.5$$

$$Fz = Vy + Lx$$

$$y = -\frac{Lx}{V} + \frac{Fz}{V}$$

$$\frac{V}{F} = 0.5$$

$$Fz = Vy + Lx$$

$$150 \text{ kmol/h} (0.6) = V(0.5) + L(0.5)$$

$$90 \text{ kmol/h} = 0.5(V + L)$$

$$i. \text{ Vapor flow rate} = \frac{90 \text{ kmol/h}}{0.5} = 180 \text{ kmol/h}$$

$$\text{liquid flow rate} = \frac{90 \text{ kmol/h}}{0.5} = 180 \text{ kmol/h}$$

$$V = 75 \text{ kmol/h}$$

$$L = 75 "$$

$$\text{slope} = -1$$

$$(0.45, 0.75)$$

1/5

Based on the Raoult's law equation value, take another guess for T_{dew} that would result in the Raoult's law equation to be zero. Repeat as needed



Alternatively, you can use Newton method to find an educated guess for T_{dew} where

$$df = \frac{(K_i - 1)^2 z_i}{(1 + (K_i - 1))^2}$$

After find the T_{dew} temperature that would result in the Raoult's law equation to be zero, we now have our T_{dew} value.



Using the T_{dew} value and mass balance equation we can find V/F , x_i , y_i L and K_i

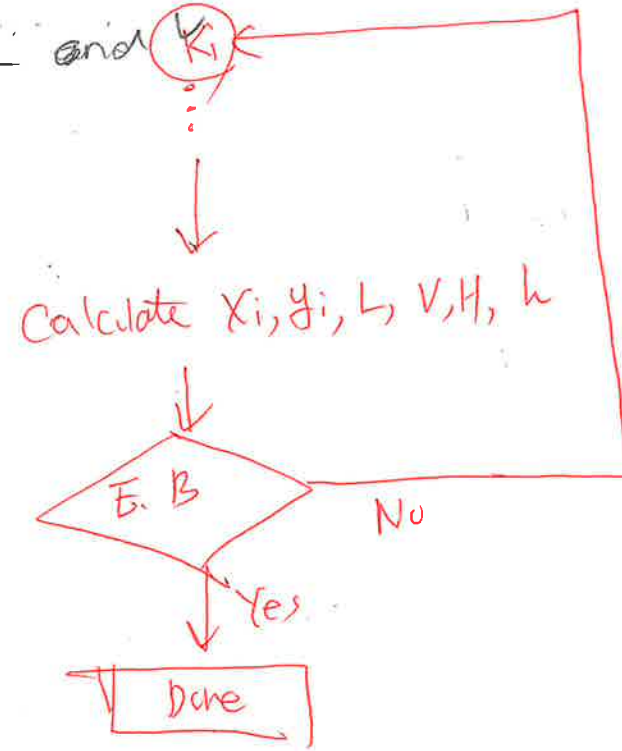
where $\Rightarrow F = L + V$

$\Rightarrow Fz = Lx_i + Vy_i$

$y = K_i x_i$

$x_i =$

7/10



$$(4) \quad y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$$

$$\text{slope} \Rightarrow \frac{L}{V}$$

$$y\text{-intercept} \Rightarrow \left(1 - \frac{L}{V}\right)x_D \quad (0.8, 0.8)$$

$$y = x_D = 0.6 \Rightarrow (0.6, 0.6)$$

$$(5) \quad y = \left(\frac{L}{V}\right)x - \left(\frac{L}{V} - 1\right)x_B$$

$$\text{slope} \Rightarrow \frac{L}{V}$$

$$y\text{-intercept} \Rightarrow \left(\frac{L}{V} - 1\right)x_B$$

$$y = x_B = 0.4 \Rightarrow (0.4, 0.4)$$

$$(0.1, 0.1)$$

$$(6) \quad y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}, \quad q_i = \frac{L_{Fi}}{F_i}$$

Using the above feed line equations, we can draw feed lines F_1 & F_2 , where $\frac{q_i}{q_i - 1}$ is the slope and $\frac{z_i}{1 - q_i}$ is the y-intercept.

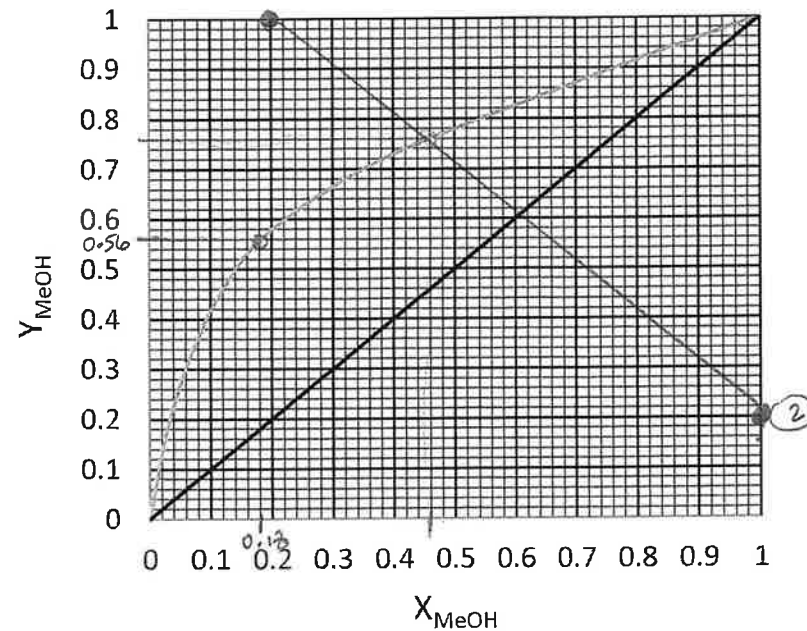
(7) Overall mass balance

$$\Rightarrow F = V + L$$

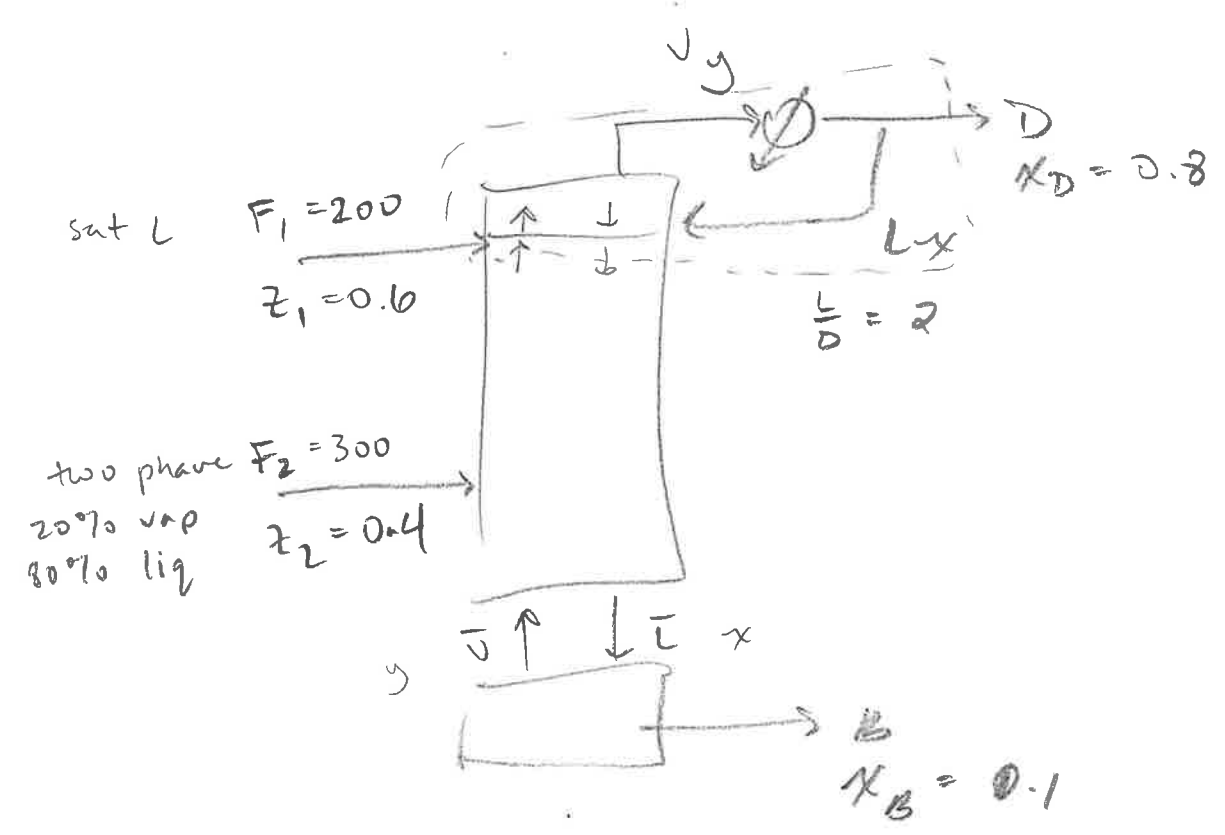
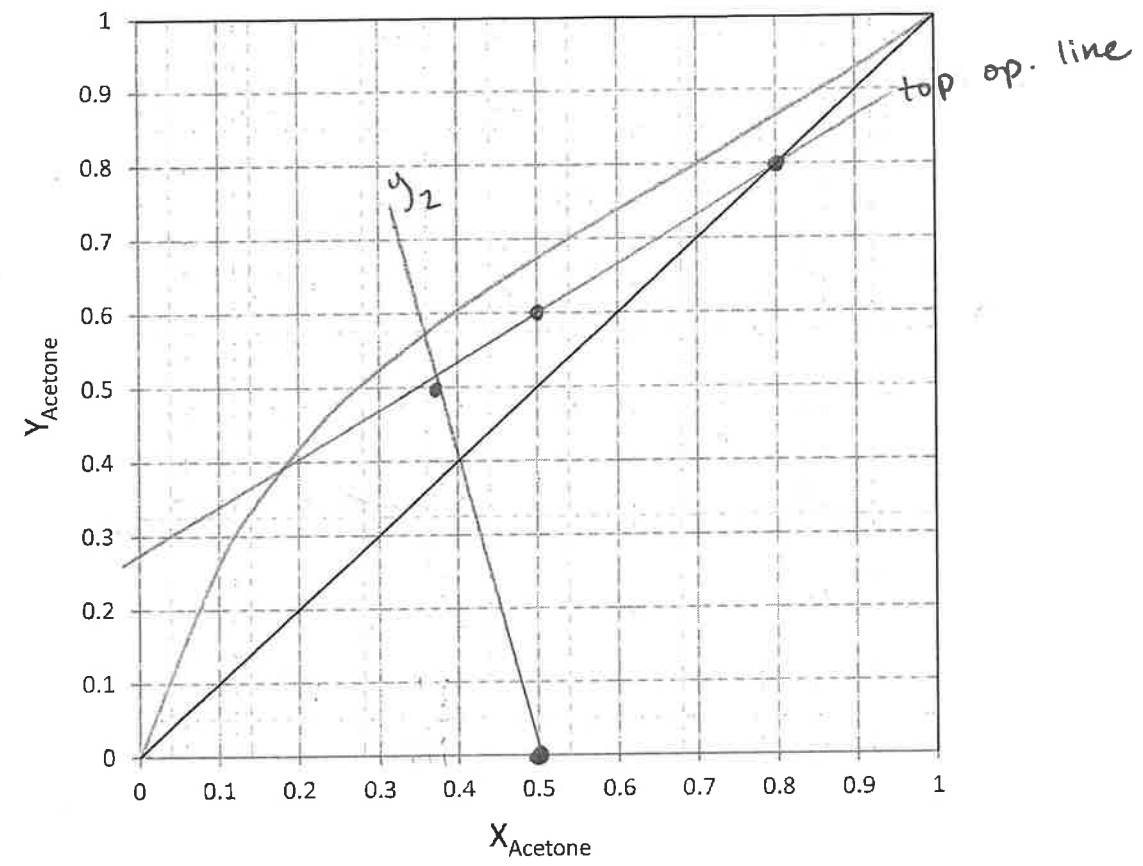
component M.B

$$\Rightarrow Fz = Vy + Lx$$

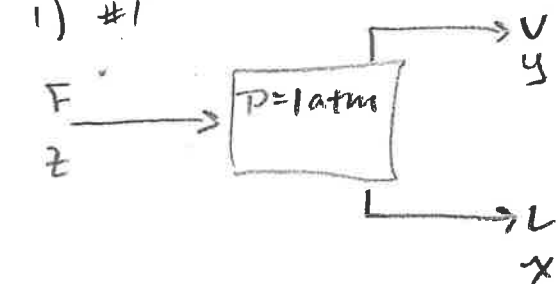
1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V



1) #1



overall MB: $F = V + L$

comp. MB: $Fz = Vy + Lx$

op eq:

$$y = \frac{F}{V} z - \frac{L}{V} x$$

1) 3/3

2) 5/5

3) 10/10

#2 $F = 150 \text{ kmol/h}$

$z = 0.6$

$\frac{V}{F} = 0.5$

$V = 0.5F$

$150 = 0.5(150) + L$

$V = 75 \text{ kmol/h}$

$L = 75 \text{ kmol/h}$

op eq: $y = 2(0.6) - x$

$y = 1.2 - x$

$y = 1 = 1.2 - x \Rightarrow x = 0.2$

$(0.2, 1)$

$y = 1.2 - 1.0 \Rightarrow y = 0.2$

$(1, 0.2)$

@ intersection of VLE
& op eq:

$x = 0.46$

$y = 0.76$

lig. mol frac.

vap. mol frac.

#3 $z = 0.3$

$x = 0.18$

$F = 150 \text{ kmol/h}$

$y = \frac{F}{V} z - \frac{L}{V} x$

$F = V + L$

when $x = 0.18$, $y = 0.56$

$y = \frac{F}{V} z - \frac{F-V}{V} x = \frac{F}{V} z - \left(\frac{F}{V} - 1\right)x$

$0.56 = \frac{150}{V}(0.3) - \left(\frac{150}{V} - 1\right)(0.18)$

$0.56 = \frac{45}{V} - \frac{27}{V} + 0.18$

$0.38 = \frac{18}{V}$

$V = 47.37 \text{ kmol/h}$

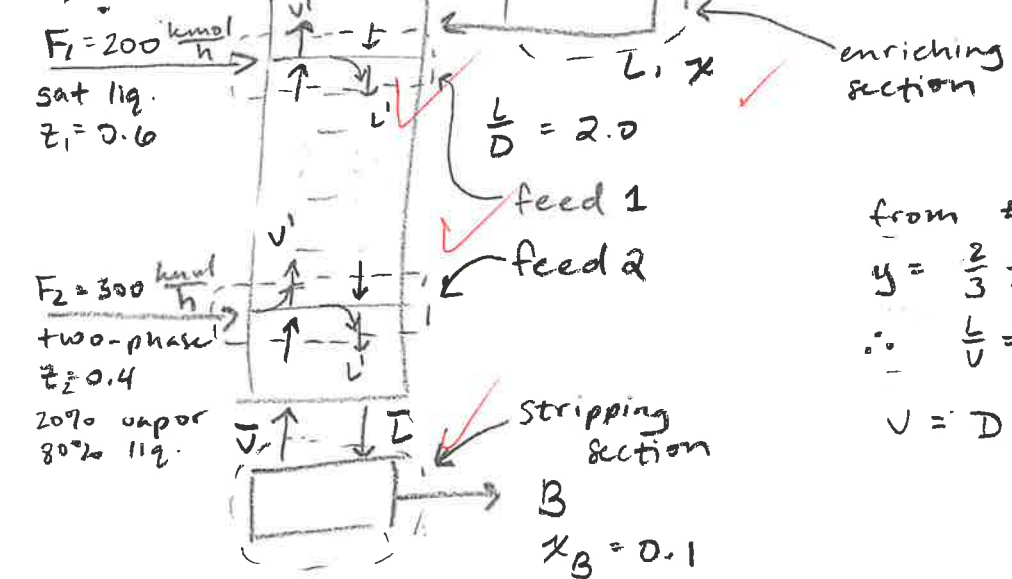
$L = 102.63 \text{ kmol/h}$

$\frac{V}{F} = \frac{47.37}{150}$

$\frac{V}{F} = 0.316$

Naya Majanukas

3) #1 & 2



enriching section 1) 3/3
from #4: 2) 8/10
3) 10/15

$$y = \frac{2}{3}x + 0.267$$

$$\therefore \frac{L}{V} = \frac{2}{3} \quad V = D + L$$

$$V = D + \frac{2}{3}V \quad \frac{1}{3}V = D = 271.4$$

$$V = 814.3 \frac{\text{kmol}}{\text{h}}$$

$$L = 2D \quad \text{! this makes sense!}$$

$$L = 542.9 \frac{\text{kmol}}{\text{h}}$$

#3

$$F_1 + F_2 = D + B \rightarrow 200 + 300 = 500 = D + B$$

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B \rightarrow 200(0.6) + 300(0.4) = 0.8D + 0.1B$$

#4

$$V = D + L \quad \& \quad \frac{L}{D} = 2$$

$$Vy = Dx_D + Lx$$

$$y = \frac{L}{V}x + \frac{D}{V}x_D$$

$$y = \frac{L}{D+L}x + \frac{D}{D+L}x_D = \frac{4D}{D+4D}x + \frac{D/D}{D/D+4D/D}x_D = \frac{2}{3}x + \frac{1}{3}x_D$$

$$240 = 0.8(500 - B) + 0.1B$$

$$240 = 400 - 0.8B + 0.1B$$

$$B = 228.6 \text{ kmol/h}$$

$$D = 271.4 \text{ kmol/h}$$

$$y = \frac{2}{3}x + 0.267 \rightarrow \text{top op. line}$$

$$\hookrightarrow y = 0.5 = \frac{2}{3}x + 0.267 \quad x = 0.35$$

$$x = 0.5 \quad y = \frac{2}{3}(0.5) + 0.267 = 0.6$$

slope: $\frac{2}{3}$
y-int: 0.26667
 $y = x @ (0.8, 0.8)$

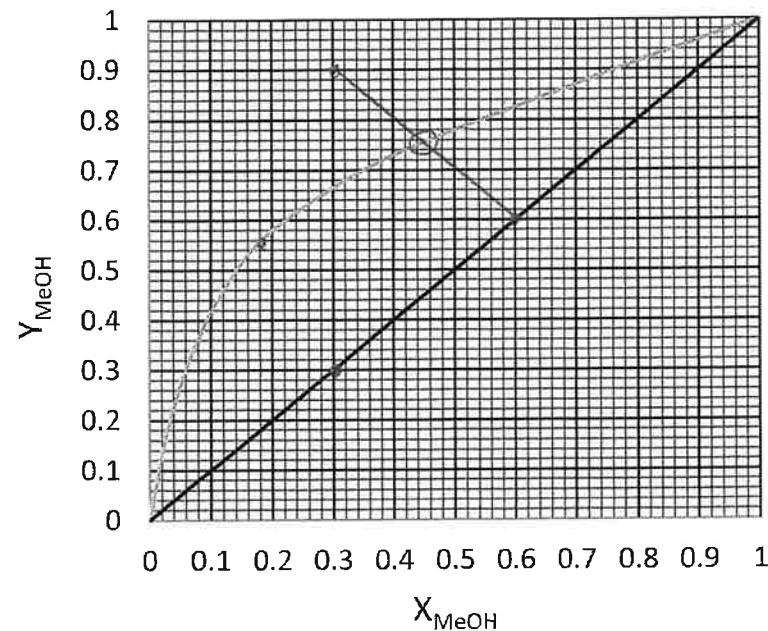
$$x = \frac{2}{3}x + 0.67$$

$$\frac{1}{3}x = 0.267 \quad x = 0.8$$

4) 8/8
5) 8/8

→

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

- 1st step: guess T_{drum} 2. DePriester chart for K
 3. Guess $\frac{V}{F}$
 4. RR newtonian procedure
 5. $RR < .005$
 6. x_i, y_i
 7. Energy balance

$$RR: 0 = \sum \frac{(K_i - 1) z_i}{1 + (K_i - 1) \frac{V}{F}}$$

$$\frac{dRR}{d\frac{V}{F}}: 0 = \sum \frac{(K_i - 1) z_i}{[1 + (K_i - 1) \frac{V}{F}]^2}$$

Aliya Kareemullah Exam 1

1.) Overall mass balance: $F = L + V$

component balance : $Fz = Lx + Vy$

operating eq $Vy = Fz - Lx$

$$y = \frac{F}{V}z - \frac{L}{V}x$$

3/3

2. $F = 150 \frac{\text{kmol}}{\text{h}}$ $z = .6$ $\frac{V}{F} = .5$ $\frac{L}{F} = .5$ $\frac{L}{V} = \frac{L/F}{V/F} = 1$

op eq: $y = \frac{1}{.5}x \cdot .6 - x$

$y = -x + 1.2$ check $-.6 + 1.2 = .6 \checkmark$

$(.6, .6)$ and $(.3, .9)$

$-.3 + 1.2 = .9$

From graph $x_m = .45$ $y_m = .76$

$L = F - V$

$V(.76) = 150 \frac{\text{kmol}}{\text{h}} \times .6 - (150 - V) \times .45$

$.76V - .45V = 22.5 \frac{\text{kmol}}{\text{h}}$

$V = 72.58 \frac{\text{kmol}}{\text{h}}$ $L = 150 - 72.58 = 77.42 \frac{\text{kmol}}{\text{h}}$

$V = 150 \times 0.5 = 75 \text{ kmol/h}$
 $L = 150 - 75 = 75$

3/5

3. $z = .3$ want $x_m = .18$

one pt is $(.3, .3)$. Based on $x_m = .18$, other pt is $(.18, .56)$ $y_m = .56$

slope from op eq = $-\frac{L}{V} = -\frac{L/F}{V/F} = -\frac{1 - V/F}{V/F} = \frac{.3 - .56}{.3 - .18} = -2.167$

$1 - \frac{V}{F} = \frac{V}{F} \cdot 2.167$

$1 = \frac{V}{F}(1 + 2.167)$

$\frac{V}{F} = .316$ $V = .316 \times 150 \frac{\text{kmol}}{\text{h}} = 47.4 \frac{\text{kmol}}{\text{h}}$

$L = 150 - 47.4 = 102.6 \frac{\text{kmol}}{\text{h}}$

from earlier $x_m = .18$ $y_m = .56$

10/10

3.) overall MB: $F_1 + F_2 = D + B$

Component: $F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$B = F_1 + F_2 - D$

$F_1 z_1 + F_2 z_2 = D x_D + (F_1 + F_2 - D) x_B$

$240 = D \cdot 8 + (500 \times 1) - D \cdot 1$

$190 = .70$

$D = 271.43 \frac{\text{kmol}}{\text{h}}$

$B = 500 - 271.43 = 228.57 \frac{\text{kmol}}{\text{h}}$

Top: $V = D + L$

$\frac{L}{D} = 2$

$L = 2 \times 271.43 = 542.86 \frac{\text{kmol}}{\text{h}}$

$V = D + L = 271.43 + 542.86 = 814.59 \frac{\text{kmol}}{\text{h}}$

Feed 1:

$F_1 + V' + L = V + L'$

$F_1 z_1 + V' y' + L x = V y + L' x'$

$x_D = x = y = .8$

$L' = F_1 + V' + L - V$

$200 \times .6 + V' y' + 542.86 \times .8 = 814.59(.8) + (F_1 + V' + L - V) x'$

$L' - L = F_1$ $L' = L + F_1 = 542.86 + 200 = 742.86 \frac{\text{kmol}}{\text{h}}$

$V' = V - F_1 + F_1 = 814.59 \frac{\text{kmol}}{\text{h}}$

Feed 2

$F_2 + \bar{V} + L' = V' + \bar{L}$

$\frac{V'}{F_2} = .2$

$V_{F_2} = \bar{V} - V' = .2 \times 300 = 60$

$L_{F_2} = 240 = L' - \bar{L}$

$300 + 60 + 742.86 = \bar{L} = 1102.86 \frac{\text{kmol}}{\text{h}}$

$\bar{V} = 60 + 814.59 = 874.59 \frac{\text{kmol}}{\text{h}}$

14/15

4.) slope = $\frac{L}{V} = \frac{L}{L+D} = \frac{40}{40+1} = \frac{2}{3}$
 $y_{int} = (1 - \frac{2}{3})x_D = \frac{1}{3} \cdot 8 = .2667$
 2 pts: $(8, .8)$ and $(0, .2667)$

8/8

5.) slope = $\frac{L}{V} = \frac{1102.86}{754.59} = 1.46$

1.3

@ $y=1$ $1 = 1.46x - .46x \cdot 1$
 $x = .716$

Method is ok!

7/8

2 pts: $(0, .1)$ and $(.716, 0)$

6.) a.) $q_1 = 1$ (saturated liquid)
 Thus slope is ∞ thru $x_1 = z_1 = .6$

✓

b.) $q_2 = .8$ slope = $\frac{.8}{-.2} = -4$

x_{int} $0 = -4x + \frac{.4}{.2}$ $x = 1$ 0.5

2 pts: $(.4, .4)$ and $(1, 0)$

6/8

7.) $F_1 z_1 + V_1 y_1 + Lx = L'x' + Vy_2$

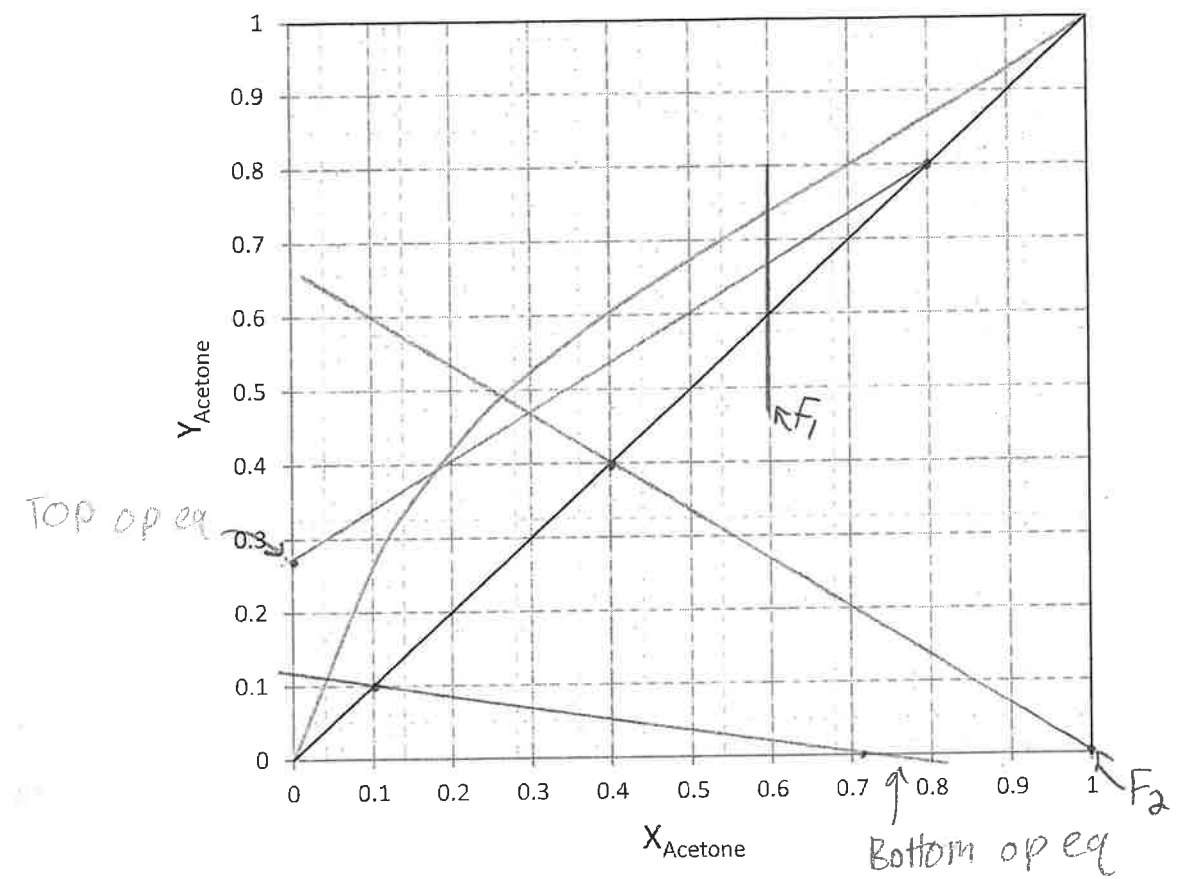
$F_1 + V_1 + L = L' + V$

$V_1 y_1 =$

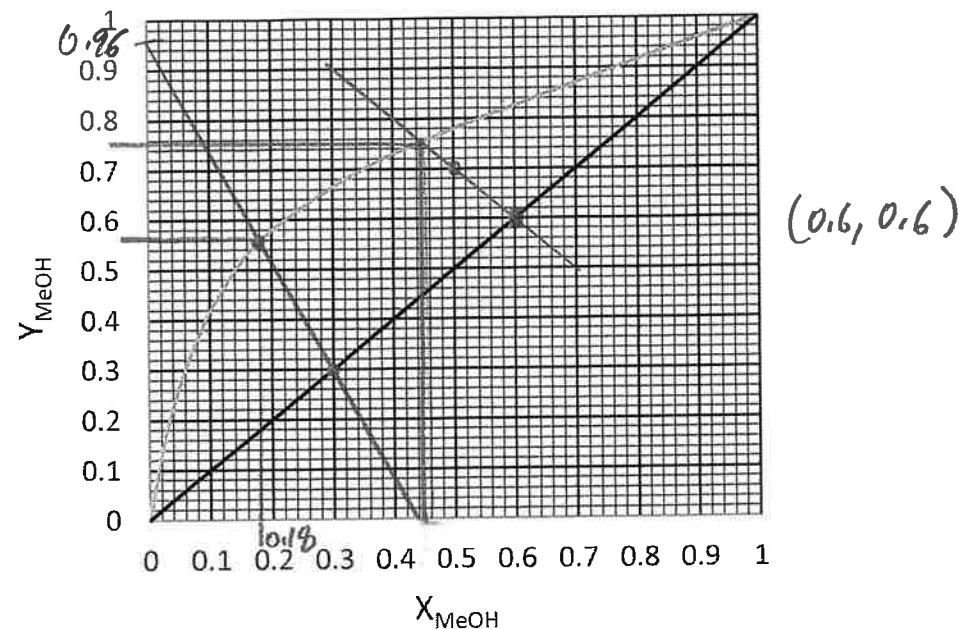
$F_2 + V_2 = L' + D$

$F_2 z_2 + V_2 y_2 = L'x + Dx_D$

1/10



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
- overall $F = V + L$ $Vy = Lx + Fz$
 component $Fz = Vy + Lx$ $y = \frac{L}{V}x + \frac{F}{V}z$
- 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

guess $\frac{V}{F}$ and T_{drum} to find x_i

- get K values

- use xyT table

$x_i = \frac{z_i}{1 + (K_i - 1)\frac{V}{F}}$

to find y_i

$y_i = K_i x_i$

$\downarrow$

$0 = \sum_{i=1}^n \frac{(K_i - 1) z_i}{1 + (K_i - 1)\frac{V}{F}}$

use $\left(\frac{V}{F}\right)_{K+1} = \left(\frac{V}{F}\right)_K - \frac{f_K}{\left(\frac{df_K}{d\frac{V}{F}}\right)}$ to find new $\frac{V}{F}$ values

1. overall: $F = V + L$

component: $Fz = Vy + Lx$

operation: $y = -\frac{L}{V}x + \frac{F}{V}z$

3/3

2) $F = 150 \frac{\text{kmol}}{\text{h}}$

$(0.6, 0.6) \quad x=y=z$

$z_M = 0.6$

$V = 150(0.5) = 75 \frac{\text{kmol}}{\text{h}} \quad V$

$L = 150 - 75 = 75 \frac{\text{kmol}}{\text{h}} \quad V \quad y = -\frac{75}{75}x + \frac{150}{75}(0.6)$

$y = -1x + 1.2 \quad (0.5, 0.7)$

$-1(0.5) + 1.2 = 0.7$

5/5

from graph

$x = 0.45 \quad \checkmark$

$y = 0.75 \quad \checkmark$

3. $F = 150 \frac{\text{kmol}}{\text{h}}$

$z = 0.3$

$y = -\frac{L}{V}x + \frac{F}{V}z$

$(0.3, 0.3) \quad x=y=z$

$(0.18, 0.56)$

graph

$x = 0.18 \rightarrow y = 0.56$

slope: $\frac{0.56 - 0.3}{0.18 - 0.3} = \frac{0.26}{-0.12} = -2.1\bar{6}$

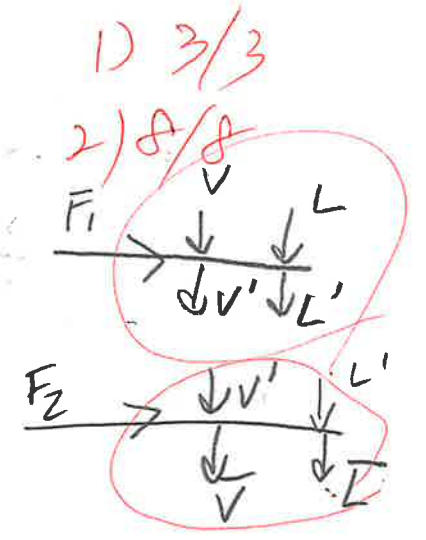
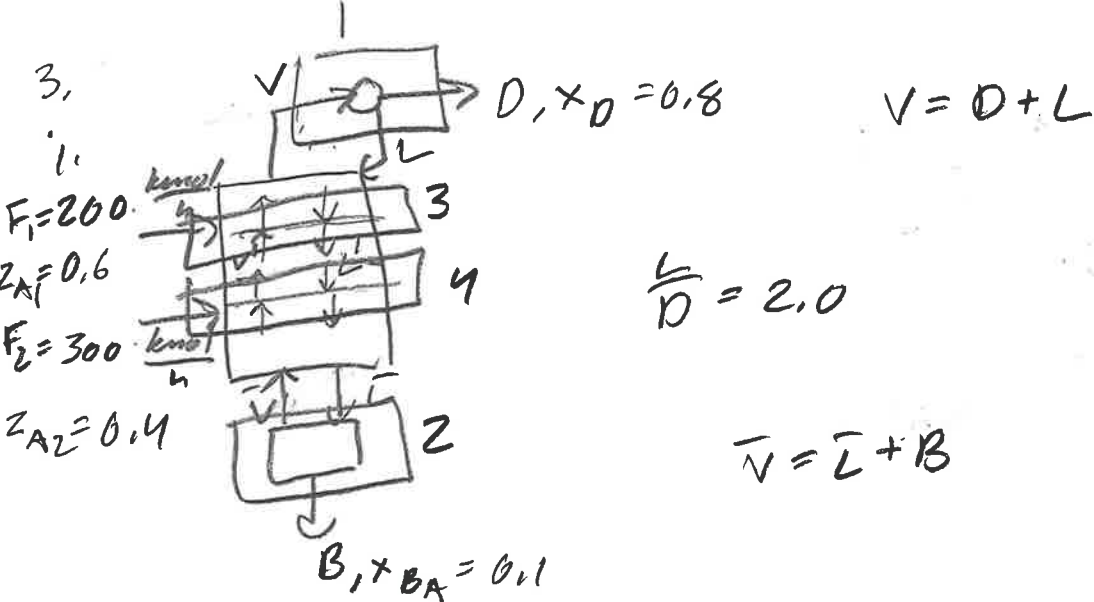
y-intercept: $0.96 = \frac{F}{V}z \quad \checkmark \quad z = 0.3$

$\frac{0.96}{0.3} = \frac{F}{V} = 3.2$

10/10

$\frac{F}{V} = 3.2 \quad V = \frac{F}{3.2} = \frac{150}{3.2} = 46.875 \frac{\text{kmol}}{\text{h}}$

$L = 150 - 46.875 = 103.125 \frac{\text{kmol}}{\text{h}}$



2. i. enricher
 $V = D + L$
 $V y_1 = D x_D + L x_1$
 2. stripping
 $\bar{V} = \bar{L} + B$
 $\bar{V} y_2 = \bar{L} x_2 + B x_B$

3. Feed
 $F_1 = L - L'$
 $F_1 + L + V' = V + L'$
 $F_1 z_1 + L x_1 + V' y_3 = V y_1 + L' y_3$
 4. Feed
 $F_2 + L' + \bar{V} = V' + \bar{L}$
 $F_2 z_2 + L' x_3 + \bar{V} y_2 = V' y_3 + \bar{L} x_2$
 $L' \neq \bar{L} \neq L$
 $V' \neq \bar{V} \neq V$

4. $\bar{V} = D + L$ $\frac{L}{D} = 2$ $x_D = 0.8$

$$\frac{y}{D} = \frac{D}{D} + \frac{L}{D}$$

$$\frac{y}{V} = \frac{L}{D+L} = \frac{\frac{L}{D}}{\frac{D}{D} + \frac{L}{D}} = \frac{2}{1+2} = \frac{2}{3}$$

7/8 $\frac{2}{3}$ - slope $x_D = x_1 = y_1$
(0.8, 0.8)

$$y = \frac{2}{3}x + (1 - \frac{2}{3})(0.8)$$

$$y = \frac{2}{3}x + 0.2\bar{6} - D \text{ line}$$

$$\frac{2}{3}(0.7) + 0.2\bar{6} = 0.7\bar{3}\bar{3}$$

(0.7, 0.7 $\bar{3}$)

Supply, y-intercept = (0, 0.27)

$$6. \quad y = \frac{q_1}{q_1 - 1} x + \frac{z_1}{1 - q} \quad q_i = \frac{L F_i}{F_i} \quad 6) \quad 6/10$$

a. F_1

$$y_1 = \frac{\cancel{1}}{\cancel{1-1}} x + \frac{0.6}{1-1}$$

$$y = 0$$

$$(0.6, 0.6)$$

$$z_{A1} = 0.6$$

$$F_1 = 200 \frac{\text{kmol}}{\text{h}}$$

$$\text{sat. liq. } q = 1$$

$$z_{A2} = 0.4$$

$$F_{A2} = 300 \frac{\text{kmol}}{\text{h}}$$

b. F_2

$$y = \frac{0.8}{0.8 - 1} x_2 + \frac{0.4}{1 - 0.8}$$

$$\frac{L F_2}{F_2} = \frac{1 - V F_2}{F_2} = 1 - 0.2$$

$$\frac{L F_2}{F_2} = 0.8$$

$$y_2 = -4x_2 + 2$$

$$x_2 = 0.3 \rightarrow y = 0.8$$

$$(0.4, 0.4)$$

$$(0.3, 0.8)$$

$$\left. \begin{array}{l} z = y = x \\ \text{plot} \end{array} \right\}$$

$$x\text{-intercept} = (0.5, 0)$$

$$7. \quad V + L' = L + V' + F_1$$

$$F_2 + V' = L' + D$$

$$7) \quad 1/10$$

$$V y_1 + L' x_3 = L x_1 + V' y_3 + F_1 z_1$$

$$F_2 z_2 + V y = L' x + D x_D$$

$$y = \frac{L'}{V} x + \frac{D x_D - F_2 z_2}{V}$$

$$3.3. \quad F_1 = L - L'$$

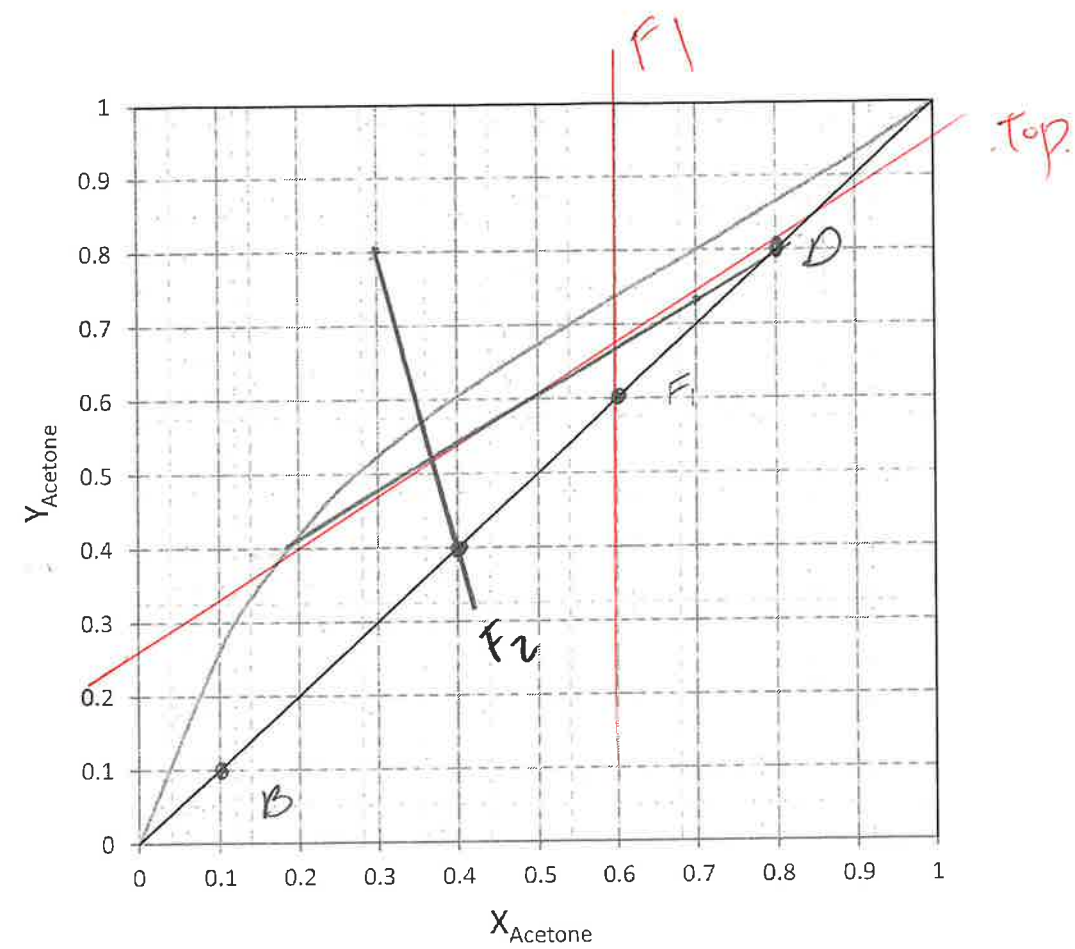
$$F_1 + L + V' = V + L'$$

$$F_{21} = L x_1 - L' x_3$$

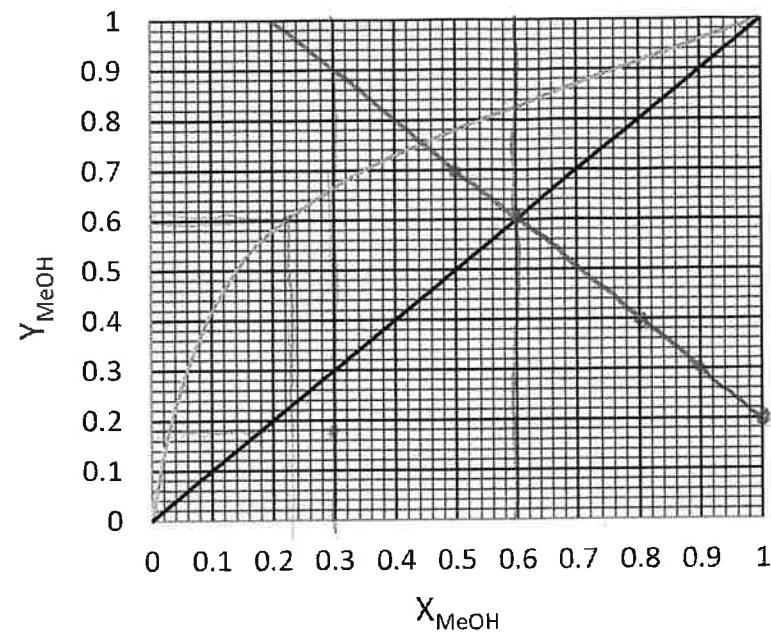
$$V' = V \quad - \text{no vapor going in}$$

$$V = D + L$$

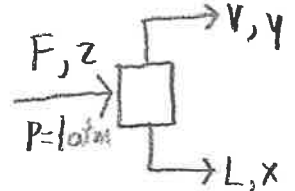
$$V y_1 = D x_D + L x_1$$



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1.1)  $F = V + L$; $Fz_1 = Vy_1 + Lx_1 \Rightarrow y_1 = \frac{Fz_1 - Lx_1}{V}$ $y = -\frac{L}{V}x_1 + \frac{F}{V}z_1$

3/3

2) $F = 150 \frac{\text{kmol}}{\text{hr}}$; $z_1 = .6$; $V = 75 \frac{\text{kmol}}{\text{hr}}$ $L = (5)(150 \frac{\text{kmol}}{\text{hr}}) = 75 \frac{\text{kmol}}{\text{hr}}$ $m = -\frac{L}{V} = -\frac{75}{75} = -1$
 $V = .5(150 \frac{\text{kmol}}{\text{hr}})$

$y(x=0) = 0 + \frac{150}{75}(.6) = 1.2 \Rightarrow (0, 1.2)$ $y(x=1) = -1 + 1.2 = .2$

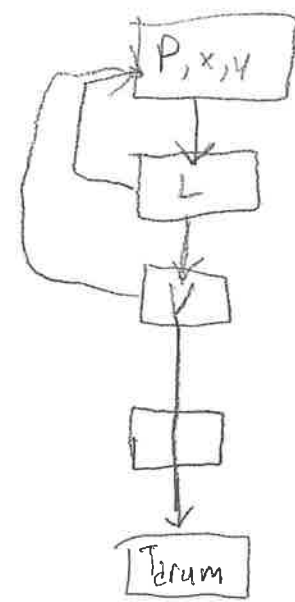
~~$x_1 = .23$~~ ~~$y_1 = .60$~~ (0.45, 0.75), .2

3/5

3) $z_1 = .3$ $F = 150 \frac{\text{kmol}}{\text{hr}}$ $L = 60 \frac{\text{kmol}}{\text{hr}}$ $V = 90 \frac{\text{kmol}}{\text{hr}}$ $x_1 = .18$ $y_1 =$
 $x_1 = .18$ $\frac{V}{F} = \frac{.18}{.30}$ $y_1 = -\frac{L}{V}(.18) + \frac{150}{90}(.3)$

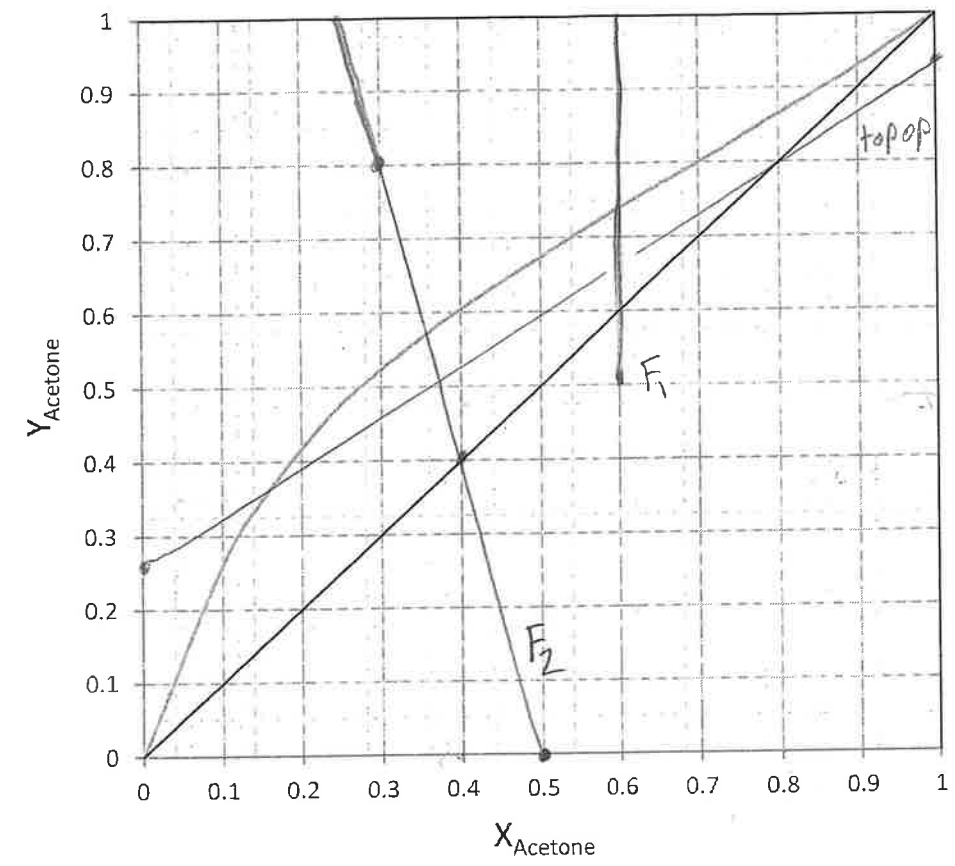
1/10

2.

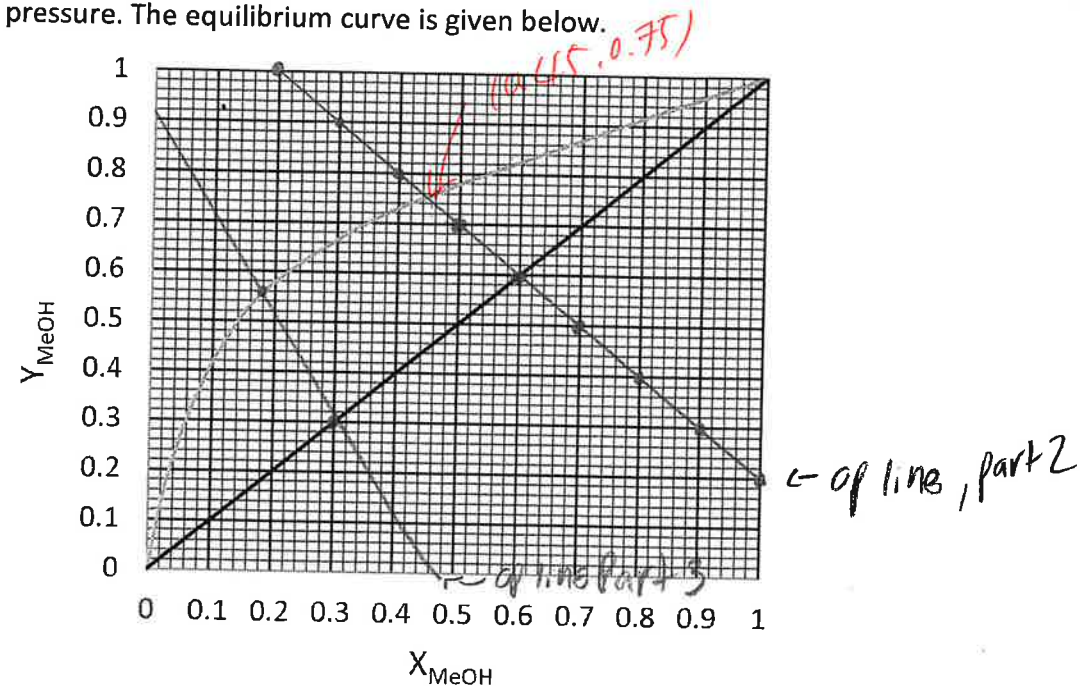


1/10

Daniel Brouwer

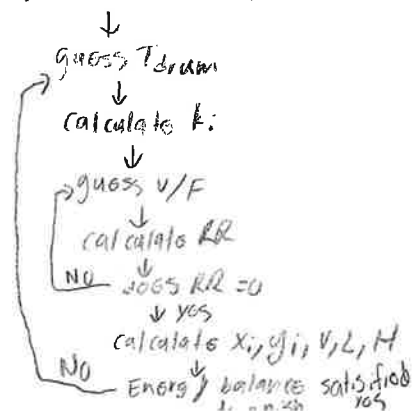


1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.

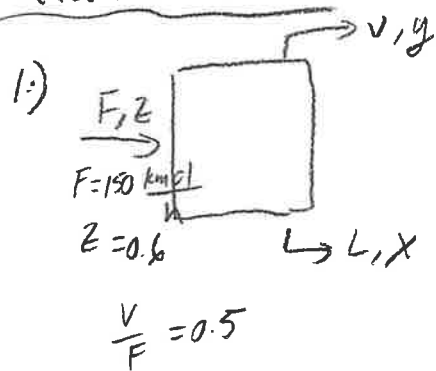


- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
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Input spec condition / Eq Data



Question 1



Overall MB: $F = V + L$

Component MB: $Fz = Vy + Lx$

Operating Eqn: $Fz - Lx = Vy$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

3/3

2.) $\frac{V}{F} = 0.5 \rightarrow V = 0.5(F) \Rightarrow 0.5(150) = 75 \frac{\text{kmol}}{\text{hr}} = V$ ✓

$L = F - V \Rightarrow L = 150 - 75 = 75 \frac{\text{kmol}}{\text{hr}}$ ✓

Operating line $\rightarrow y = -\frac{75}{75}x + \frac{150}{75}(0.6)$

$y = -x + 1.2$

$y \text{ intercept} = (0, 1.2)$

$x = 0.45$
 $y = 0.75$

5/5

X	y
0	1.2
0.1	1.1
0.2	1
0.3	0.9
0.4	0.8
0.5	0.7
0.6	0.6

10/10

$y = -\frac{L}{V}x + \frac{F}{V}z$

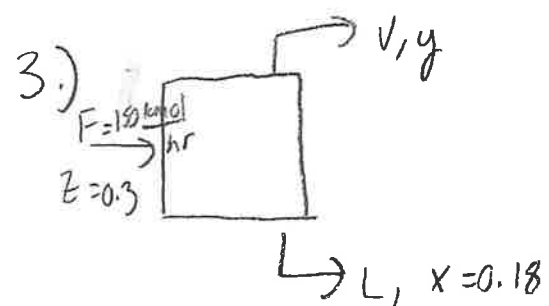
$y \text{ intercept using graph @ } (0, 0.9)$

$\frac{F}{V}z = 0.9 \rightarrow \frac{F}{V}(0.3) = 0.9$

$\frac{F}{V} = 3 \rightarrow \frac{V}{F} = \frac{1}{3}$ ✓

$V = \frac{1}{3}(F) = \frac{1}{3}(150) = 50 \text{ kmol/hr}$

$L = F - V = 150 - 50 = 100 \frac{\text{kmol}}{\text{hr}}$



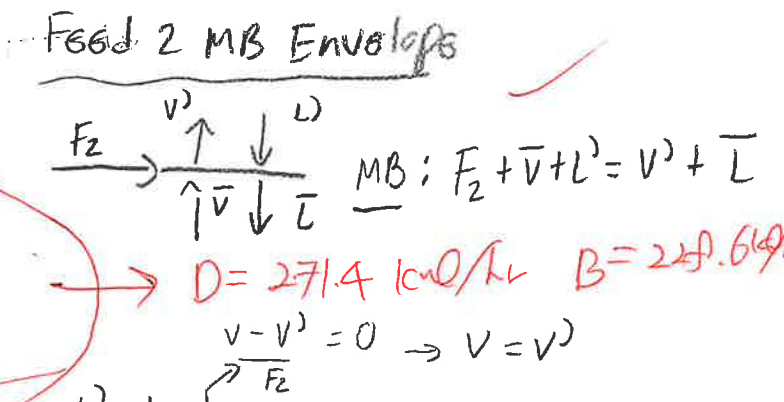
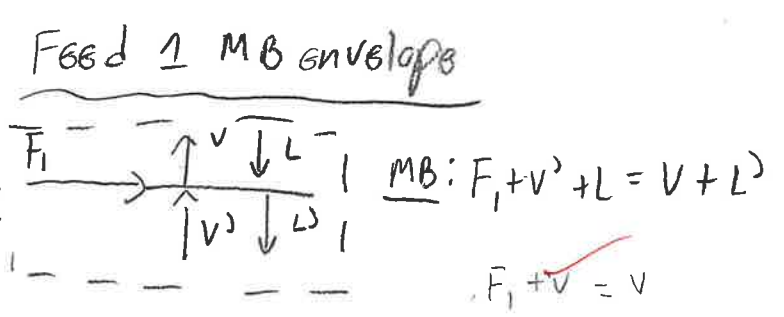
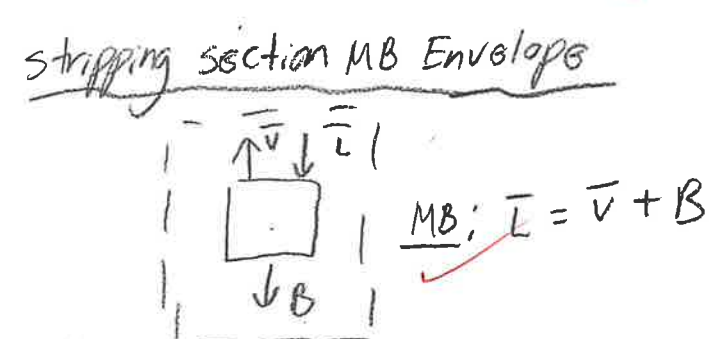
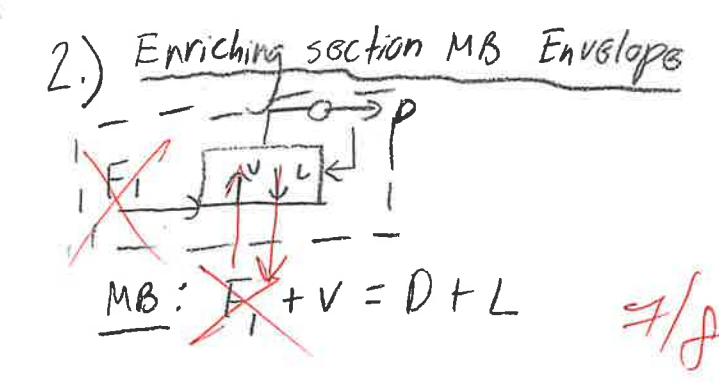
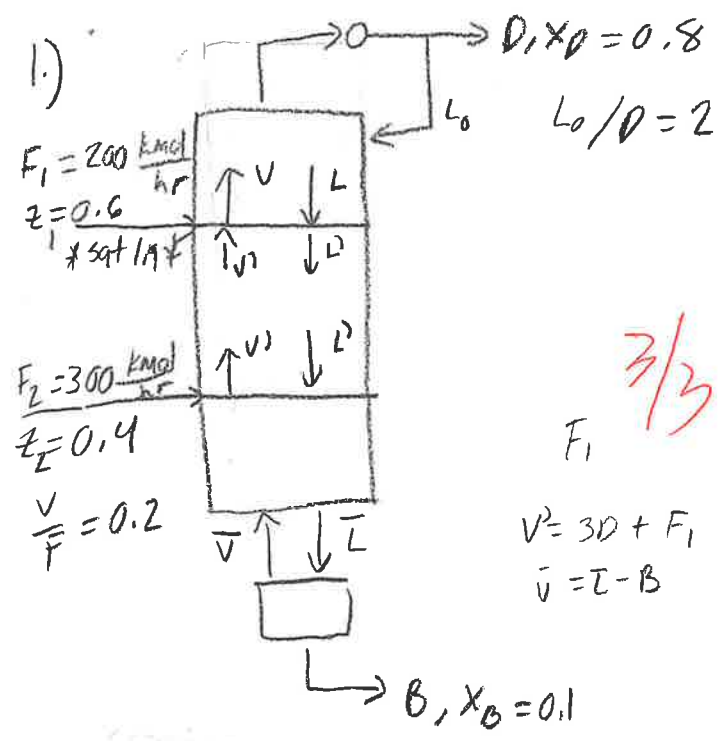
using VLE data
 $(0.18, 0.56)$

$x = 0.18$
 $y = 0.56$

Question 3

out-in

* Remember feed 1 is sat liq *



3.) overall MB: $F_1 + F_2 = D + B$
 comp MB: $F_1 z_1 + F_2 z_2 = D x_D + B x_B$

We know Feed 1 is sat liq $\rightarrow q_1 = 1 = \frac{L_F}{F_1} = \frac{L' - L}{F_1} \rightarrow L' - L = F_1 = 200$

We know Feed 2 $\frac{V_F}{F} = 0.2 \rightarrow \frac{V_{F2}}{F_2} = \frac{V' - \bar{V}}{F_2} = 0.2 \rightarrow V' - \bar{V} = 0.2(F_2) = 60$

$F_1 + V' + L = V + L'$
 $V = V' + L - F_1$
 $\bar{V} = L - B$
 $F_1 + V' - L = 200 + V$

$\frac{L_{F2}}{F_2} = 0.8 = \frac{L - L'}{F_2} \rightarrow L - L' = 0.8(300) = 240$

$D = 271.4 \text{ kmol/hr}$
 $B = 228.6 \text{ kmol/hr}$



Question 3, Pt 3 continued...

$$V' - \bar{V} = 60$$

$$\bar{L} - L' = 240$$

$$L' - L = 200$$

$$V' - V = 0 \rightarrow V' = V$$

$$\bar{V} = V' + 60$$

$$F_2 + \bar{V} = \bar{L} - L' + V'$$

$$300 + V' - 60 = 240 + V'$$

$$\begin{array}{l} \bar{V} = -60 \frac{\text{kmol}}{\text{hr}} \\ V' = 0 \frac{\text{kmol}}{\text{hr}} \\ V = 0 \frac{\text{kmol}}{\text{hr}} \end{array}$$

$$F_1 + V = D + L \rightarrow F_1 = 3L \rightarrow 200 = 3L \rightarrow L = 66.67 \frac{\text{kmol}}{\text{hr}}$$

$$L = 2D \rightarrow D = \frac{L}{2} = \frac{66.67}{2} = 33.33 \frac{\text{kmol}}{\text{hr}} = D$$

$$L' - L = 200 \rightarrow L' = 200 + L = 200 + 66.67 = 266.7 \frac{\text{kmol}}{\text{hr}} = L'$$

$$\bar{L} - L' = 240 \rightarrow \bar{L} = 240 + L' = 240 + 266.7 = 506.7 \frac{\text{kmol}}{\text{hr}} = \bar{L}$$

$$B = F_1 + F_2 - D = 200 + 300 - 33.33 = 466.7 \frac{\text{kmol}}{\text{hr}} = B$$

5/15

4.) Top of line

$$\text{slope} = \frac{L}{V} = \frac{66.67}{0} = \infty$$

$$\text{coordinate on } y=x \rightarrow (0.8, 0.8)$$

$$y \text{ intercept} = (1 -$$

2/8

5.) Bottom of line

$$\text{slope} = \frac{\bar{L}}{V} = \frac{506.7}{-60} = -8.445$$

$$\text{coordinate } y=x \rightarrow (0.1, 0.1)$$

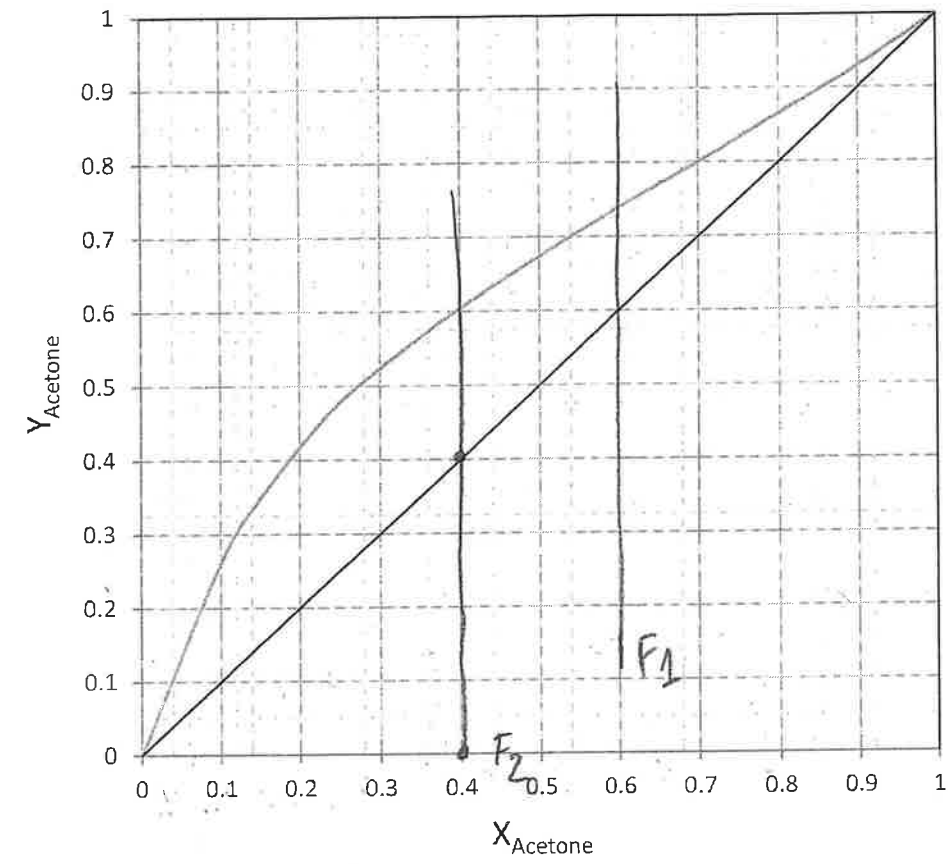
$$y \text{ intercept} = -\left(\frac{\bar{L}}{V} - 1\right) x_B = -(-8.445 - 1) 0.1 = 0.94 \rightarrow (0, 0.94)$$

3/8

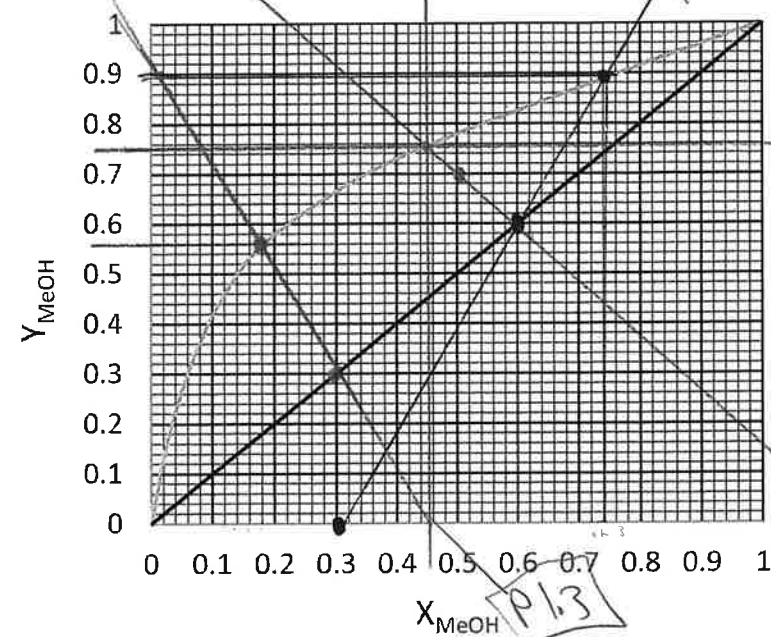
0.79



Iliana Perez



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
- 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
- 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

RR

k-values, V/F , $z_i \rightarrow$ RR eqn

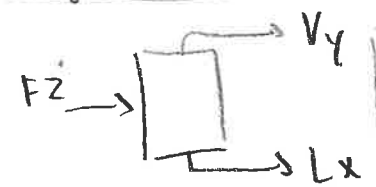
$$F = V + L$$

$$Fz = Vy + Lx$$

$$y = Kx$$

$$Fz = V(Kx) + Lx \rightarrow$$

$$z = \frac{V}{F} Kx + \left(1 - \frac{V}{F}\right) x$$



1.1.)

Overall MB: $F = V + L \rightarrow 150 = V + L$
 Comp. MB: $Fz = Vy + Lx \rightarrow 150 \cdot 0.6 = Vy + Lx$
 $\hookrightarrow 90 = Vy + Lx$

$Fz = Vy + Lx$
 $Vy = Fz - Lx$
 $y = \frac{Fz}{V} - \frac{Lx}{V}$

operating eqn $Fz = Vy + Lx \rightarrow Vy = Fz - Lx \rightarrow y = \frac{Fz}{V} - \frac{Lx}{V}$ $F = V + L \rightarrow \frac{F}{V} = 1 + \frac{L}{V}$

$y = \frac{Fz}{V} - \frac{Lx}{V} \rightarrow y = \frac{Fz}{V} - (\frac{F}{V} - 1) \cdot x$ $\frac{F}{V} = \frac{1}{f} \rightarrow y = \frac{z}{f} - (\frac{1}{f} - 1) \cdot x$ $\frac{L}{V} = \frac{F}{V} - 1$

opn. eqn.

$z = 0.60, f = \frac{V}{F} = 0.5 \rightarrow y = \frac{0.60}{0.50} - (\frac{1}{0.5} - 1) \cdot x$
 $\hookrightarrow \frac{F}{V} = 2$

$\hookrightarrow y = 1.2 - x$

3/3

1.2 $x=0 \rightarrow y=1.2$ ~~(0, 1.2)~~ $z=0.6 \rightarrow (0.6, 0.6)$ $\hat{z}(0.5, 0.7)$

$y = 1.2 - 0.5 = 0.7 \rightarrow$ intersect $\rightarrow x=0.45, y=0.75$

$90 = 0.75V + 0.45L$

$150 = V + L \rightarrow V = 150 - L$

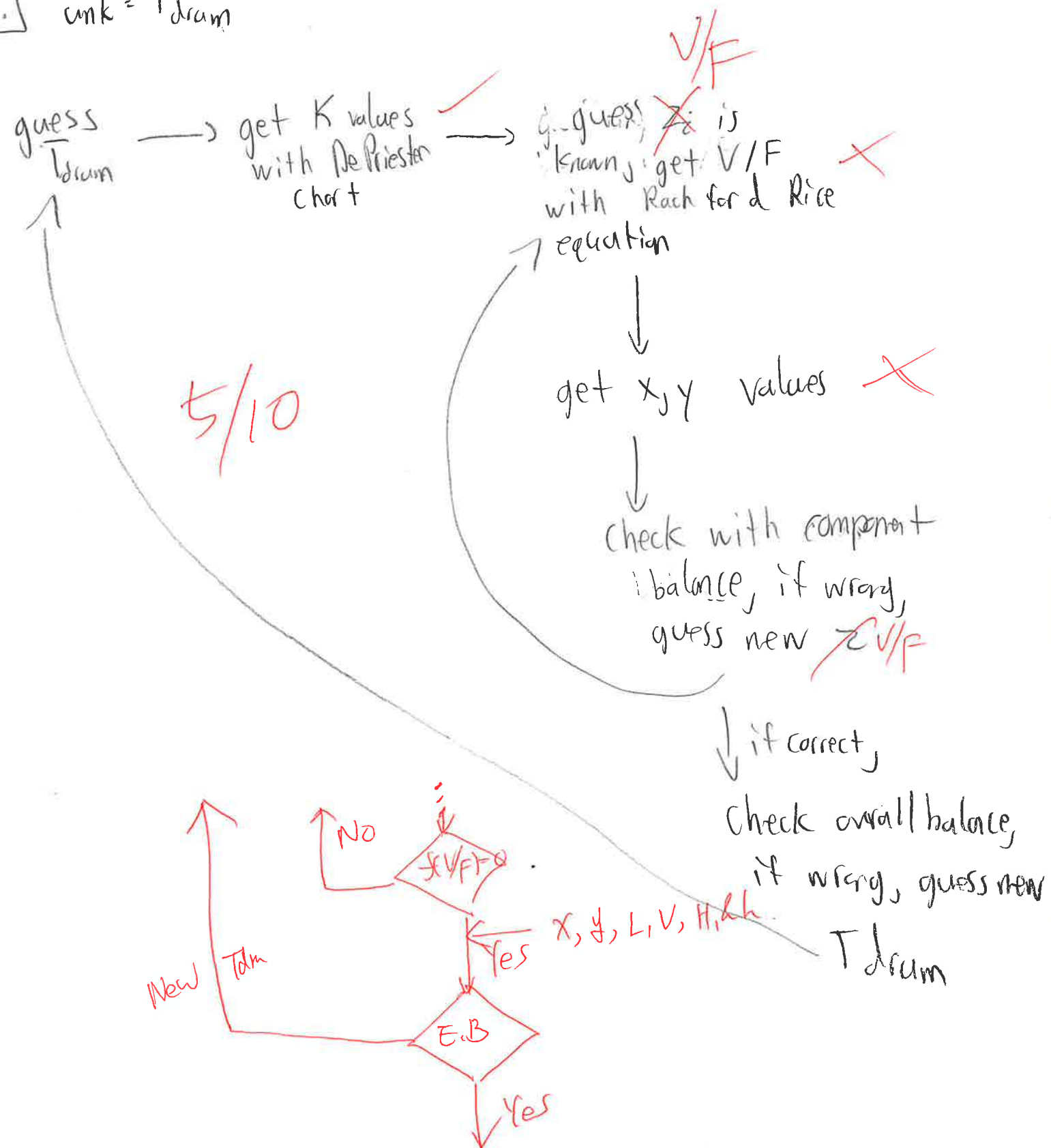
$90 = 0.75(150 - L) + 0.45L \rightarrow 90 = 112.5 - 0.75L + 0.45L$

$L = 75 \rightarrow V = 75$

5/5

$L = 75 \text{ kmol/hr}$
 $V = 75 \text{ kmol/hr}$

2. unk = T_{drum}



D, B, L, V, V', L', V, L

3.3 Overall MB: $F_1 + F_2 = D + B$
 comp: $F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$$500 = D + B \rightarrow D = 500 - B$$

$$240 = 0.8D + 0.10B$$

$$240 = 0.80(500 - B) + 0.10B$$

$$240 = 400 - 0.80B + 0.10B$$

$$B = 228.57 \text{ mol/hr}$$

$$D = 271.43 \text{ kmol/hr}$$

enriching m MB: $V = D + L$

a comp: $V y = D x_D + L x$

$$L/D = 2.0 \rightarrow L = 2D$$

$$V = D + 2D \rightarrow V = 814.29 \text{ kmol/hr} \rightarrow L = 542.86 \text{ kmol/hr}$$

stripping MB
+ Feeds

$$\bar{L} = \bar{V} + B$$

Feed $F_1 + F_2 + L = V + B$

$$F_2 + \bar{L}' + \bar{V} = \bar{V}' + \bar{L}$$

$$\bar{L} = \bar{V} + B$$

$$\bar{L}' = \bar{V}' + B$$

$$\bar{V}' = \bar{V} + F_1 + L - \bar{L}$$

$$\bar{L} = \bar{V} + B$$

$$F_2 + \bar{L}' + \bar{V} = \bar{V}' + \bar{L}$$

$$\bar{L} = \bar{V} + 228.57$$

$$300 + 742.86$$

$$\bar{V}' = 814.29$$

$$\bar{L}' = 742.86$$

$$\bar{L} = \bar{V} + B = 982.86$$

$D = 228.57 \text{ kmol/hr}$	$L = 542.86 \text{ kmol/hr}$	$L' = 742.86 \text{ kmol/hr}$	$\bar{V} = 754.29 \text{ kmol/hr}$
$B = 271.43 \text{ kmol/hr}$	$V = 814.29 \text{ kmol/hr}$	$V' = 814.29 \text{ kmol/hr}$	$\bar{L} = 982.86 \text{ kmol/hr}$

3.4 $y = \left(\frac{542.86}{814.14} \right) x + \left(1 - \frac{542.86}{814.14} \right) \cdot 0.80$

$y = 0.67x + 0.267$

slope: 0.67

y-int = 0.267

xy coord: (0.80, 0.80)

8/8

3.5 $y = \left(\frac{491.86}{754.14} \right) x + \left(\frac{491.86}{754.14} - 1 \right) \cdot 0.10$

$y = 1.3x + 0.030$

slope = 1.3

x-int at $y=1$: 0.75

xy coord: (0.10, 0.10)

$\rightarrow \frac{1 - 0.03}{1.3} = 0.75$

8/8

3.6 $q_1 = 0 \rightarrow F_1$ slope is ∞

a.) xy coord: (0.6, 0.6)

b.) $q_1 = 1 - 0.2 = 0.8 \rightarrow \text{slope} = \frac{0.8}{0.8 - 1} = -4$

xy coord: (0.4, 0.4)

x-int: 0.5

10/10

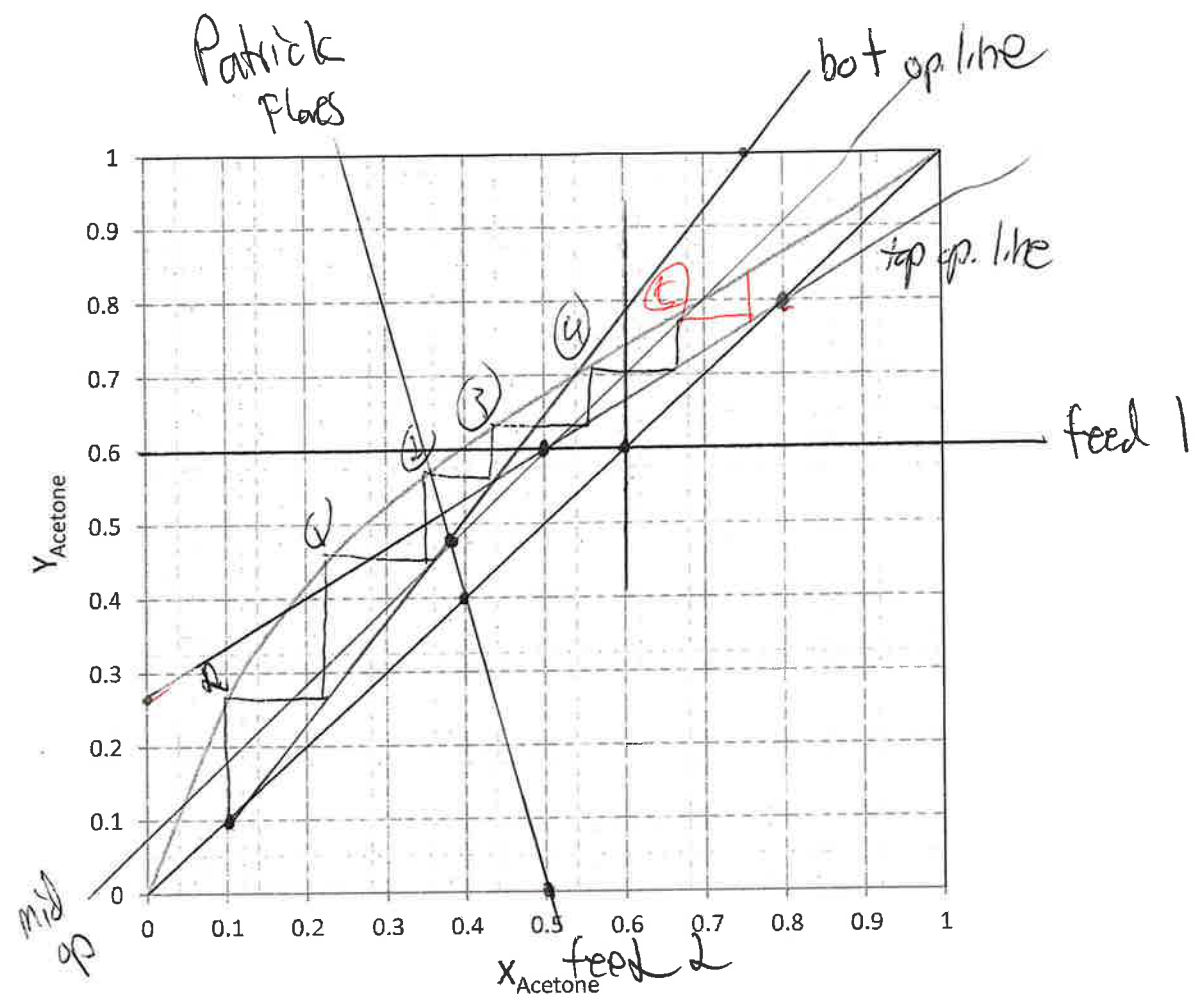
3.8 ≈ 4.5 stages 5 stages + PB

4/5

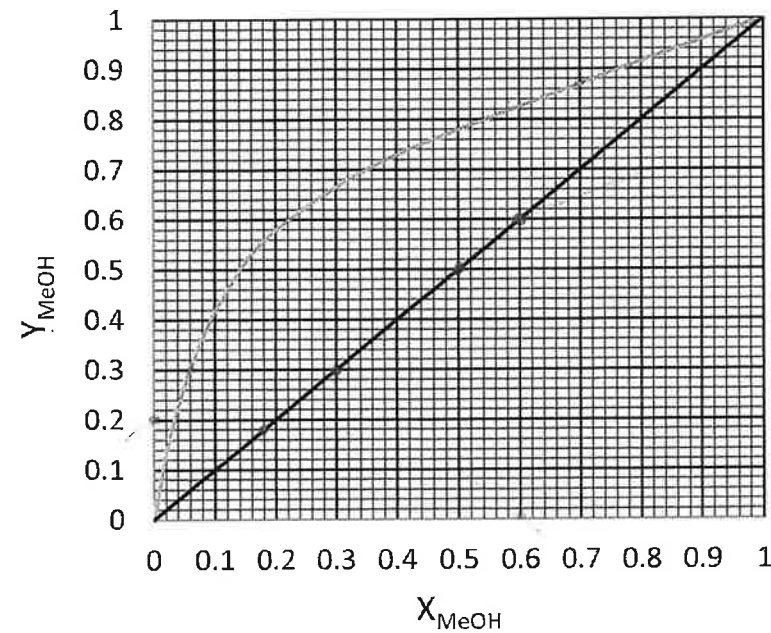
3.9 optimal location is stage 2 for feed 2, stage 3 for feed 1

2/3

4



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



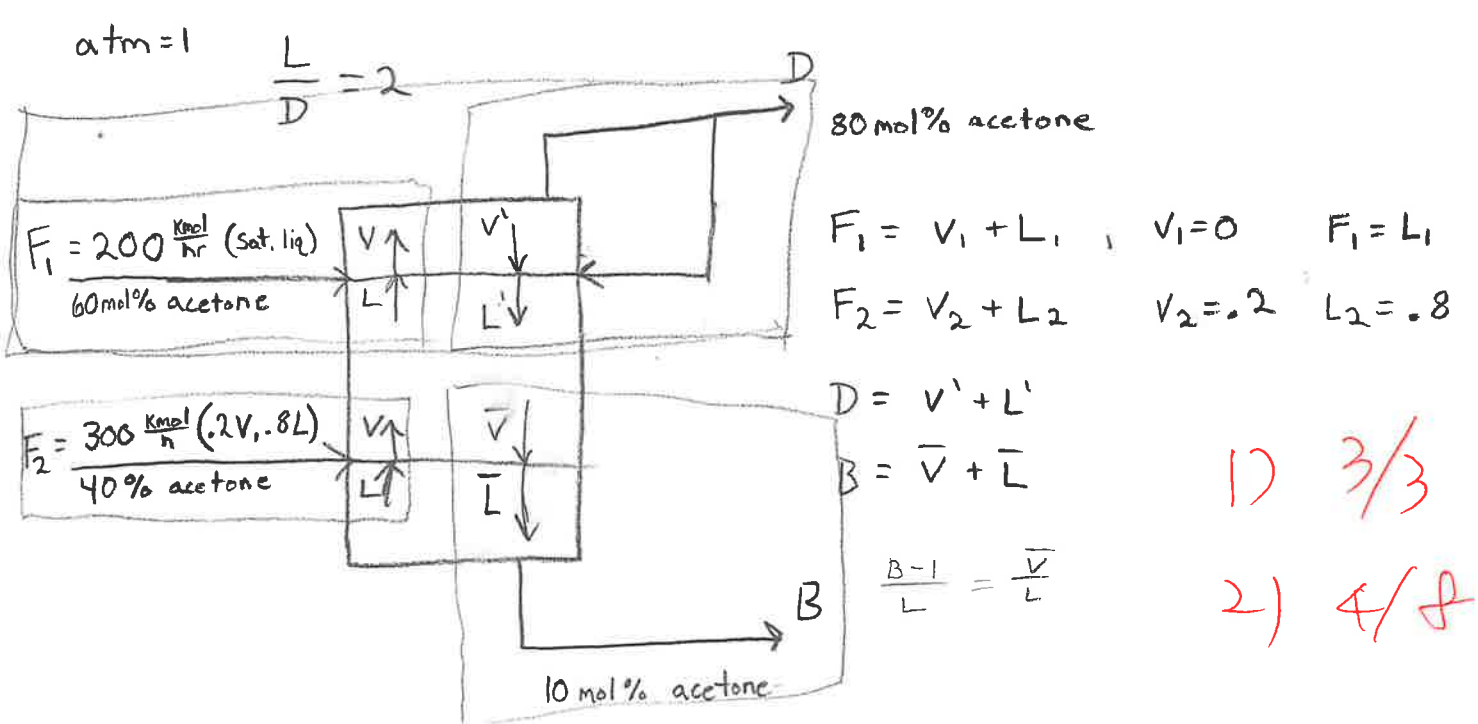
11/100

- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.

$$-L + F = V + L, \quad Fz = Vy + Lx, \quad y = -\frac{L}{V}x + \frac{F}{V}z$$
- 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction. $y = -\frac{.3}{.3}x + \frac{.6}{.3}z$ $x = .3$
 $y = .7$
- 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.

$$\frac{V}{F} = L - 1$$

2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V



1) 3/3
2) 4/8

$F = F_1 + F_2$
 $F = 500 \text{ kmol/h}$

$200(.6) + 300(.4)$
 $= 240 \frac{\text{kmol}}{\text{h}}$ acetone
 32.712 vap acetone
 207.288 liq

$300(.2) = 60 \frac{\text{kmol}}{\text{h}}$ (vap)
 $440 \frac{\text{kmol}}{\text{h}}$ (liq)
 $z = .1363$ vap

Top $y = \left(\frac{207.288}{32.712} \right) x + (1 - 6.33) x_D$

bot $y =$

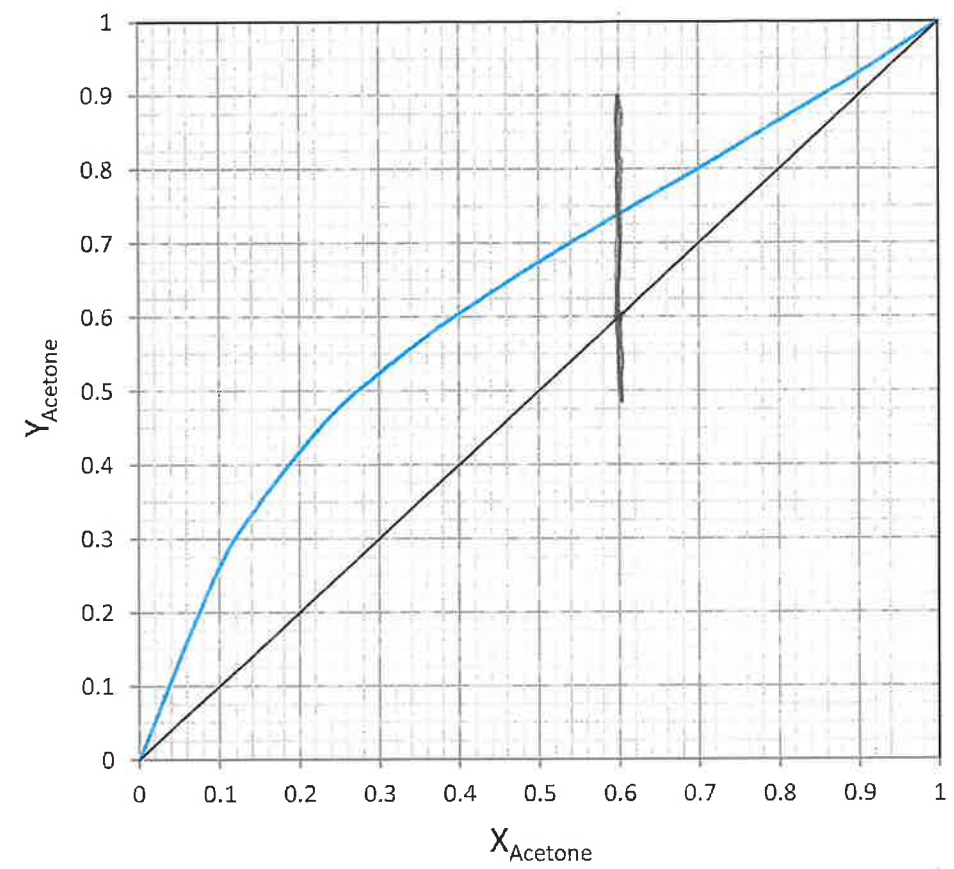
$q_i = \frac{L F_i}{F_i}$ $q_1 = 1$ $q_2 = .8$

$y_1 = \frac{1}{6} x + 0$ (.6, .6) slope ∞

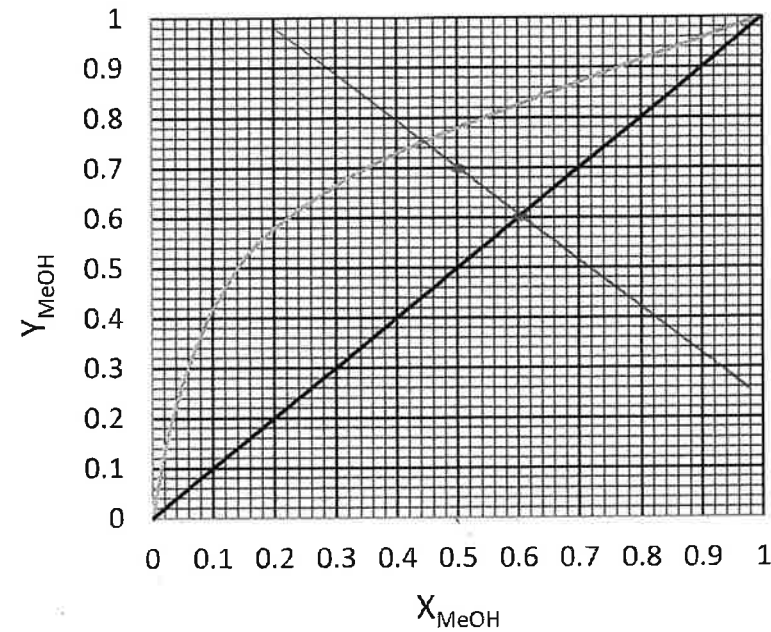
$y_2 = \frac{.8}{-.2} x + \frac{.2}{.2}$
 $-4x +$

4/10

$$240 = .8D + .1B$$

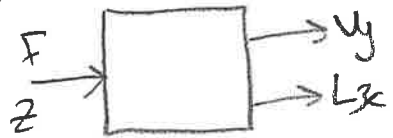


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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1.)



$$F_z = V_y + L_x$$

$$F = V + L$$

$$V_y = F_z - L_x$$

$$y = -\frac{L_x}{V} + \frac{F_z}{V}$$

3/3

2.) $F = 150 \text{ kmol/h}$

$$z = .6$$

$$V/F = 50\%$$

$$L = 150(.5) = 75 \text{ kmol/h}$$

$$V = 150(.5) = 75 \text{ kmol/h}$$

$$y = -\frac{75}{75}x + \frac{150(.6)}{75} = -x + 1.2$$

$$(.5, .7) \quad (.6, .6)$$

$$x = .43$$

$$y = .15$$

5/5

3.) $z = .3$

$$x = .18$$

$$F = 150$$

$$y = -\frac{L_x}{V} + \frac{F_z}{V}$$

$$F = V + L$$

$$150 = V + L$$

$$z = .3$$

$$V = F - L$$

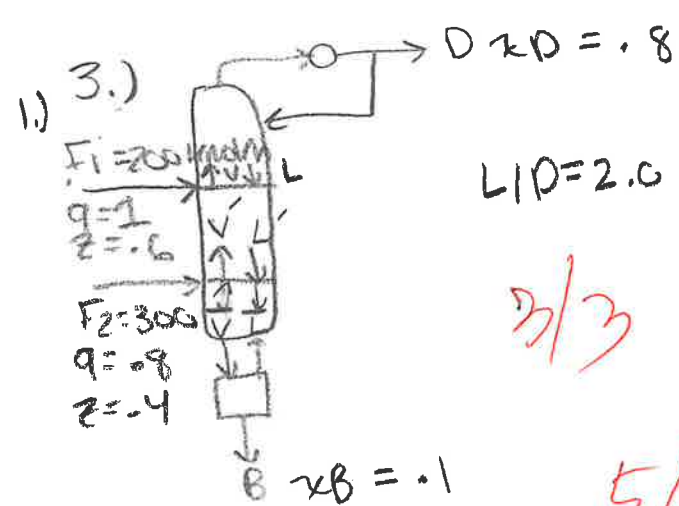
$$y = -\frac{L_x}{V} + \frac{150(.3)}{V}$$

$$V = 1 - L$$

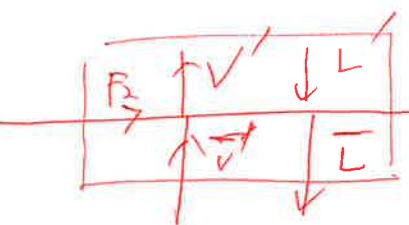
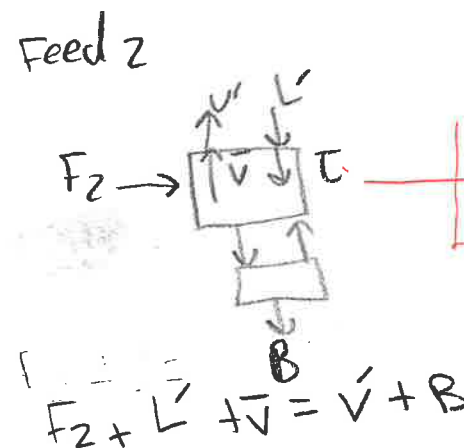
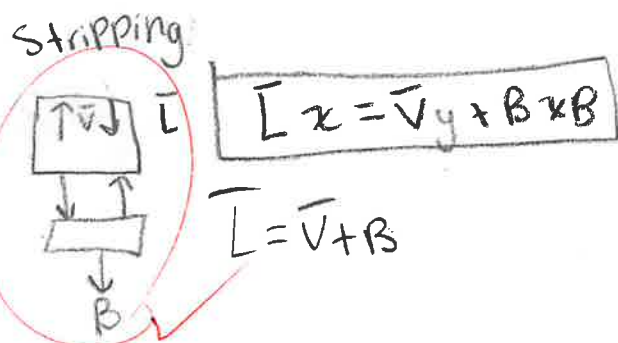
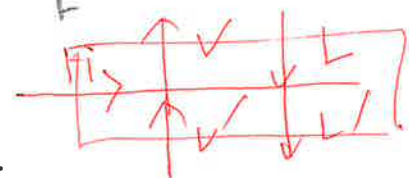
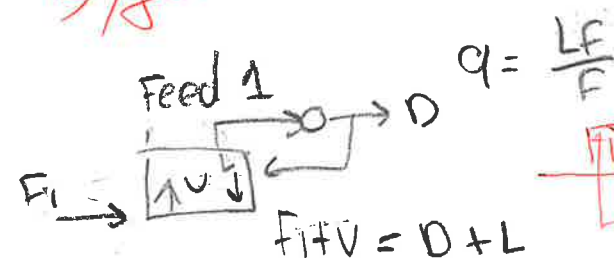
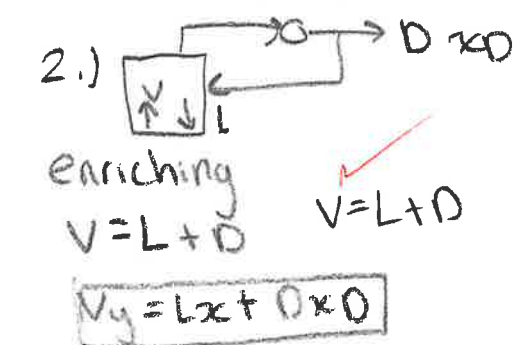
$$y = \frac{-L_x}{1-L} + \frac{150(.3)}{1-L}$$

Mustafa
Mezher

4/10



$$F_1 + F_2 = D + B$$



3.) overall

don't know D or B ??

$$F_1 + F_2 = D + B$$

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B$$

$$q = \frac{L - \bar{L}}{F_2}$$

$D = 271.4 \text{ kmol/hr}$
 $B = 228.6 \text{ kmol/hr}$

$240 = 66.67(0.8) + B(0.1)$

$B = 1866.64 \text{ kmol/hr}$

↑ makes no sense

$D = 1/2 L$

$F_1 = L + D$

$200 = L + D$

$D = 1/2 L$

$200 = 3/2 L$

$L = 133.3 \text{ kmol/hr}$

$D = 66.67 \text{ kmol/hr}$

likely wrong, no V

3/15

3.)

4.) $y = \frac{R}{R+1}x + \frac{x_0}{R+1}$ 6/8

$$y = \frac{2}{3}x + \frac{.8}{3}$$

$$y = \frac{2}{3}x + .27$$

(0.8, 0.8) on y=x
not (0.9, 0.9)

7.)

$$\sqrt{y} = L'x + D x_0$$

$$y = \frac{L'x}{\sqrt{}} + \frac{D x_0}{\sqrt{}}$$

5.)

starts *
at x_d

$$y = \frac{\bar{L}}{\bar{V}}x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

↑
don't know $\frac{\bar{L}}{\bar{V}}$ can't find

8.)

do steps
and can't
all of them

b.) $F_1 = \frac{1}{x-1} + \frac{2}{x-1}$

(2, 2)

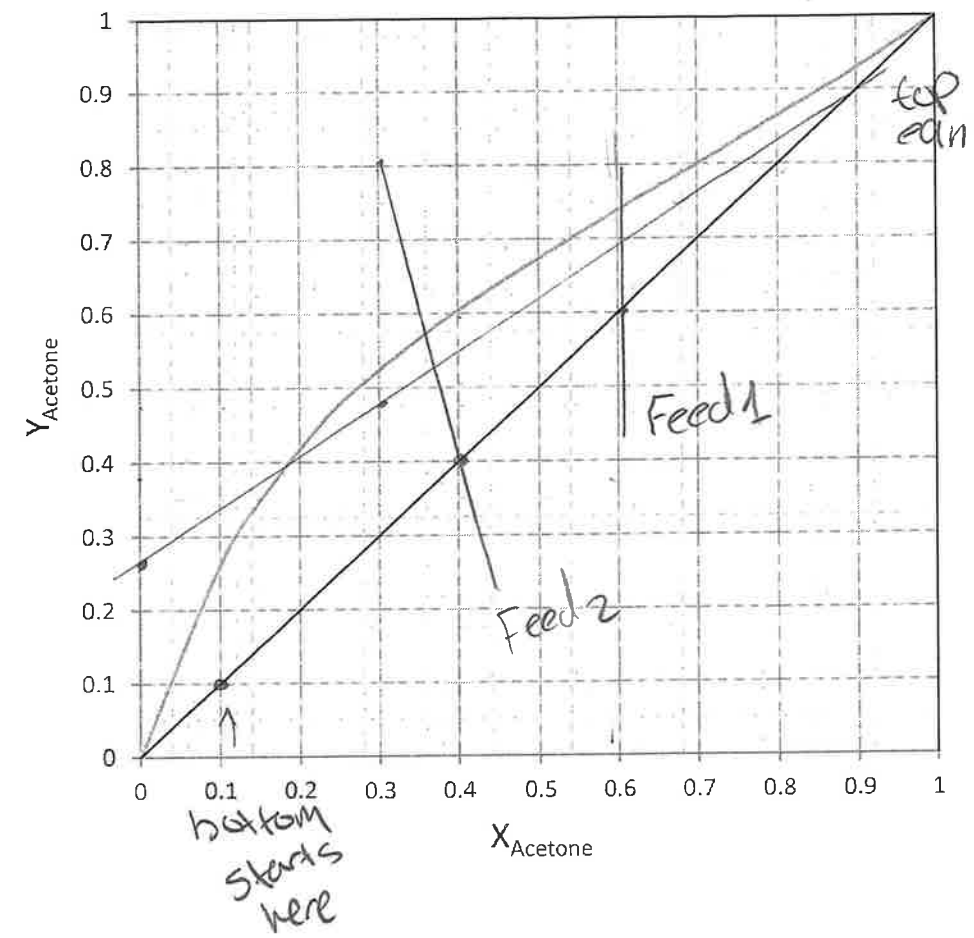
(.6, .6) Vertical line ✓

$$F_2 = \frac{.8}{.8-1}x + \frac{.4}{1-.8}$$

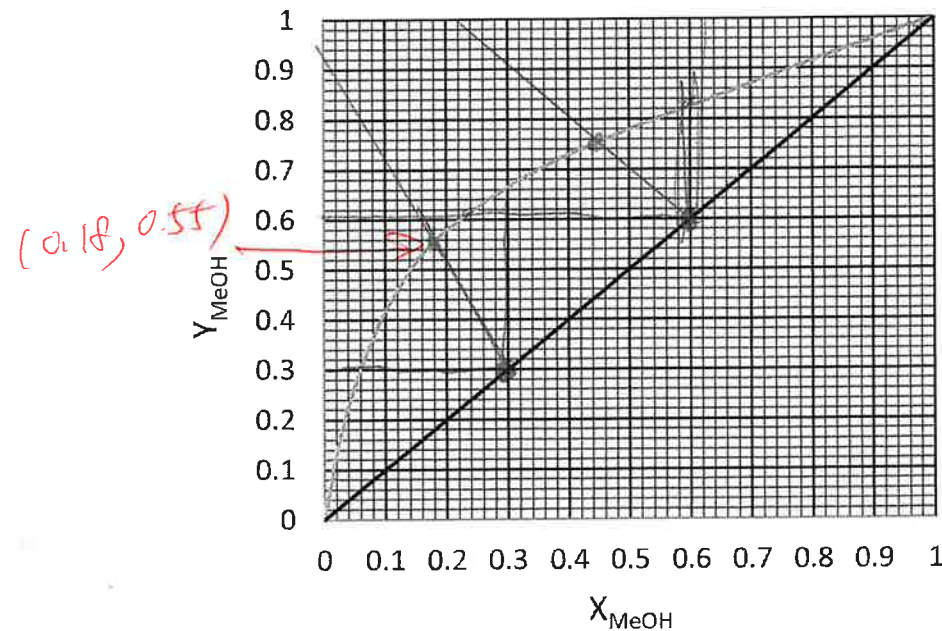
$$y = -4x + 2$$

(.4, .4)

10/10



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.

$$FZ = XB + YD \quad Y = \frac{q}{q-1}X + \frac{g}{q-1}$$

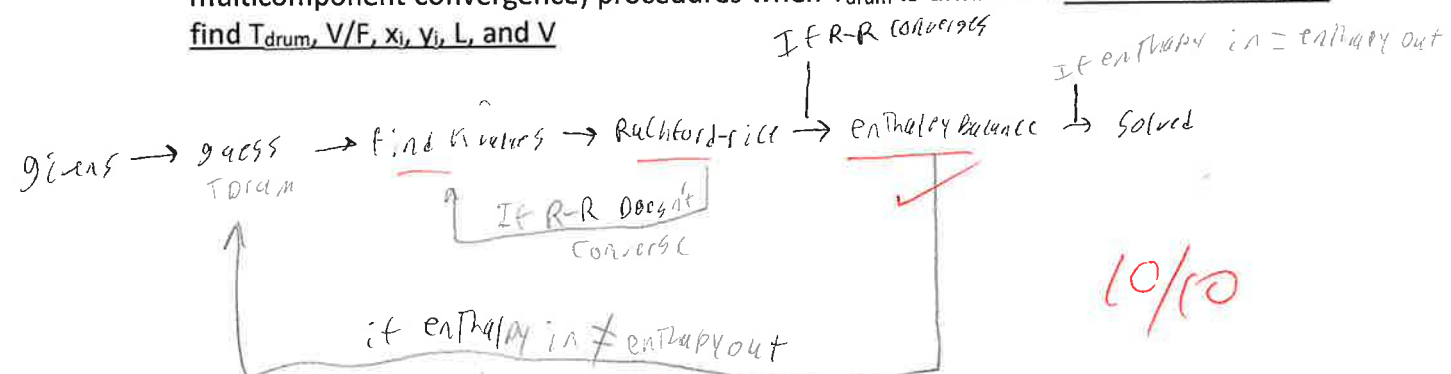
- 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.

$$X = 0.45 \quad Y = 0.75$$

- 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.

$$\frac{V}{F} = 0.5 \quad B = 75 \frac{\text{kmol}}{\text{hr}} \quad D = 75 \frac{\text{kmol}}{\text{hr}} \quad X = 0.18 \quad Y = 0.42$$

2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F, x_i , y_i , L, and V



1) -

2) $F = 150 \frac{\text{kmol}}{\text{h}}$

$z = 60\%$

$\frac{V}{F} = 5$

$x_B =$
 $b =$

Next time,
write down
your answers
clearly
for each question

Op-line

$z \rightarrow (.6, .6)$

$y\text{-int} \rightarrow y = \frac{-0.5}{.5-1} x + \frac{.5}{.5-1} (.9) =$

Op slope $\frac{q}{q-1} = -1$

3) $x_B = .18$

$R = .3$

$F = B + D$

$Fz = x_B + y_D$

$F(z) =$

$y\text{-int} \begin{pmatrix} 0, .9 \\ .3, .3 \end{pmatrix} \quad \frac{.3 - .9}{.3 - .0} = \frac{-.6}{.3} = -\frac{1}{2}$

Slope $= \frac{\frac{V}{F}}{1 - \frac{V}{F}} = -\frac{1}{2}$

$1 - \frac{V}{F} = -\frac{1}{2} \frac{V}{F}$

$1 = \frac{1}{2} \frac{V}{F}$

$\frac{1}{2} = \frac{V}{F}$

$150(.03) = 75(.18) + 75(y)$

$45 = 13.5 + 75(y)$

$31.5 = 75(y)$

$y = .42$

y-intercept: $\sim 0.9 = \frac{F}{V} z$

$\downarrow$
 $\frac{V}{F} = \frac{1}{3}$

$V = 50 \text{ kmol/h}$

$L = 100 \text{ g}$

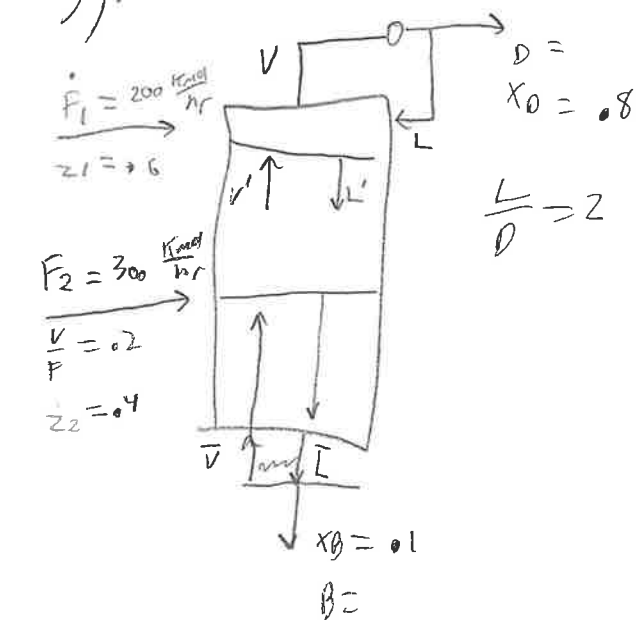
1) 3/3

2) 5/5

Overall M.B.: $F = V + L$
Cap. " : $Fz = V y + L x$

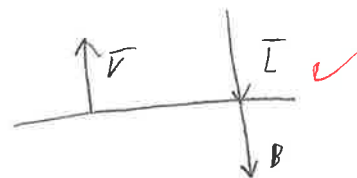
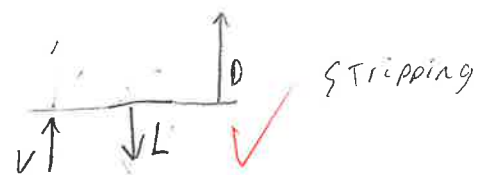
3) 3/10

3).



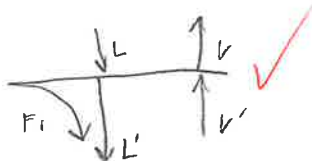
3/3

2) enrich



8/8

Feed 1



Feed 2



$$3) F_1 + F_2 = B + D \rightarrow D = F_1 + F_2 - B$$

$$F_1(z_1) + F_2(z_2) = Bx_B + Dx_D$$

$$200 \frac{\text{kmol}}{\text{hr}} (0.6) + 300 \frac{\text{kmol}}{\text{hr}} (0.4) = B(0.1) + (F_1 + F_2 - B)(0.8)$$

$$240 \frac{\text{kmol}}{\text{hr}} = B(0.1) + 440 \frac{\text{kmol}}{\text{hr}} - B(0.8)$$

$$-200 \frac{\text{kmol}}{\text{hr}} = -B(0.7)$$

228.6

$$289.7 \frac{\text{kmol}}{\text{hr}} = B$$

$$D = 500 \frac{\text{kmol}}{\text{hr}} - 289.7 \frac{\text{kmol}}{\text{hr}}$$

$$\bar{V} = V \cdot 0.2 F_2$$

$$= 754.2 \frac{\text{kmol}}{\text{hr}}$$

$$D = 214.3 \frac{\text{kmol}}{\text{hr}}$$

$$271.4$$

→

$$\frac{L}{D} = 2$$

$$L = 214.3 \cdot 2 = 428.6 \frac{\text{kmol}}{\text{hr}}$$

$$542.8$$

$$\bar{V} + L' = L' + \frac{4}{5} F_2$$

$$V' - \frac{1}{5} F_2 = \bar{V}$$

$$\bar{V} = 582.9 \frac{\text{kmol}}{\text{hr}}$$

$$L = 168.6 \frac{\text{kmol}}{\text{hr}}$$

$$F_1 + L' + V = V' + L$$

$$V' = V = 582.9 \frac{\text{kmol}}{\text{hr}}$$

814.3

$$L = L' + F_1$$

$$L' = L - F_1$$

$$L' = (168.6 - 200) \frac{\text{kmol}}{\text{hr}}$$

$$L' = -31.4 \frac{\text{kmol}}{\text{hr}}$$

742.8

Tyler Eaton

3-6)
a) $\frac{L_1}{F_1} = 1 = a_1$

$y = \frac{1}{1}x$

$z = .6$
 $(.6, .6)$

b) $\frac{\frac{4}{5}}{\frac{5}{5}} = \frac{4}{5} = a_2$

$z = (.4, .4)$

$y = \frac{.8}{-.2}x + \frac{.4}{1-.4}$

$0 = \frac{.8}{-.2}x + \frac{.4}{.2}$

$-2 = -4x$

$x = .5$ $(.5, 0)$

10/10

3-7)

$F_1 + V' + L = v + L'$

$F_1(z) + v'(x_0) + L(x_0) = v(x_0) + L'(x_1)$

$F_1 + V' = L' + D$

$F_1 z_1 + v' y = L' x + D x_0$

$y = \frac{L'}{v'}x + \frac{D x_0 - F_1 z_1}{v'}$

4/10

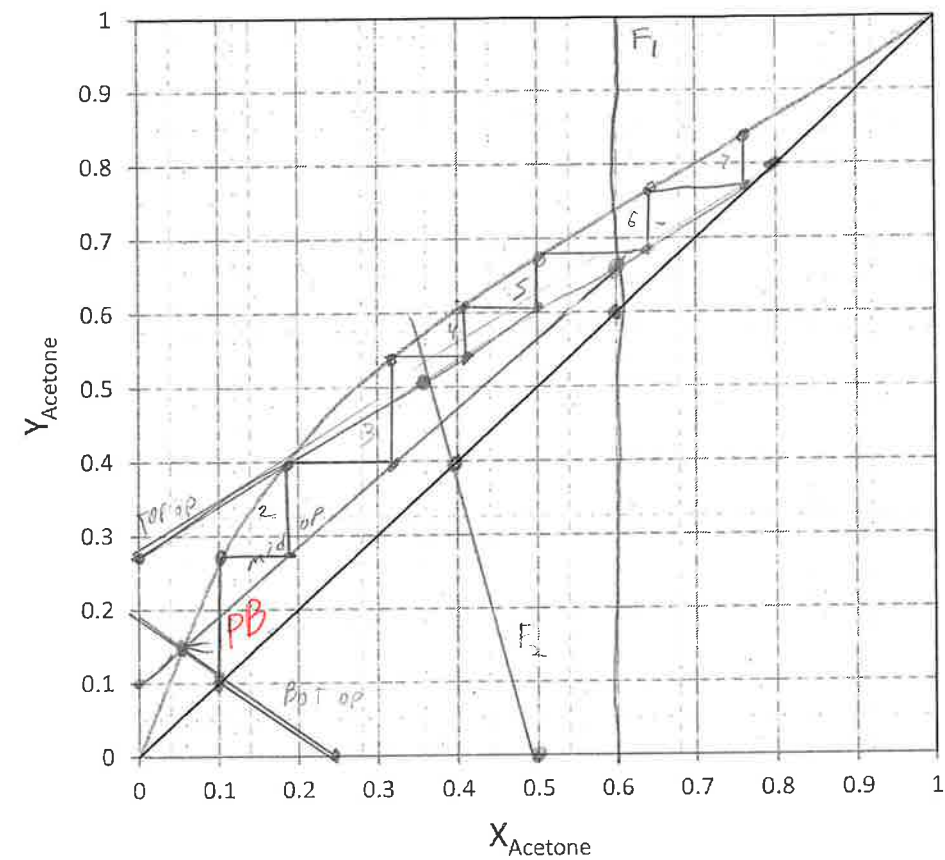
Top opp Intersects F_1

Bot opp Intersects equilb

$y_{int} = (0, 1)$
 $(.5, .65)$

slope = $\frac{1 - .65}{0 - .6} = .42$

Tyler E 1011



8) STEPS (11) Total 6 = 5 stages + PB 3/5
7 stages

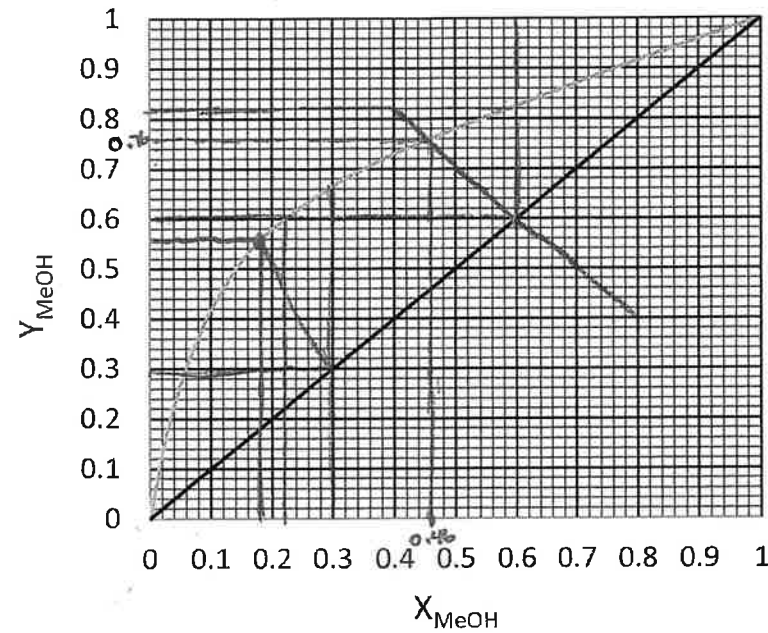
9) Feed 1 between stage ~~5 and 6~~ 4
Feed 2 between stage ~~3 and 4~~ 2 2/3

10) Stage 3 is $y=0.55$
 $x=0.31$ 1/2

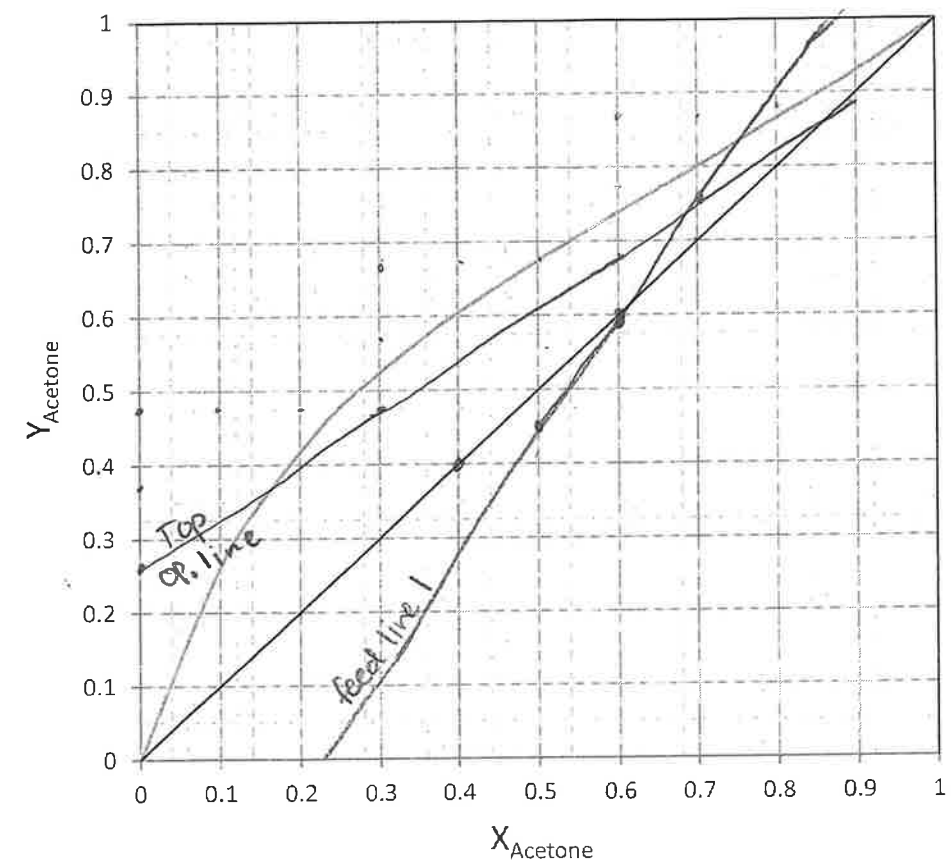
Andrew Hawth

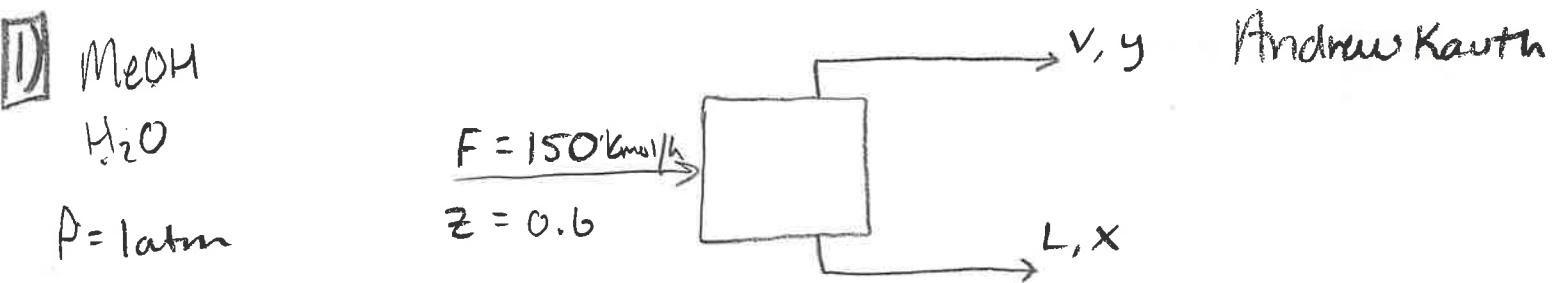
668817735

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V





① $F = L + V$ overall

$Fz = Lx + Vy$ component

$y = -\left(\frac{L}{V}\right)x + \left(\frac{F}{V}\right)z$ operating eqn

② $150 = L + V$

$V = 0.5(F) = (0.5)(150) = 75 \text{ kmol/h}$

$L = F - V = 150 - 75 = 75 \text{ kmol/h}$

Slope: $-\frac{75}{75} = -1$

$X_{\text{MeOH}} = 0.46, Y_{\text{MeOH}} = 0.76$

③ $z = 0.3$

$X = 0.18 \rightarrow Y = 0.56$

$(150)(0.3) = L(0.18) + (150 - L)(0.56)$

$45 = 0.18L + 84 - 0.56L$

$-39 = 0.18L - 0.56L$

$\frac{-39}{-0.38} = \frac{-0.38L}{-0.38}$

$L = 102.63 \text{ kmol/h}$

$V = F - L \quad V = 150 - 102.63 = 47.37$

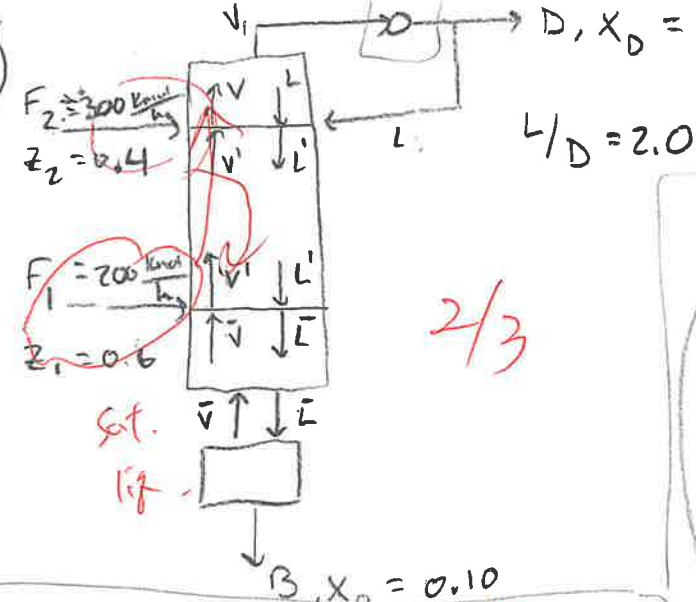
$V = 47.37 \text{ kmol/h}$

Slope: $-\frac{L}{V}, -\frac{102.63}{47.37}$

$= -2.167$

3

1



2/3

2

Overall: $F_1 + F_2 = D + B$

enriching: $L + V' = L' + V + D$

stripping: $\bar{V} + \bar{L}' = \bar{L} + V' + B$

feed 1: $F_1 + \bar{V} + \bar{L}' = \bar{L} + V'$

feed 2: $F_2 + V' + L = V + L'$

3

$F_2 + F_1 = D + B$ $B = 500 - D$

$F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$500 =$

$(200)(0.6) + (300)(0.4) = D(0.8) + (500 - D)(0.1)$

$120 + 120 = 0.8D + 50 - 0.1D$

$\frac{190}{0.7} = \frac{0.7D}{0.7}$

$D = 271.43 \text{ kmol/h}$

$B = 228.57 \text{ kmol/h}$

$B = 500 - D$

$B = 500 - 271.43$

3 continued

$\frac{L}{D} = 2.0$

$\frac{L}{271.43} = 2.0$

$L = (2.0)(271.43)$

$L = 542.86 \text{ kmol/h}$

$V = D + L$

$V = 271.43 + 542.86$

$V = 814.29 \text{ kmol/h}$

$\bar{L} = B + \bar{V}$

$\bar{L} - \bar{V} = 228.57$

$228.57 = F_1 + L' - V'$

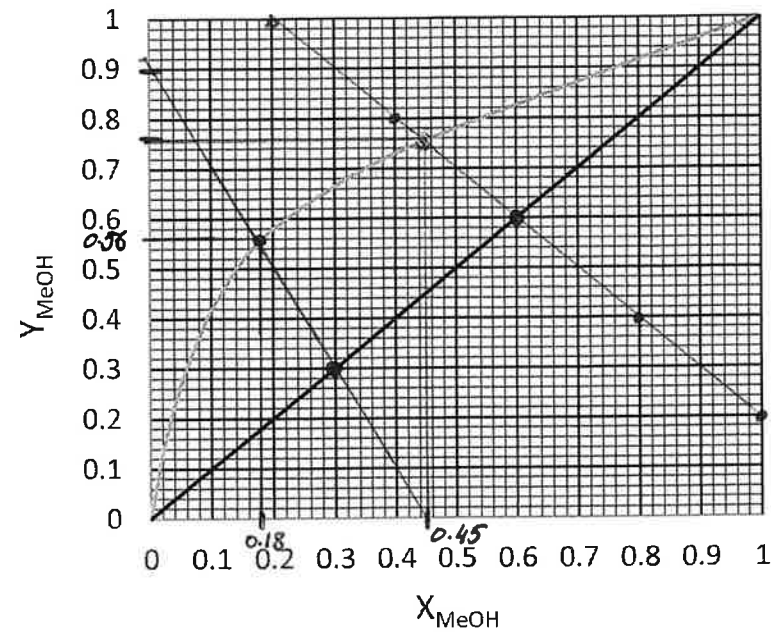
$228.57 - 200 = L' - V'$

$L' - V' = 28.57$

$V' + L = L + V + D$

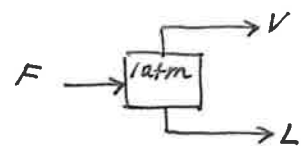
9/15

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1. (1)



$$F = V + L$$

$$Fz = Vy + Lx$$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

3/3

(2) $F = 150 \text{ kmol/h}$, $z = 0.60$, $V/F = 0.50$

$$V = 0.50 \cdot 150 \text{ kmol/h} = 75 \text{ kmol/h}$$

$$L = F - V = (150 - 75) \text{ kmol/h} = 75 \text{ kmol/h}$$

$$\text{slope} = -\frac{L}{V} = -1$$

$$y\text{-int} = 0.60 \left(\frac{150}{75} \right) = 1.2$$

$$y = 0.76 \text{ \& } x = 0.45$$

5/5

(3) $\left. \begin{array}{l} \text{slope} = -2 \\ y\text{-int} = 0.9 \end{array} \right\} \text{ from graph}$

$$x = 0.30$$

$$x = 0.18$$

$$\text{slope} = -\frac{L}{V} \rightarrow -2 = -\frac{L}{50} \rightarrow L = 100 \text{ kmol/h}$$

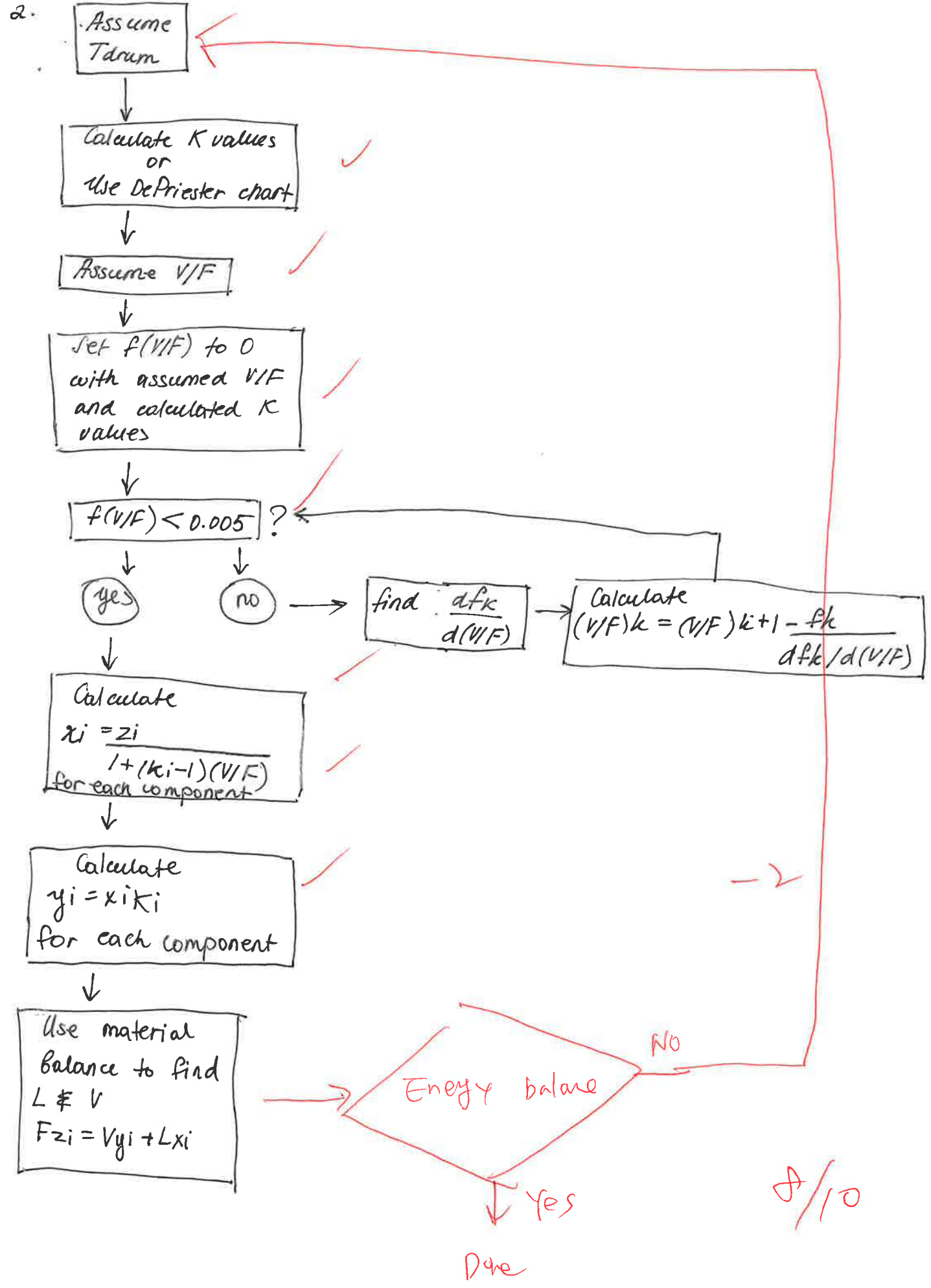
$$y\text{-int} = \frac{F}{V}z$$

$$0.9 = \frac{150}{V}(0.3) \rightarrow 3 = \frac{150}{V} \rightarrow V = 50 \text{ kmol/h}$$

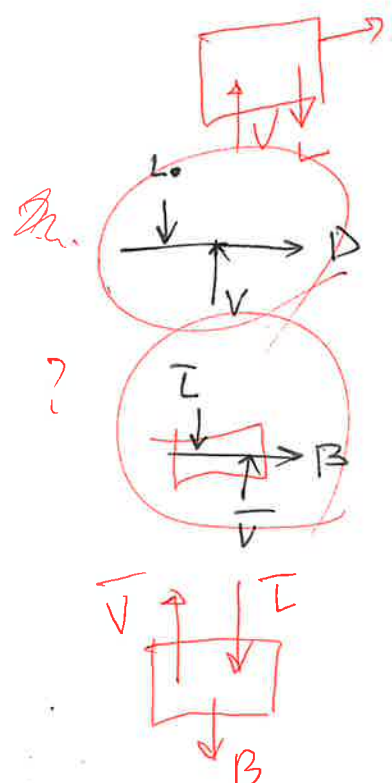
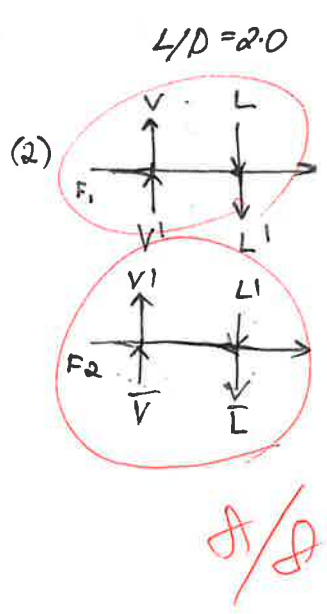
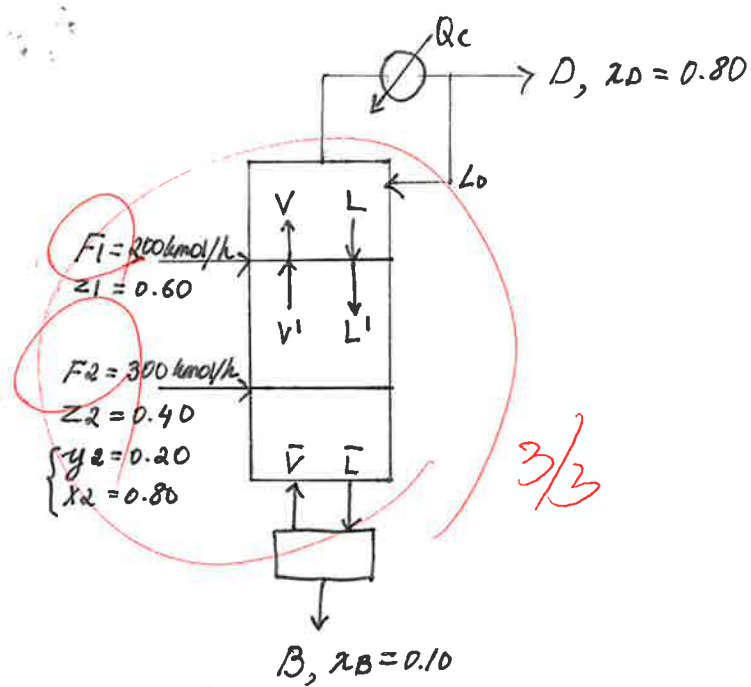
$$V/F = (50/150) = 1/3$$

$$y = 0.56 \text{ \& } x = 0.18 \text{ (given)}$$

10/10



3. (1)



(3) External M.B.:

$$\begin{aligned} F_1 + F_2 &= B + D \\ F_1 z_1 + F_2 z_2 &= B x_B + D x_D \\ 240 \text{ kmol/h} &= 0.10 B + 0.80 D \\ B &= F_1 + F_2 - D \\ 240 &= 0.10 (500 - D) + 0.80 D \\ 240 &= 50 - 0.10 D + 0.80 D \\ 190 &= 0.70 D \\ D &= 271.43 \text{ kmol/h} \\ B &= 228.57 \text{ kmol/h} \end{aligned}$$

$L/D = 2.0$ $B = 500 - 271.43$

$L = 2.0 (271.43) = 542.86 \text{ kmol/h}$

$V = L + D = 814.29 \text{ kmol/h}$

$F_1 + V' + L = V + L'$

$$\begin{aligned} -71.43 + V' &= L' \\ -71.43 &= L' - V' \\ 71.43 &= V' - L' \end{aligned}$$

(4) Top M.B.:

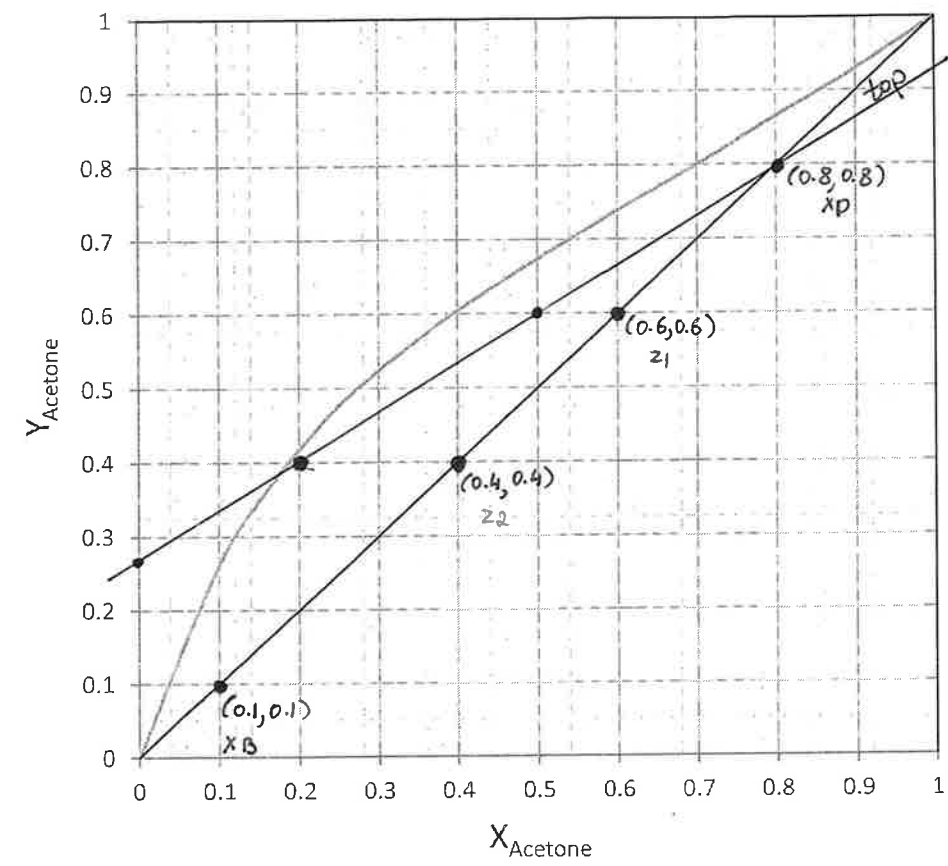
$$\begin{aligned} V &= L + D \\ V y &= L x + D x_D \\ \text{slope} &= \frac{L}{V} = 0.66 = \frac{2}{3} \\ y_{int} &= (1 - \frac{2}{3})(0.80) = 0.267 \end{aligned}$$

(7) $F_1 + V' + L = L' + D$

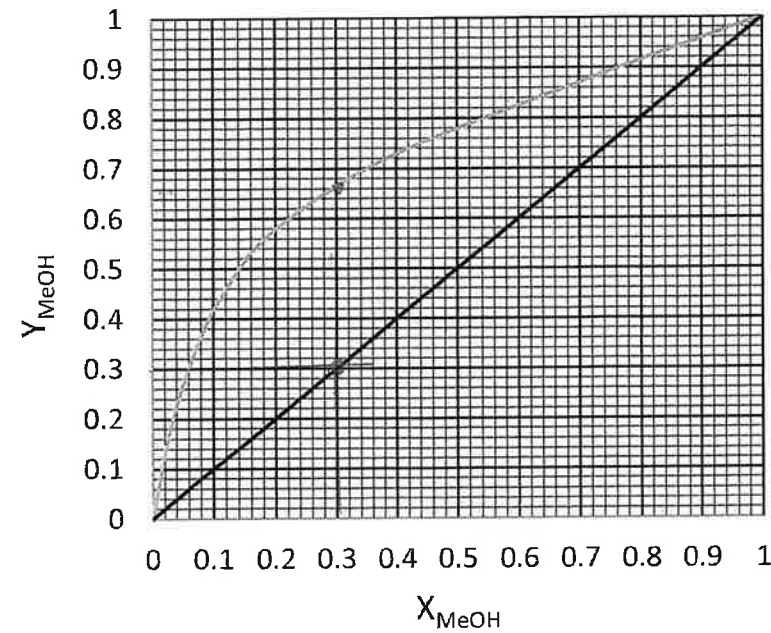
(6) a) $q_1 = 1$ (sat. liq.)

1/10

8/15



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

- ① Estimate T_{drum}
- ② Find K -value based on T_{drum}
- ③

3. $q = 1 - f$

$f = \frac{L}{F} = 0.18$

$q = \frac{V}{F} = 1 - 0.18 = 0.82$

$V = 123 \text{ kmol/h}$

$L = 27 \text{ kmol/h}$

2/10

of line passes

$(0.18, 0.55) \text{ \& } (0.3, 0.3)$

$y\text{-intercept} = 0.9$

$\frac{F}{V} z = 0.9 \quad \frac{V}{F} = \frac{1}{3}$

$V = 50 \text{ kmol/h}$

$L = 100 \text{ "}$

2. Set up the problem

↓

Estimate T_{drum}

↓

Based on T_{drum} find K -values

↓

Guess $\frac{V}{F}$

↓

Solve RR

↓

IS $RR < 0.001$? No?

↓ Yes

Find $x_i, y_i, L, V, \text{Energy } (H_{T_{\text{drum}}})$

↓

Find $T_{\text{drum } 2}$

↓

IS $|T_{\text{drum } 2} - T_{\text{drum } 1}| < 0.2^\circ$? No?

↓

Yes? Finish

10/10

✓

3. Overall Mass Balance

$$F_1 + F_2 = D + B$$

Component Mass Balance

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B$$

$$F_1 = 200 \text{ kmol/hr}$$

$$F_2 = 300 \text{ kmol/hr}$$

$$\frac{\bar{V}}{F_2} = 0.2$$

$$q_2 = 0.8$$

$$\frac{L}{D} = 2$$

$$x_D = 0.8$$

$$x_B = 0.1$$

$$q_2 = \frac{L_{F_2}}{F_2} = \frac{\bar{L} - L'}{F_2}$$

$$240 = \bar{L} - L'$$

$$0.8 = \frac{460 - L'}{F_2}$$

$$L' = 220 \text{ kmol/h}$$

$$V' = F_2 + \bar{V} + L' - \bar{L}$$

$$V' = 120 \text{ kmol/h}$$

F_1 = Saturated liquid so $q_1 = 1$

$$L_{F_1} = F_1 = 200 \frac{\text{kmol}}{\text{h}}$$

$$\frac{L}{D} = 2$$

$$L = 200 \frac{\text{kmol}}{\text{h}}$$

$$\frac{L}{2} = \frac{2D}{2}$$

$$D = \frac{L}{2} = 100 \text{ kmol/h}$$

$$V = D + L$$

$$\bar{V} = 200 + 100 = 300 \text{ kmol/h}$$

$$F_1 + F_2 = D + B$$

$$200 + 300 = 100 + B$$

$$B = 400 \text{ kmol/h}$$

$$\bar{V} = 0.2(300) = 60 \text{ kmol/h}$$

$$\bar{L} = \bar{V} + B = 60 \text{ kmol/h} + 400 \frac{\text{kmol}}{\text{h}}$$

$$\bar{L} = 460 \frac{\text{kmol}}{\text{h}}$$

$$L = \left(\frac{L}{D}\right) \cdot D = 542.8 \text{ kmol/h}$$

$$V = L + D = 642.8$$

$$V' = V = "$$

$$L' = L + F_1 = 742.8$$

$$\bar{L} = L' + 0.8 F_2 = 982.8$$

$$\bar{V} = V - 0.2 F_2 = 754.2$$

5/15

4 top operating line

$$(0.8, 0.8) \quad y = x$$

$$y = \frac{200}{300} x + \left(1 - \frac{200}{300}\right) (0.8)$$

$$y = \frac{2}{3} x + 0.27$$

$$\text{Slope: } \frac{1}{v} = \frac{2}{3}$$

$$y\text{-intercept: } 1 - \frac{L}{V} x_D = 0.27$$

b. $y = \frac{\bar{L}}{V} x - \left(\frac{\bar{L}}{V} - 1\right) x_B$

$$\text{Slope: } \frac{\bar{L}}{V} = \frac{460}{60} = 7.6$$

@ $y = 1$

$$y = 7.6x - 0.67$$

$$x = 0.22$$

$$y = x \quad (0.1, 0.1)$$

Method 3 o/c

6/8

b. $q_2 = 1 - f = 1 - \frac{\bar{V}}{F} = 0.8$

$$y = x \quad (0.4, 0.4)$$

$$\text{Slope} = \frac{0.8}{0.8 - 1} = -4$$

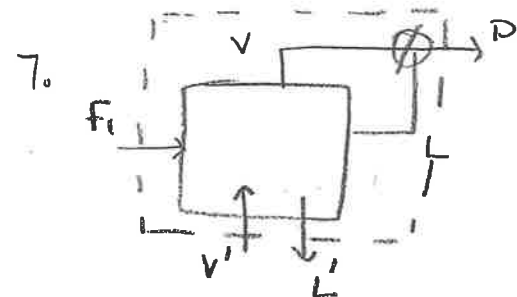
X-intercept

$$y = -4x + \frac{0.4}{1 - 0.8}$$

$$y = -4x + 2$$

$$x = 0.5$$

when $y = 0$



$$F_1 + V' = L' + D$$

$$F_1 z_1 + V' y = L' x + D x_D$$

$$y = \frac{L'}{V'} x + \frac{D x_D - F_1 z_1}{V'}$$

$$\text{Slope} = \frac{L'}{V'} = \frac{220}{120} = 1.8$$

0.9

y-intercept:

$$\frac{100 \times 0.8 - 200 \times 0.6}{120}$$

$$= -0.33$$

0.12

Method 3 o/c

8/10

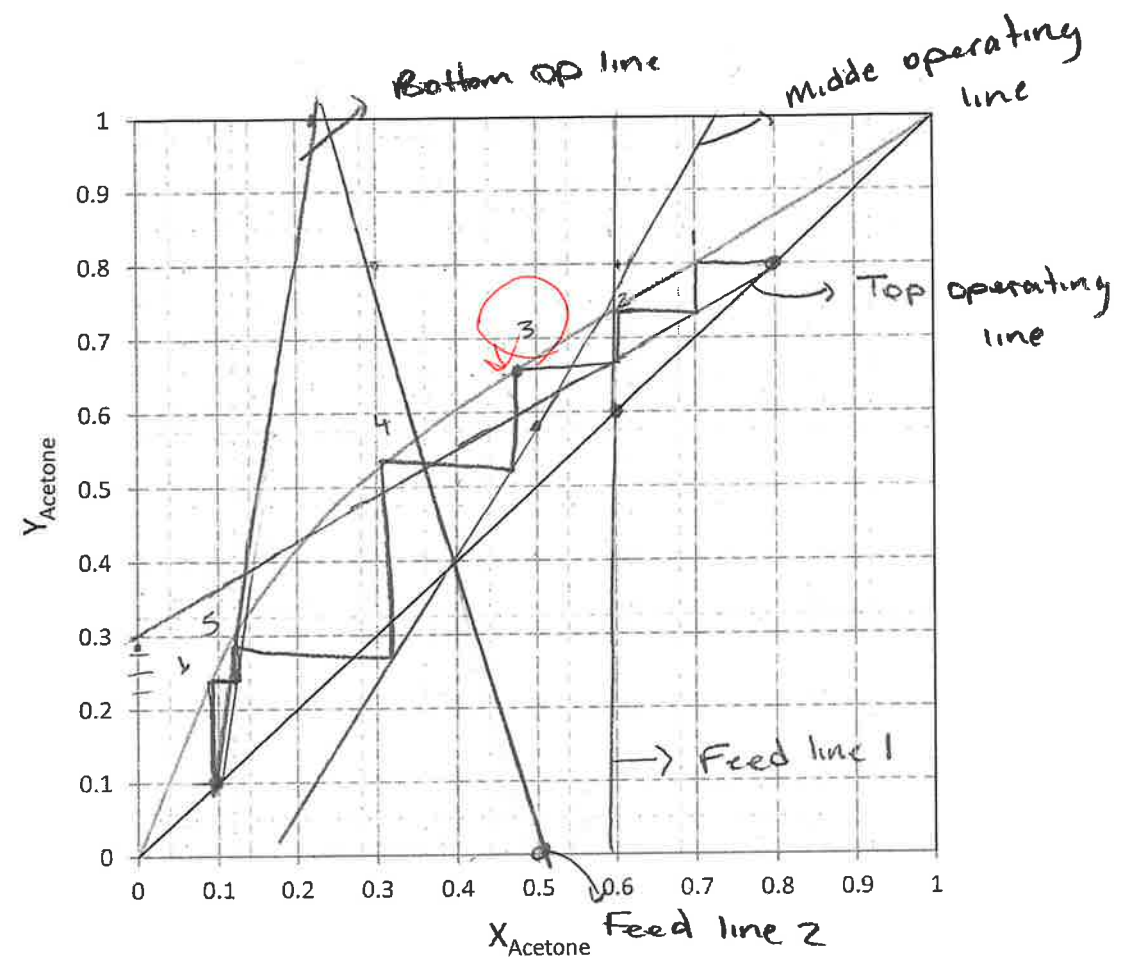
b. a. $q_1 = 1$ because F_1 is saturated liquid

$$y = x \quad (0.6, 0.6)$$

$$\text{Slope} = \frac{q_1}{q_1 - 1} = \frac{1}{1 - 1} = \frac{1}{0} = \infty$$

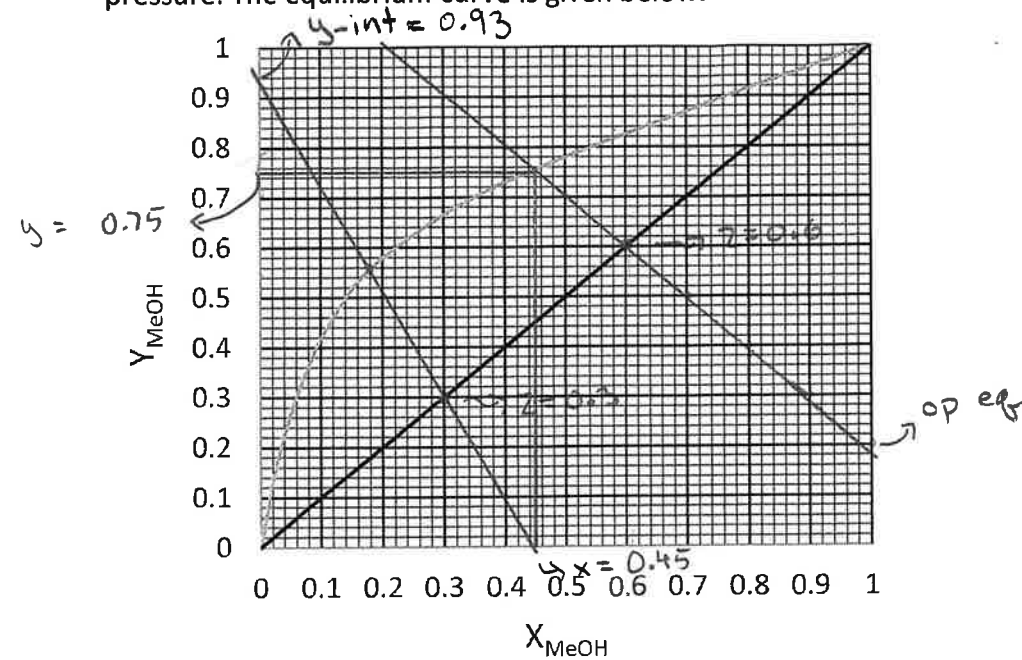
10/10

$$y = 0.6$$



- 8) 3/5 ← Look like you know how to step-off stages!
- 9) 2/3
- 10) 2/2

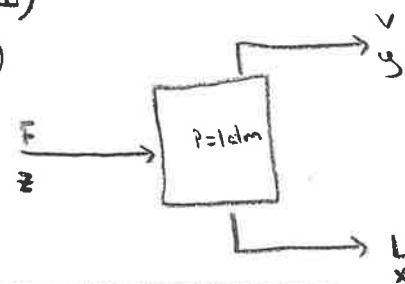
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Q1)

1)



OMB: $F = V + L$

OMB: $F \cdot z = V \cdot y + L \cdot x$

OEQ: $y = -\frac{L}{V} \cdot x + \frac{F}{V} \cdot z$

3/3

2) $F = 150 \text{ kmol/h}$ $z_m = 0.6$ $\frac{V}{F} = 0.5$

$V = F \cdot 0.5 = 75 \text{ kmol/h}$ $L = F - V = 75 \text{ kmol/h}$

$\hookrightarrow y = -\frac{15}{75} \cdot x + \frac{1}{0.5} \cdot 0.6 \rightarrow y = -1x + 1.2$

point $\rightarrow (0.6, 0.6) \rightarrow$ draw on graph with slope ($m = -1$)

$\hookrightarrow$ from graph $\rightarrow y_m = 0.75$ & $x_m = 0.45$

5/5

3) $z_m = 0.3$ $x_m = 0.18$

$F = 150 \text{ kmol/h}$

$y = -\frac{L}{V}x + \frac{F}{V}z$

 $\hookrightarrow$ 2 points $\rightarrow (0.3, 0.3)$ & $x = 0.18$ on equilibrium curve
 $\hookrightarrow y = 0.56$ $\rightarrow$ Graph it

$\hookrightarrow y\text{-int} = 0.93$

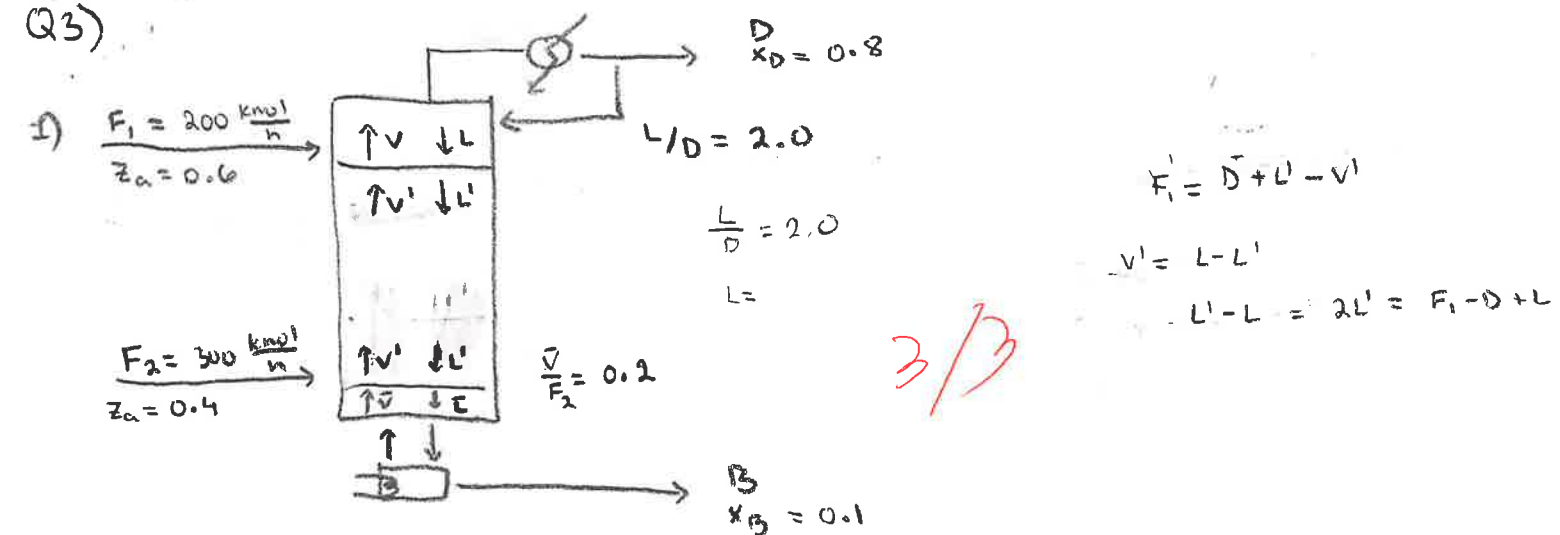
$\hookrightarrow 0.93 = \frac{F}{V} \cdot z \rightarrow \frac{F}{V} = \frac{0.93}{0.3}$

$\hookrightarrow \frac{F}{V} = 3.1 \rightarrow \frac{V}{F} = \frac{1}{3.1} = 0.32$

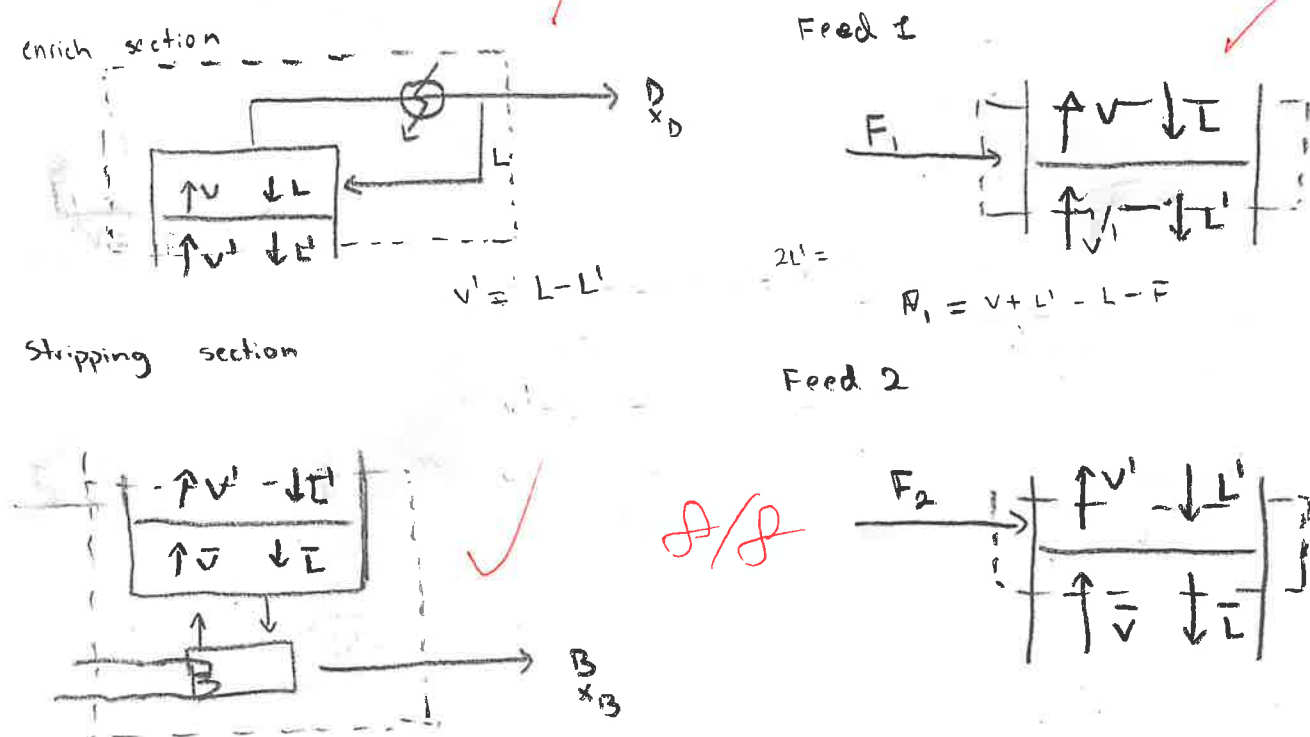
$\hookrightarrow V = 0.32 \cdot 150 = 48 \text{ kmol/h}$ & $L = 150 - 48 = 102 \text{ kmol/h}$

10/10

Q3)



2)



3) OMB: $F_1 + F_2 = D + B$
 CMB: $F_1 \cdot z_1 + F_2 \cdot z_2 = D \cdot x_D + B \cdot x_B$ $\rightarrow 2 \text{ eq, 2 unk}$

$$D = (200 + 300) \frac{\text{kmol}}{\text{h}} - B$$

$$\hookrightarrow 200 \frac{\text{kmol}}{\text{h}} \cdot 0.6 + 300 \frac{\text{kmol}}{\text{h}} \cdot 0.4 = [500 \frac{\text{kmol}}{\text{h}} - B] \cdot 0.8 + B \cdot 0.1$$

$$\hookrightarrow 240 \frac{\text{kmol}}{\text{h}} = 400 \frac{\text{kmol}}{\text{h}} - 0.7B$$

$$\hookrightarrow B = 228.57 \frac{\text{kmol}}{\text{h}}$$

$$\hookrightarrow [200 + 300 - 228.57] \frac{\text{kmol}}{\text{h}} = D$$

$$\hookrightarrow D = 271.43 \frac{\text{kmol}}{\text{h}}$$

6)

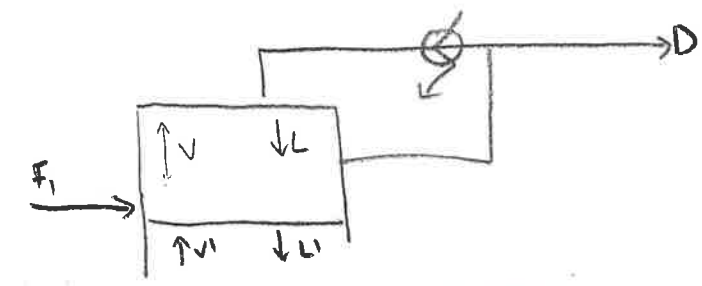
a) $x=y \rightarrow (0.6, 0.6)$, must intercept top & middle

4/10

b) $x=y \rightarrow$

b) $x=y \rightarrow (0.4, 0.4)$, must intercept bottom & middle

7)



OMB: $F_1 + V' = D + L'$

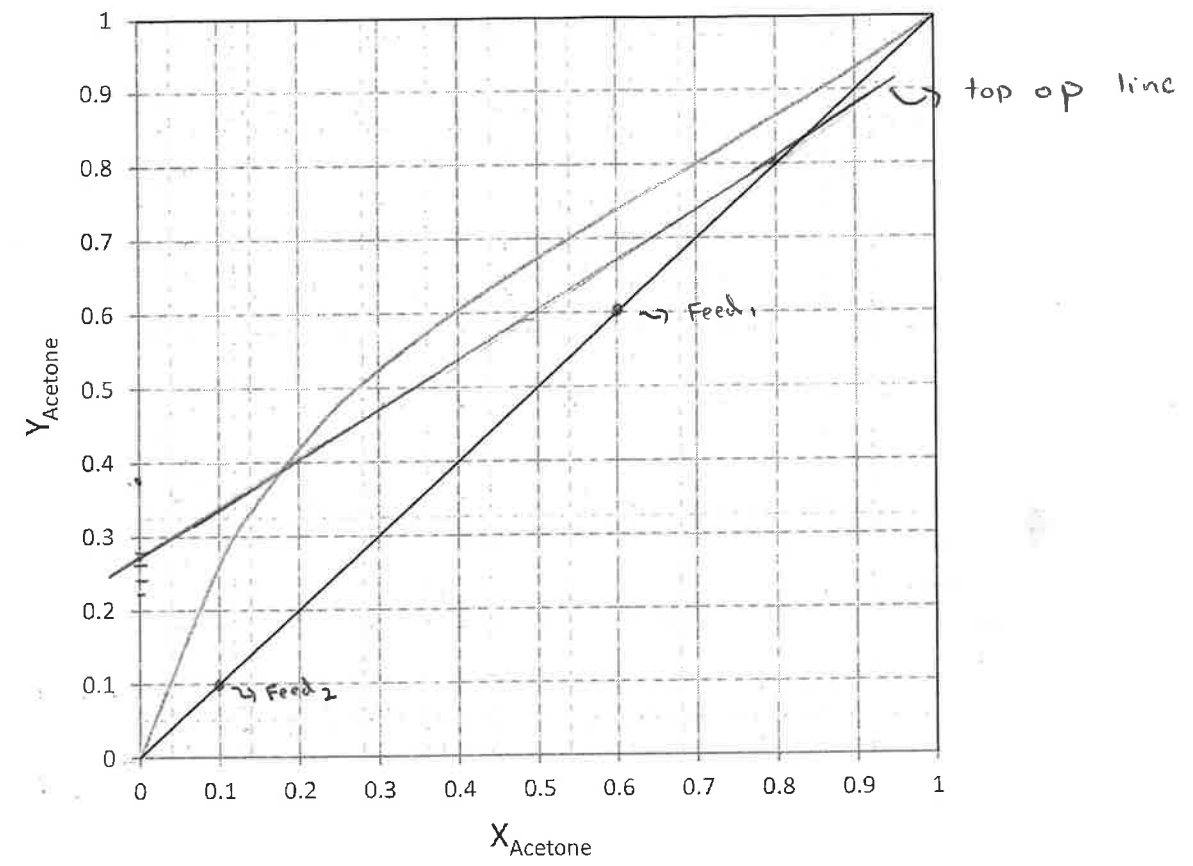
CMB: $F_1 \cdot z + V' \cdot y = D \cdot x_D + L' \cdot x$

4/10

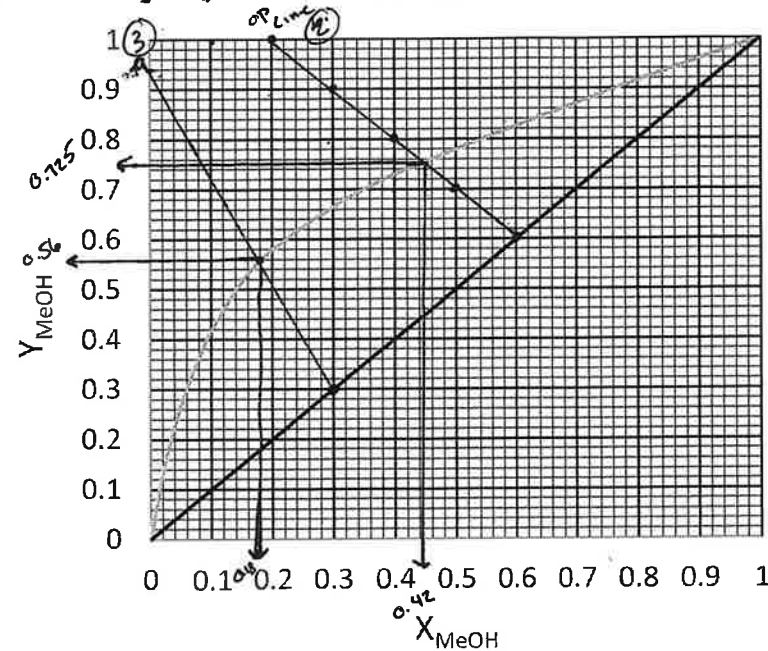
8)

9)

10)

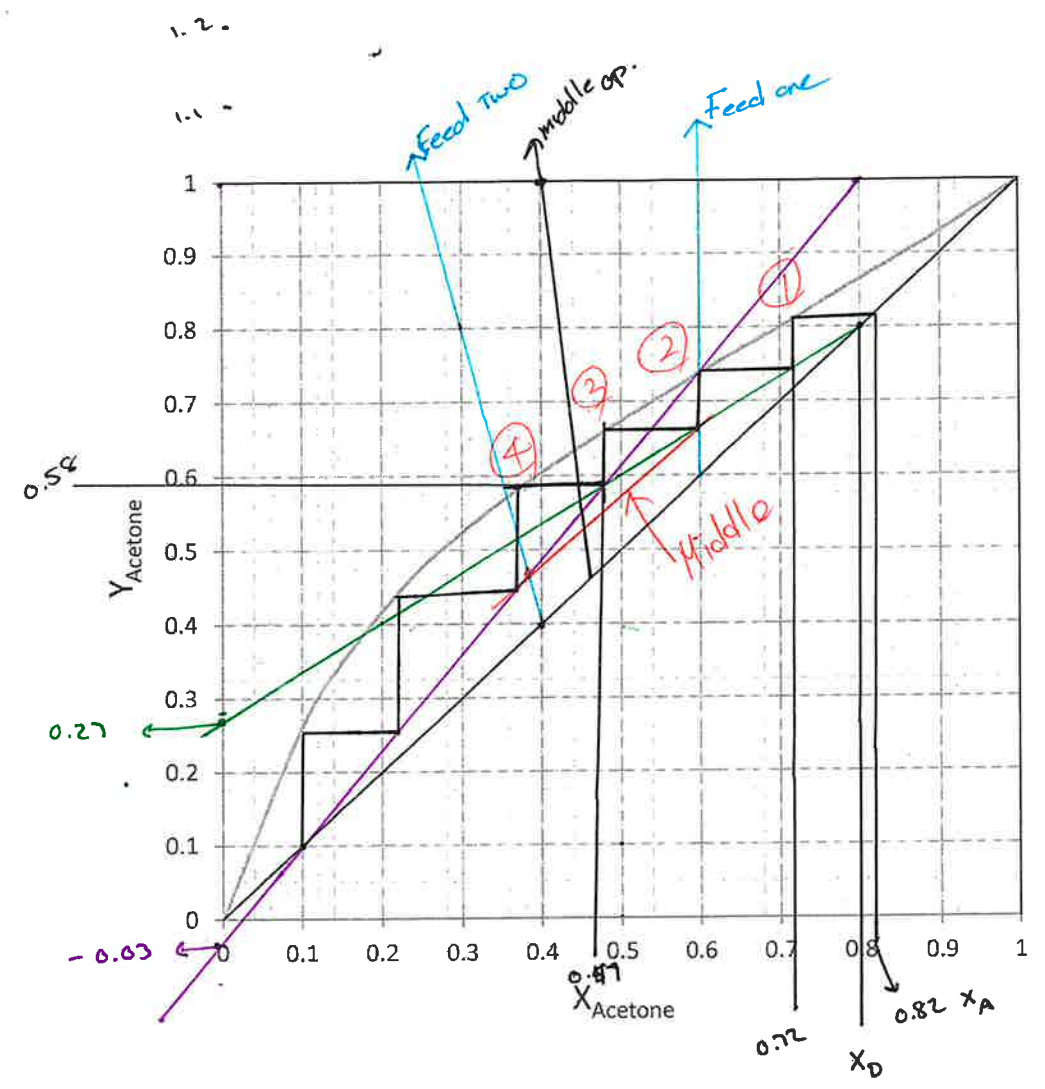


1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

12A Czeru



12A Czerwy

1.) $\frac{MB}{F} = V + L$

OP
 $y = mx + b$
 $y = \frac{-L}{V}x + \frac{F}{V}z$

1) 3/3

2) 5/5

3) 10/10

1b.) $V = 0.5F$
 $V = 0.5(150 \frac{\text{kmol}}{\text{hr}})$

$V = 75 \frac{\text{kmol}}{\text{hr}}$

$L = 75 \frac{\text{kmol}}{\text{hr}}$

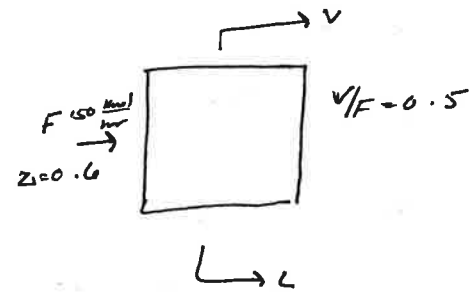
$F_z = Lx + Vy$

$-Vy = Lx - F_z$
 $-V$

$y = -\frac{L}{V}x + \frac{F}{V}z$

$y = -1x + 2(0.6)$

$y = -1x + 1.2$



using graph
 $y_v = 0.725$
 $x = 0.42$

1c.)

When $z = 0.3$ & $x = 0.18$

$0.56 = m(0.18) + 0.94$

$-0.38 = m(0.18)$

$m = -2.1 = L/V$

$F = V + L$

$F - V = L$

$F_z = Vy + Lx$

$F = 150$

$150(0.3) = V(0.56) + (F - V)(0.18)$

$150(0.3) = V(0.56) + \frac{F}{150}(0.18) - V(0.18)$

$18 = V(0.38)$

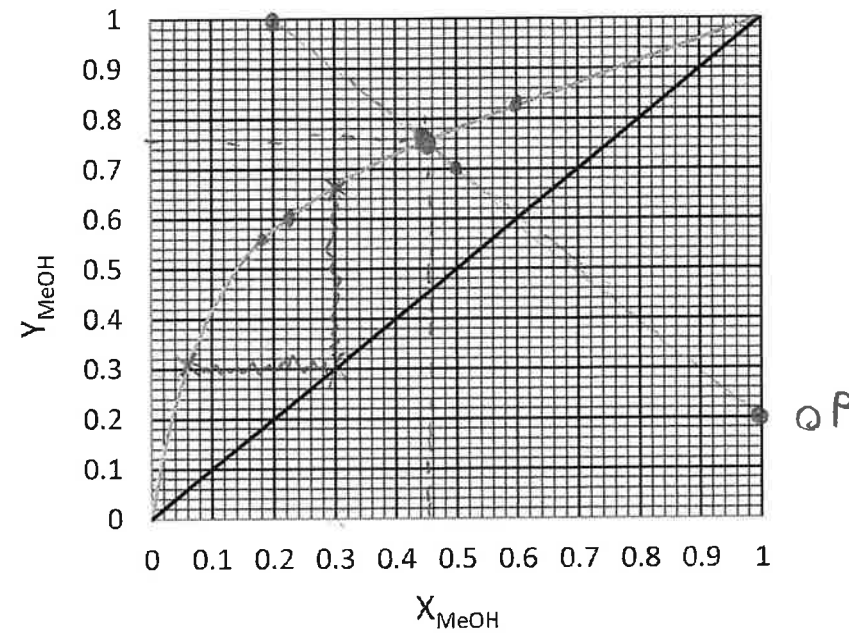
0.38

$V = 47.37 \frac{\text{kmol}}{\text{hr}}$

$150 - 47.37 = L$

$L = 102.63 \frac{\text{kmol}}{\text{hr}}$

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1a)

Overall M.B: $F = V + L$
 Comp: $Fz_i = Vy_i + Lx_i$
 OP: $Fz_i - Lx_i = Vy_i$
 $y_i = -\frac{L}{V}x_i + \frac{F}{V}z_i$ 3/3

b) Obj: Find flow rates (V and L)

$\frac{V}{F} = 0.5 \rightarrow \begin{cases} V = 0.5(150) = 75 \frac{\text{kmol}}{\text{hr}} \\ L = 150 - 75 = 75 \frac{\text{kmol}}{\text{hr}} \end{cases}$ 5/5

$y_i = -(1)x_i + 2z_i = -1x_i + 1.2$

OP intersects Data at $x \approx 0.76, y \approx 0.45$

c) Obj: Find V/F when $z_{Mc} = 0.30$ graphically

$y_i = -\frac{L}{V}x_i + \left(\frac{F}{V}\right)z_i$

$0.56 = -1(0.18) + \frac{F}{V}(0.3) \rightarrow F/V = 2.46 \rightarrow V/F = 0.405$

from graph,
 y-intercept of op line
 $0.9 \rightarrow 0.9 = \frac{F}{V}z$

3/10

3) Feed line 1 (Saturated)

6) a) slope = ∞
(0.6, 0.6) on $y=x$

(3) Feed line 1 (Saturated)

M.B: $F_1 + L + v' = v + L'$ • $L_F + L = L'$, $L_F = F$ ($q=1$)

Comp: $F_1 z_{A1} + L y + v' y = v y + L' y$ $v' = v$

$F_1 z_{A1} = (L' - L) y = L_F y \rightarrow \frac{F_1 z_{A1}}{L_F} = y \rightarrow z_{A1} = y$

2/10

b) slope: -4
(0.4, 0.4) on $y=x$
x-intercept = $x=0.5$

(4) Feed line 2 (Normal)

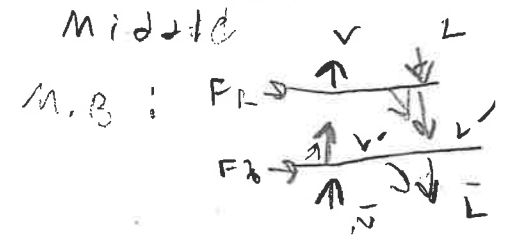
M.B: $F_2 + L' + \bar{v} = v' + \bar{L}$

Comp: $F_2 z_{A2} + L' y + \bar{v} y = v' y + \bar{L} y$ • $L_{F2} + L' = \bar{L}$

$F_2 z_{A2} = L_{F2} y + \bar{v} y - v' y$ • $\bar{v} F_2 + \bar{v} = v'$

$y = -\frac{L_{F2}}{\bar{v} F_2} x + \frac{F_2}{\bar{v} F_2} z_{A2}$

(5) Middle



3) 8/15

Overall M.B: $F_1 + F_2 = D + B = 500 \rightarrow 500 - D = B$

Comp: $F_1 z_{A1} + F_2 z_{A2} = D x_D + B x_B$

$240 = 0.8D + B(0.1)$

$240 = 0.8D + (500 - D)(0.1)$

$190 = 0.7D \rightarrow D = 271.43 \frac{\text{kmol}}{\text{hr}}$

$B = 228.57 \frac{\text{kmol}}{\text{hr}}$

Top $v - L = 271.43$

* $L/D = 2$

$L = 2(271.43) = 542.86 \frac{\text{kmol}}{\text{hr}}$

$V = 814.29 \frac{\text{kmol}}{\text{hr}}$

L, V, B, D found

$v', L', L'', \bar{L}$ un found

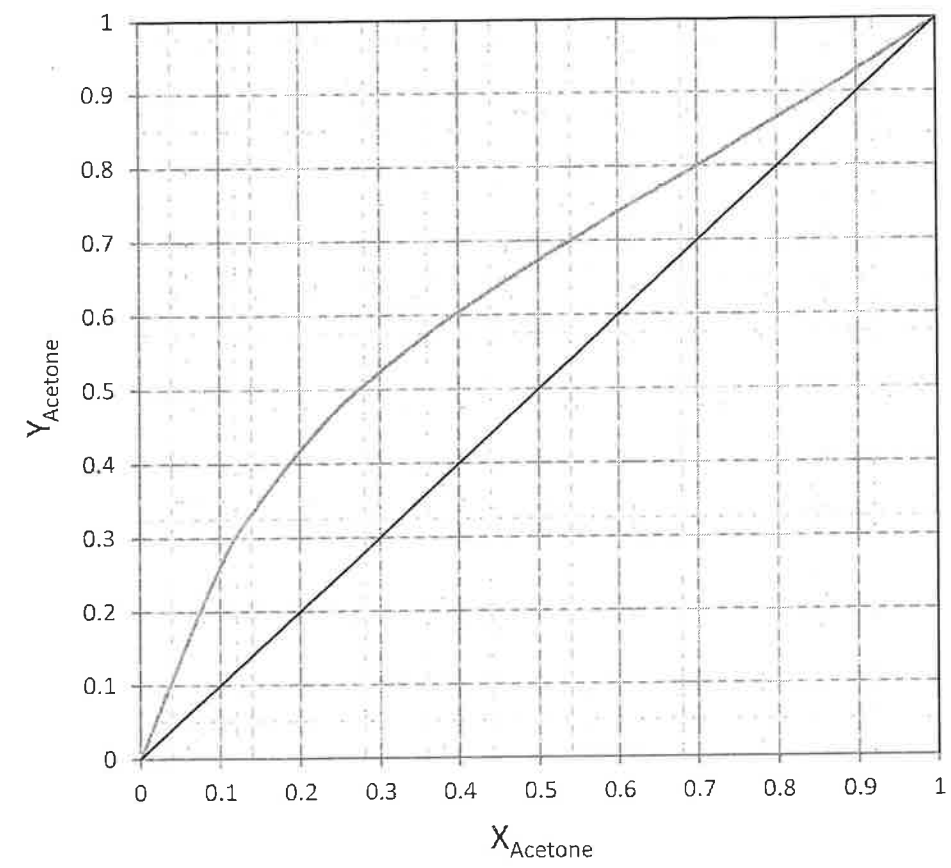
3) Finding $\bar{v}$, $\bar{v}$, v' , L'

$$L_F = 200 \Rightarrow FL1 \quad 200 + 542,86 = 742,86 = L'$$

301 $(F_1) + (L) + v' = v + (L')$

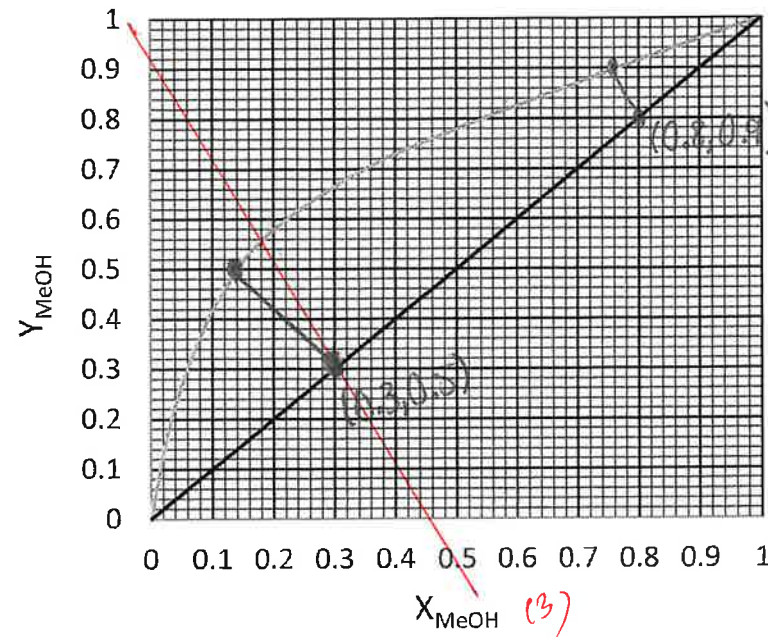
• $\bar{v} + v_{F2} = v' = v$

Δ



Jasmine Torres

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1) Mass balance

Pg 1)

i) overall mass balance: $F = V + L$

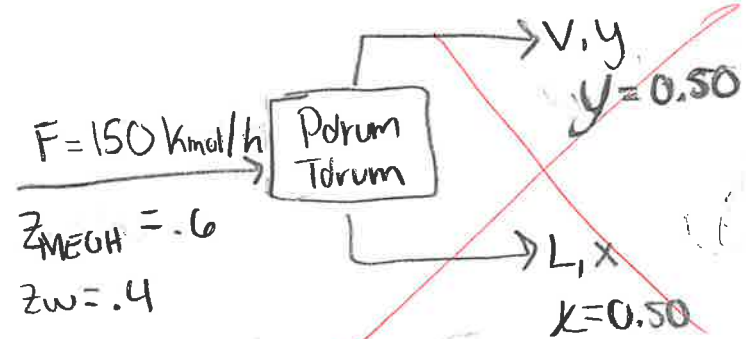
Component mass balance: $F \cdot z = V \cdot y + L \cdot x$

Op MB: $y = -\frac{L}{V}x + (1 - \frac{L}{V})z$

Water = 18 kmol

Methanol = 46 kmol

ii)



0/5

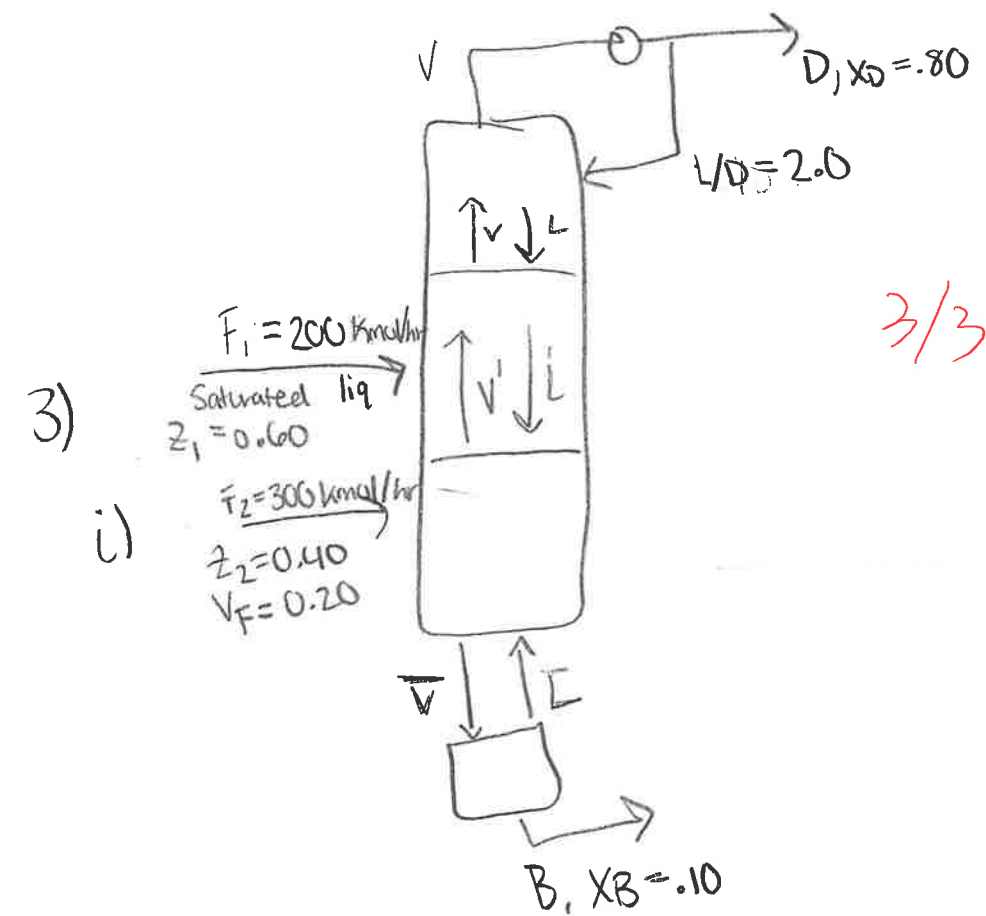
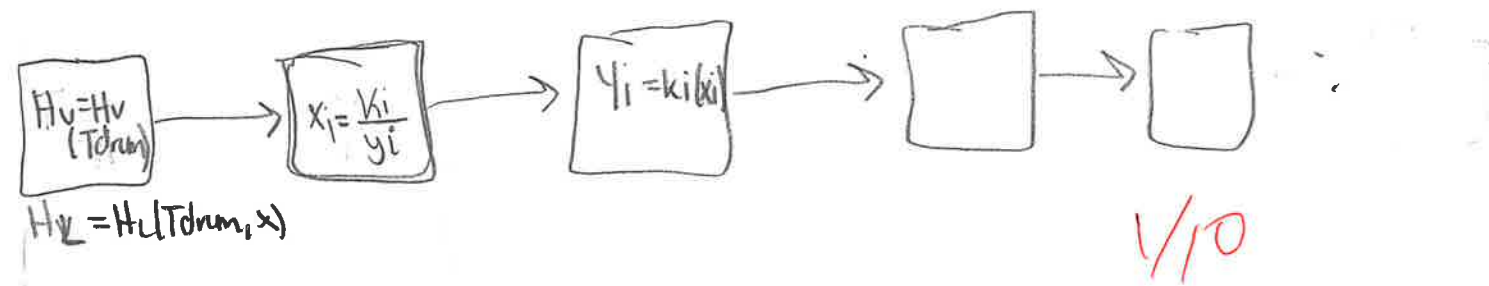
$L = V = 75 \text{ kmol/hr}$
 slope = -1
 graph
 (0.45, 0.75)

iii) on sheet.

0/10

2) Flowsheet to find T_{drum} , $\frac{V}{F}$, x_i , y_i , L , & V

Pg 2



ii) enriching section: $y_i = \left(\frac{L}{V}\right)x_i + \left(1 - \frac{L}{V}\right)x_D$

Draw M.B envelopes

stripping section: $y_i = \frac{L}{V}(x_i - 1) - \left(\frac{L}{V} - 1\right)x_B$

1/8

Feed 1: $F_1 = Vx_D + Lx_B$

Feed 2: $F_2 = Vx_D + Lx_B$

$$(ii) D = \left(\frac{z - x_B}{x_D - x_B} \right) F = D = \left(\frac{1 - (0.10)}{(0.80) - (0.10)} \right) (500) \quad \text{Pg 3)$$

$$D = \left(\frac{0.9}{0.7} \right) (500) = 642 \text{ kmol/hr}$$

$$\frac{L}{D} = 2$$

$$L = 1284 \text{ kmol/hr}$$

$$B = F - D$$

1/15

$$\frac{L}{V} = \frac{2}{3}$$

$$(v) \text{ Slope: } \frac{L}{V} = L/D + L = L/D / 1 + \frac{L}{D} = \boxed{\frac{2}{3}} \quad \begin{matrix} 3/8 \\ (0.8, 0.8) \end{matrix}$$

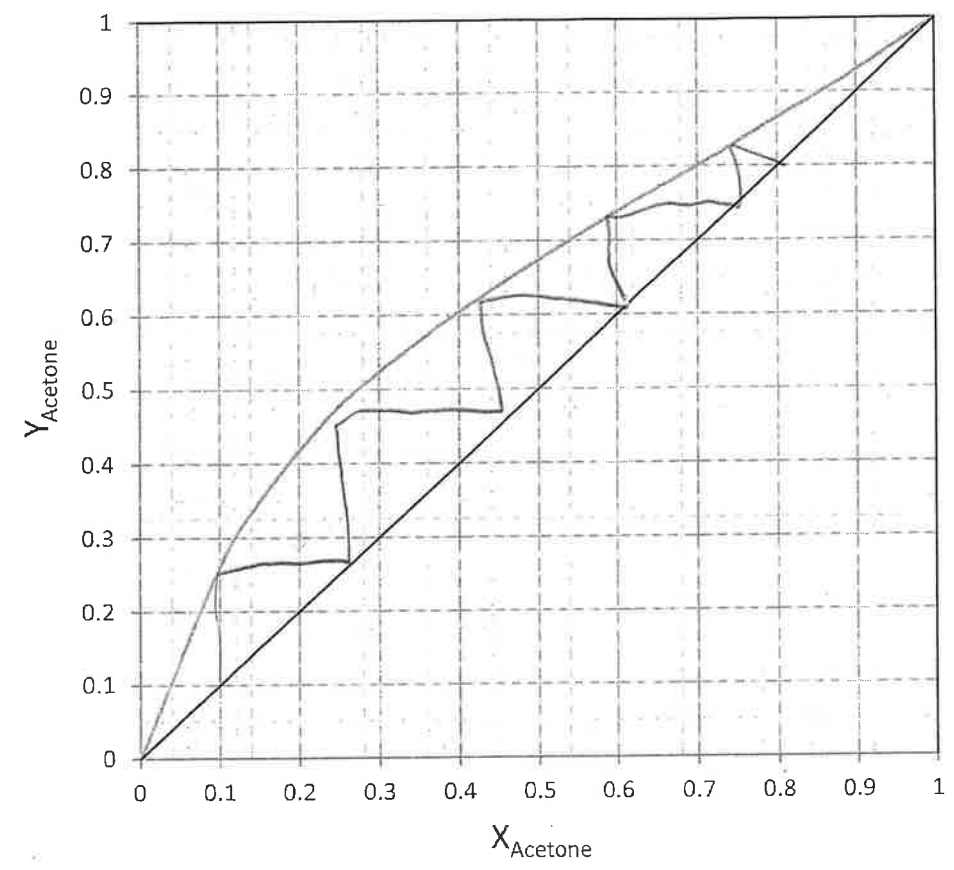
$$y = \left(\frac{L}{V} \right) x + \left(1 - \frac{L}{V} \right) x_D = y_{\text{int}} = \left(\frac{2}{3} \right) x + \left(1 - \frac{2}{3} \right) 0.80$$

y-intercept value @ $x=0$!
Simply, 0.27

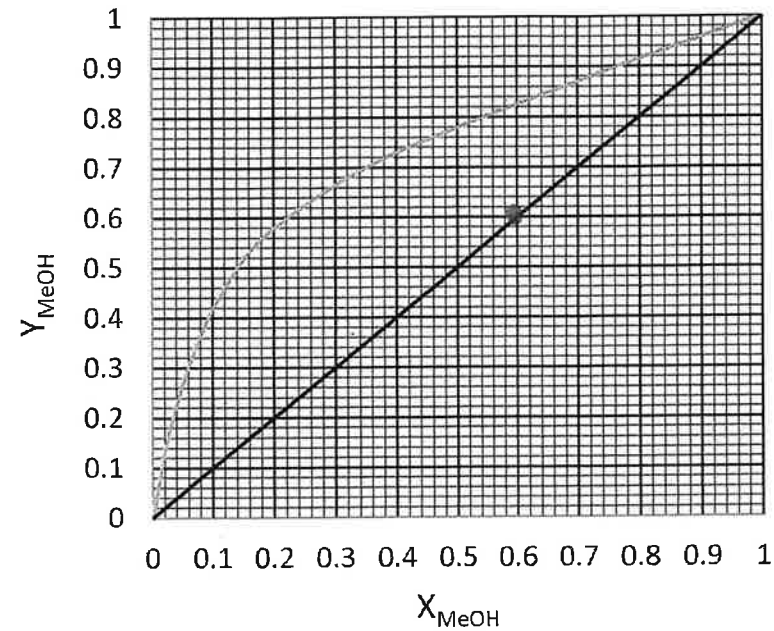
$$vi) y = \left(\frac{\bar{L}}{\bar{V}} \right) x - \left(\frac{\bar{L}}{\bar{V}} - 1 \right) x_B$$

$$vii) y = \frac{q_i}{q_{i-1}} x + \frac{z_i}{1 - q_i} \quad q_i = \frac{L_{Fi}}{F_i}$$

a)

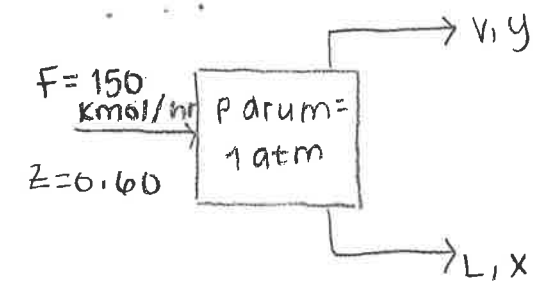


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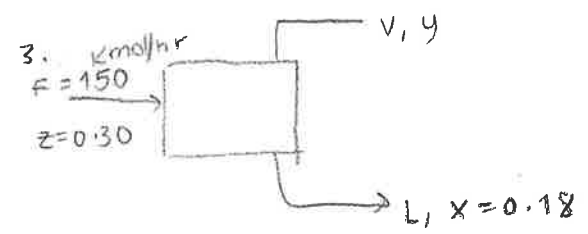
1.)



1. $F = V + L$
 $Fz = Vy + Lx$
 $y = -\frac{L}{V}x + \frac{F}{V}z$ 3/3
 $y = -\frac{L}{V}x + (1 + \frac{L}{V})z$

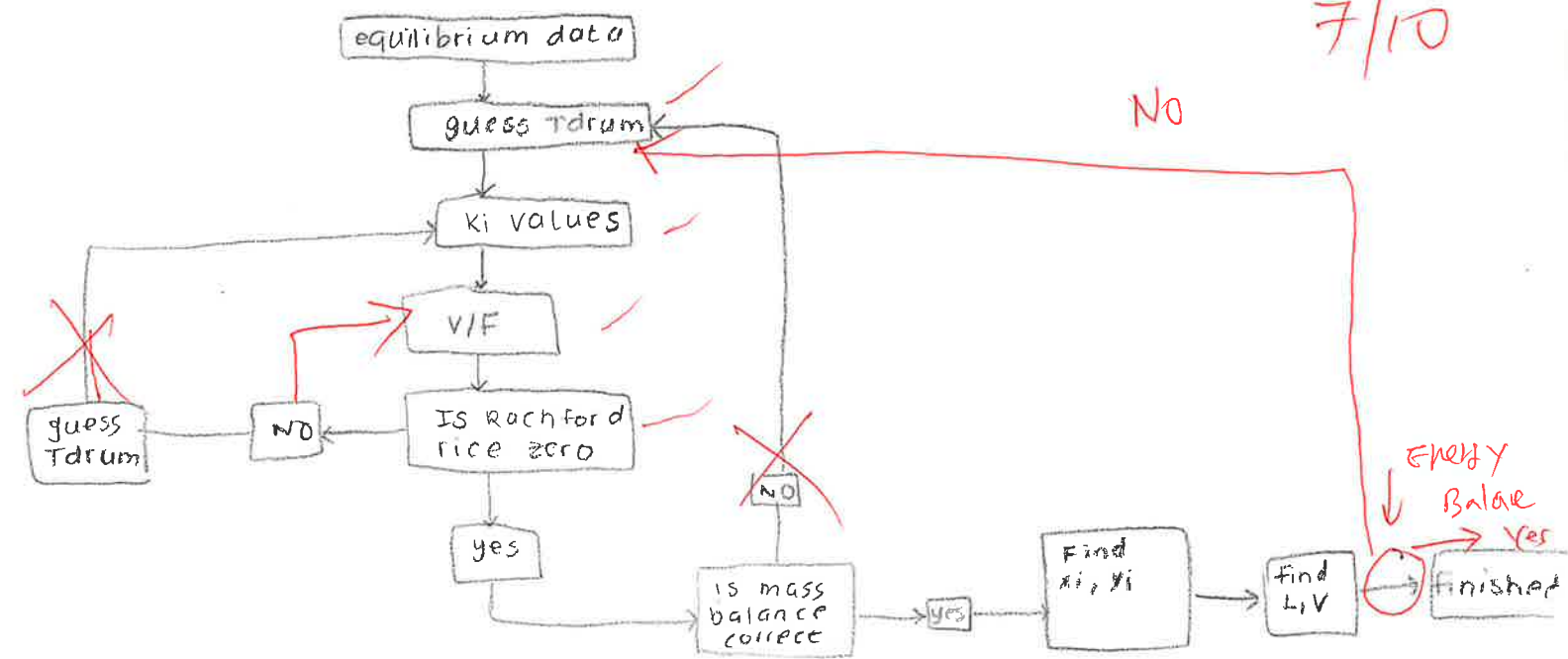
2. (0.60, 0.60)

$\frac{V_F}{F} = 0.50$
 $V_F = 75 \text{ kmol/hr}$ 2/5
 $L = 75$
 $\frac{F}{V} = \frac{150}{75} = 2$
 $(0.45, 0.75)$
 $F = fV_F$
 $V = F - L$
 $V = fV_F - L$



0/10

2.)



4. $y = \frac{L}{V} x + (1 - \frac{L}{V}) x_D$
 slope: $\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{2}{3}$
 y intercept: $(1 - \frac{2}{3}) 0.80 = 0.27$
 $(0.80, 0.80)$ ✓ $(0, 0.27)$ θ/θ

5. $y = \frac{\bar{L}}{\bar{V}} x - (\frac{\bar{L}}{\bar{V}} - 1) x_B$
 plug in $\bar{L}$ and $\bar{V}$
 after finding them $1/\theta$
 slope: 1.3
 x intercept @ $y=1$: $x=0.79$
 $(0.1, 0.1)$ on $y=x$

b. a. feed 1:
 sat liq $q=1$ vertical ✓
 $y = \frac{1}{1-1} x + \frac{1}{1-1} q_1$
 $(0.6, 0.6)$ ✓
 b. feed 2:
 $y = \frac{0.80}{0.80-1} x + \frac{0.40}{1-0.8}$
 $y = -4x + 2$
 $(0.8, 0.8)$ $(0, 2)$ not useful!
 $\theta/10$

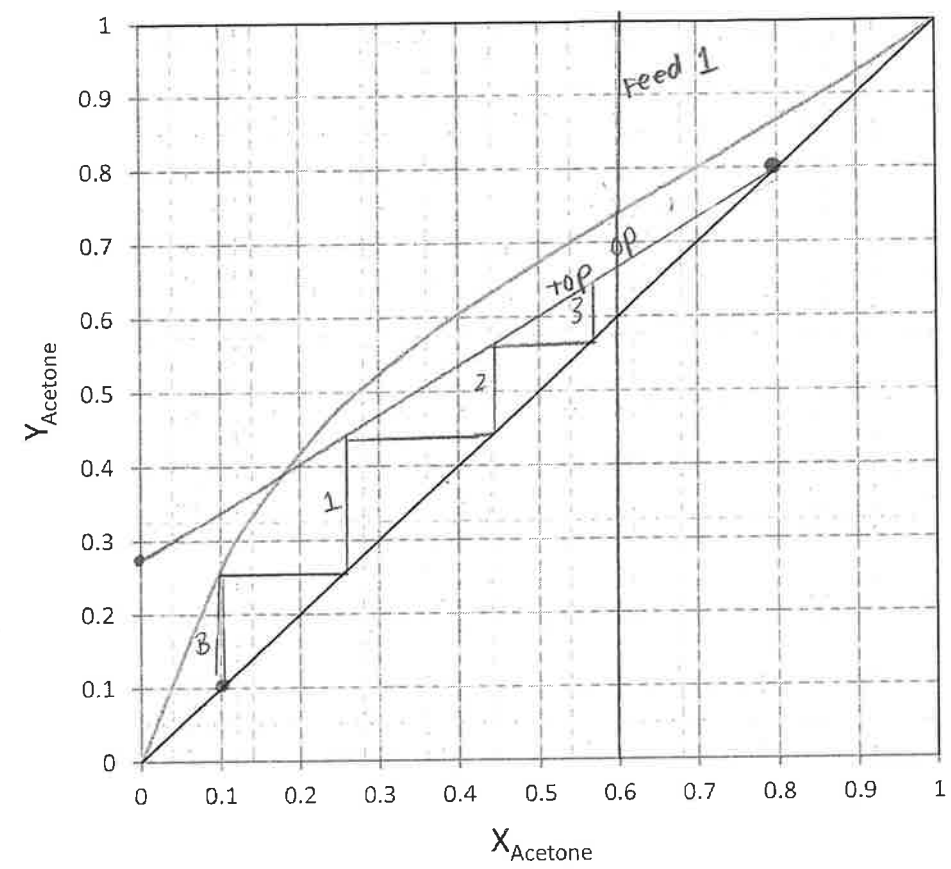
7. middle section
 ~~$F_1 + L_0 + L = V + V_0 + D$~~
 ~~$F_1 z_1 + L_0 x + L x = V y + V_0 y_0 + D x_D$~~

→ step off stages bet. equil. curve & op line

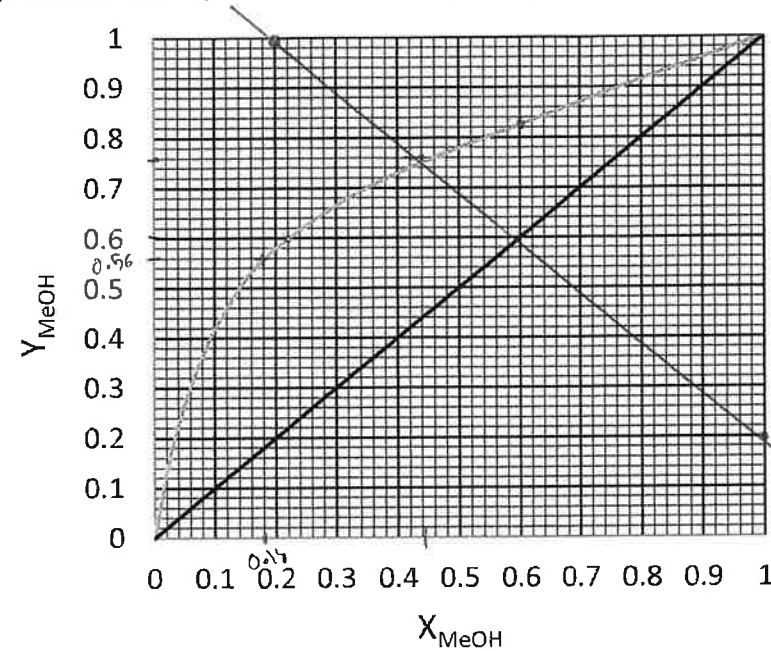
8. 5 stages + 1 reboiler $2/5$

9. optimum ~~around~~ 4 where most intersect

10. ~~$(0.56, 0.67)$~~



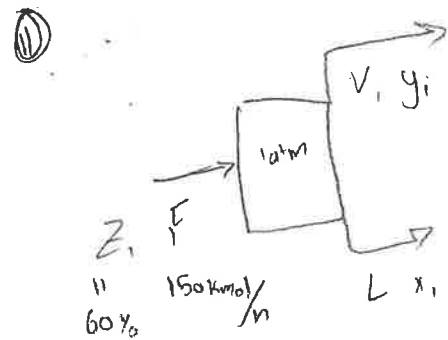
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Flow
chart



1 →

$$F = V + L$$

$$F - V = L$$

$$F \cdot Z_1 = V \cdot y_1 + L \cdot x_1$$

$$y = \frac{F \cdot Z_1 - L \cdot x_1}{V} \Rightarrow \frac{F \cdot Z_1 - \frac{L}{V} \cdot x_1}{V} = y$$

3/3

Z =

Amt of methanol coming in

$$150 \cdot 0.6 =$$

$$90 \text{ kmol/h}$$

50% = Vaporized

$$\frac{V}{F} = 0.5$$

$$V = 75 \text{ kmol/h}$$

Vapor

Liquid = 75 kmol/h

$$y = -\frac{L}{V}x + \frac{F}{V}Z = -\frac{L}{V}x + \left(\frac{150}{75}\right)(0.6) = y$$

$$1 = -\frac{F-V}{V} = \frac{1-\frac{L}{F}}{\frac{F}{V}} + 1.2$$

$$y = -1x + 1.2$$

$$y = -1(1) + 1.2$$

$$1 = -1x + 1.2$$

$$\frac{1-1.2}{-1} = x$$

$$x = 0.2$$

$$\text{Liquid} = 75 \text{ kmol/h}$$

$$x_1 = 0.44$$

$$y_1 = 0.76$$

5/5

get x, y_i values $\rightarrow x_i = \frac{z_i}{1 + (k_i - 1) \frac{V}{F}}$, $y_i = k_i x_i$

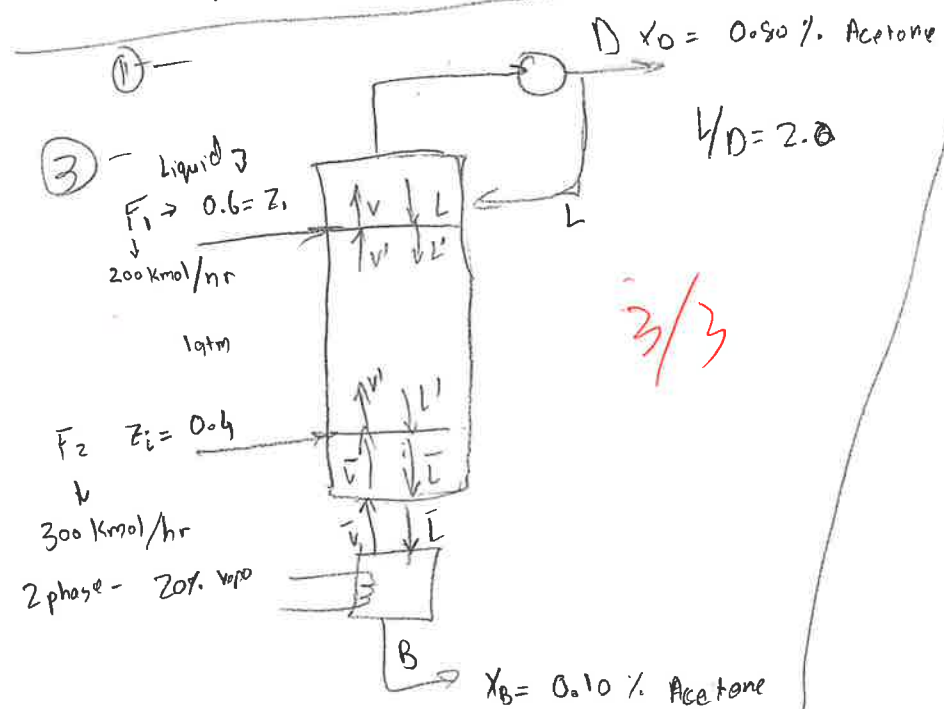


get V & L values

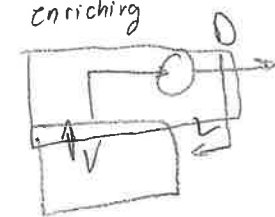


Newtonian Energy balance $\rightarrow$ Enthalpy converges to $\rightarrow \bigcirc$

get the New T and solve again

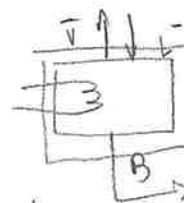


enriching



$$V = L + D$$

Strip -



$$\bar{L} = \bar{V} + B$$

8/8

② — Feed 1 F_1 $\checkmark$ $F_1 + V' + L = V + L'$

Feed 2 F_2 $\checkmark$ $F_2 + \bar{V} + L' = \bar{L} + V'$

enrich

③ $V = L + D$

$$V \cdot y = L \cdot x + D \cdot x_0$$

$$F_1 \cdot Z_1 + V' \cdot y' + L \cdot x = V \cdot y + L' \cdot x' = \text{Feed}_1$$

$$F_1 + V' + L = V + L'$$

Feed 2 - $F_2 + \bar{V} + L' = \bar{L} + V'$

$$F_2 \cdot Z_2 + \bar{V} \cdot \bar{y} + L' \cdot y' = \bar{L} \cdot \bar{x} + y' \cdot V'$$

$$F_1 = 200 \text{ kmol/hr}$$

$$Z_1 = 0.6$$

$$120 \text{ kmol/hr} = \text{Feed}_1$$

$$q = \frac{L}{F} = 0.6$$

Strip - $\bar{L} = \bar{V} + B$

$$\bar{L} \cdot \bar{x} = \bar{V} \cdot \bar{y} + B \cdot x_B$$

④ once getting ~~L & V~~ L & V - use it to get slope

Top line

$$\frac{L}{V} = \text{slope} - \left(1 - \frac{L}{V}\right) 0.8 = y \text{ intercept}$$

6/8

plot it on the graph = ~~not~~

Line goes through $(0.8, 0.8)$ & $y(\text{int})$ ✓

⑤ same as above once getting $\bar{L}$ & $\bar{V}$ use it to get slope

→
This is Bottom Line

$$\text{the } \frac{\bar{L}}{\bar{V}} - 1 \text{ for intercept}$$

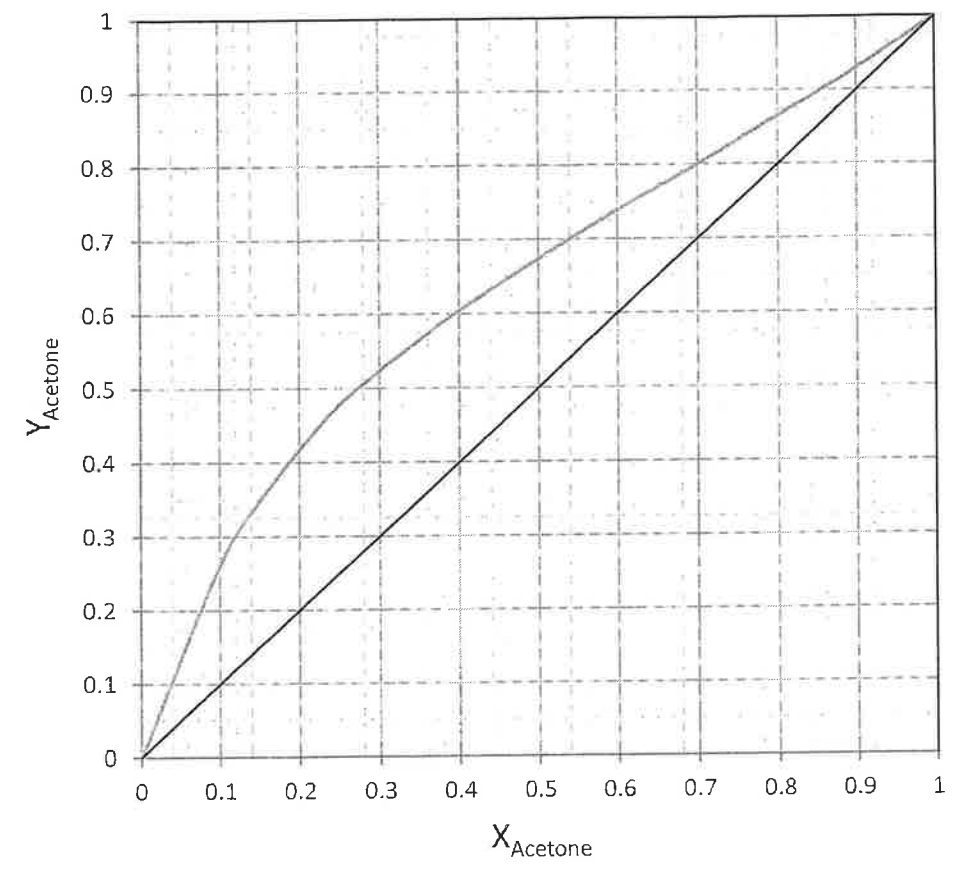
line goes through $(0.1, 0.1)$

6/8

⑥ a) Point $(0.6, 0.6)$ and $y = \frac{L F_1}{F_1}$ $q=1 \rightarrow \text{slope} = \infty$ $y = 14x + 2$

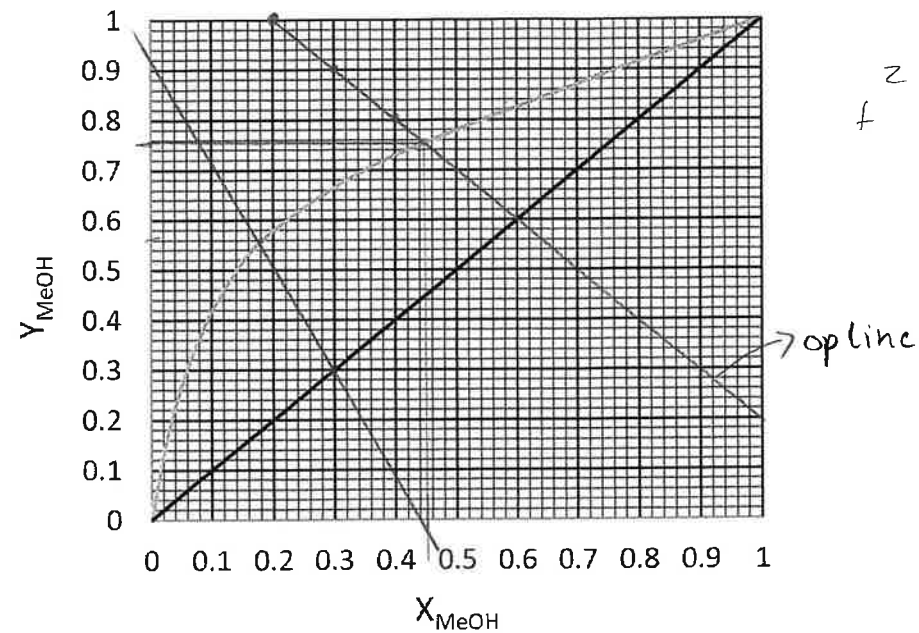
b) Points $(0.4, 0.4)$ and $y = -$ $q=0.8$ $y = -\frac{0.8}{0.8-1} + \frac{0.4}{1-0.8}$ $\text{slope} = -4$ $y \text{ int} = 2$ ~~not useful~~

6/10



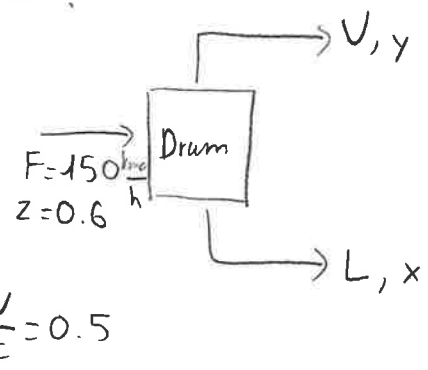
97/100

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2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1)



Overall MB: $F = V + L$

Comp MB: $Fz = Vy + Lx$

Solve for y : $Vy = -Lx + Fz$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

where $\frac{V}{F} = f$

$$\frac{L}{V} = \frac{F - V}{V} = \frac{1 - f}{f}$$

$$\Rightarrow y = -\frac{1-f}{f}x + \frac{z}{f} \rightarrow \text{Operating eq}$$

slope $-\frac{1-f}{f} = -\frac{1-0.5}{0.5} = -1$

y-intercept: $+\frac{0.6}{0.5} = +1.2$

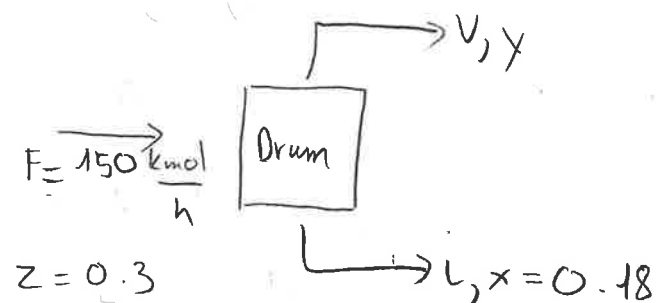
$$\Rightarrow y = -x + 1.2$$

for this $\begin{cases} y = 0.76 \\ x = 0.45 \end{cases}$

$$\frac{V}{F} = 0.5$$

$$\Rightarrow \begin{cases} V = 0.5 \times 150 = 75 \frac{\text{kmol}}{\text{h}} \\ L = 150 - 75 = 75 \frac{\text{kmol}}{\text{h}} \end{cases}$$

2) 5/5



$$\Rightarrow \frac{V}{F} = 0.326$$

$$V = 0.326 \times 150 = 48.9 \frac{\text{kmol}}{\text{h}}$$

$$L = 101.1 \frac{\text{kmol}}{\text{h}}$$

y-intercept $(0, 0.92)$

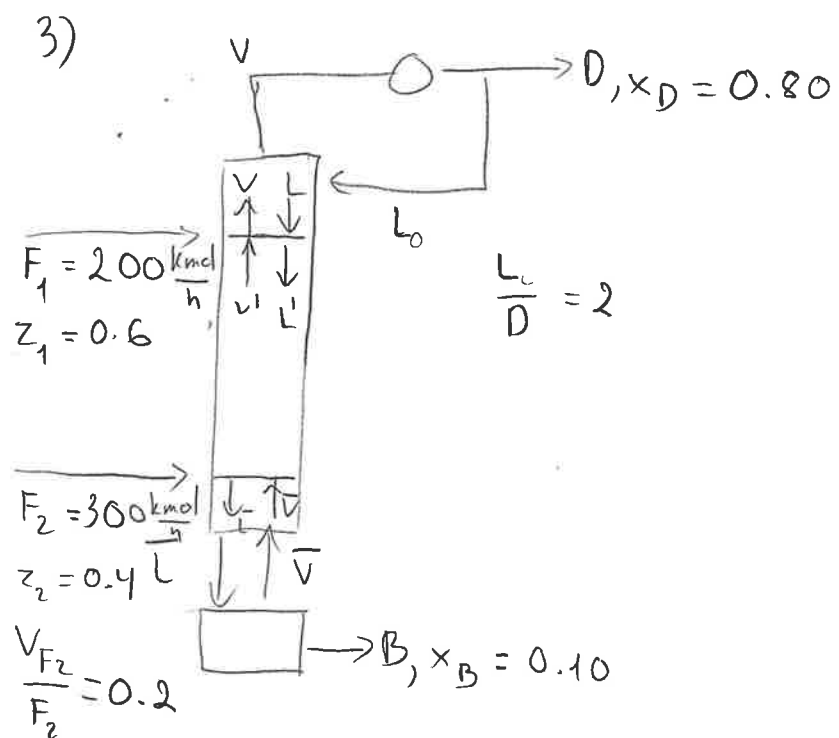
$$\Rightarrow \frac{z}{f} = 0.92$$

$$\Rightarrow f = \frac{0.3}{0.92} = 0.326$$

$$x = 0.18$$

$$y = 0.56$$

3) 10/10

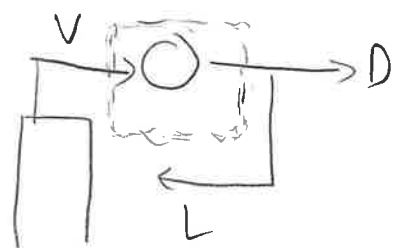


1) 3/3

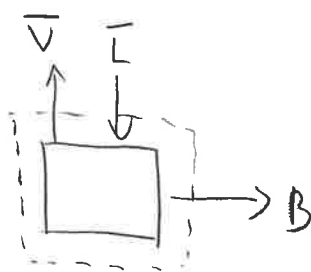
2) 8/8

3) 15/15

Enriching

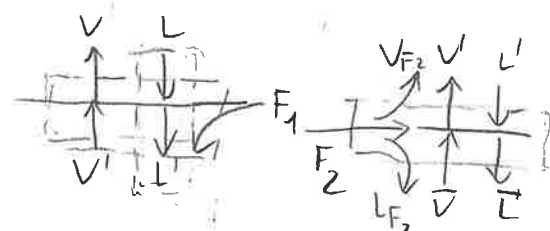


stripping



Feed 1

Feed 2



Overall MB: $F_1 + F_2 = D + B$

Comp MB: $F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$\rightarrow 200 + 300 = D + B$

$200 \times 0.6 + 300 \times 0.4 = D \times 0.80 + B \times 0.10$

$\Rightarrow D = 271.43 \frac{\text{kmol}}{\text{h}}$

$B = 228.57 \frac{\text{kmol}}{\text{h}}$ ✓

$\frac{L}{D} = 2 \rightarrow L = 2 \times 271.43 = 542.86 \frac{\text{kmol}}{\text{h}}$

$V = D + L = 271.43 + 542.86 = 814.29 \frac{\text{kmol}}{\text{h}}$ ✓

$L' = L + F_1 = 542.86 + 200$
 $= 742.86 \frac{\text{kmol}}{\text{h}}$

$V' + L + F_1 = V + L'$

$\Rightarrow V' = 814.29 \frac{\text{kmol}}{\text{h}}$

$\bar{V} = V_{F_2} + V' = 754.29 \frac{\text{kmol}}{\text{h}}$

$\bar{L} = L' + L_{F_2} = 982.86 \frac{\text{kmol}}{\text{h}}$

$$F_1 + V' = L' + D$$

$$F_1 z_1 + V'y = L'x + D x_D$$

$$V'y = L'x + D x_D - F_1 z_1$$

$$y = \frac{L'}{V'} x + \frac{D}{V'} x_D - \frac{F_1}{V'} z_1$$

$$y = \frac{L'}{V'} x + \frac{D x_D - F_1 z_1}{V'}$$

$$\text{slope} = \frac{L'}{V'} = \frac{742.86}{814.29} = 0.91$$

$$y\text{-intercept} = \frac{D x_D - F_1 z_1}{V'} = 0.12$$

6 stage (.5 stages + PR)

Feed 1 optimum 2 stage

Feed 2 optimum 4 stage

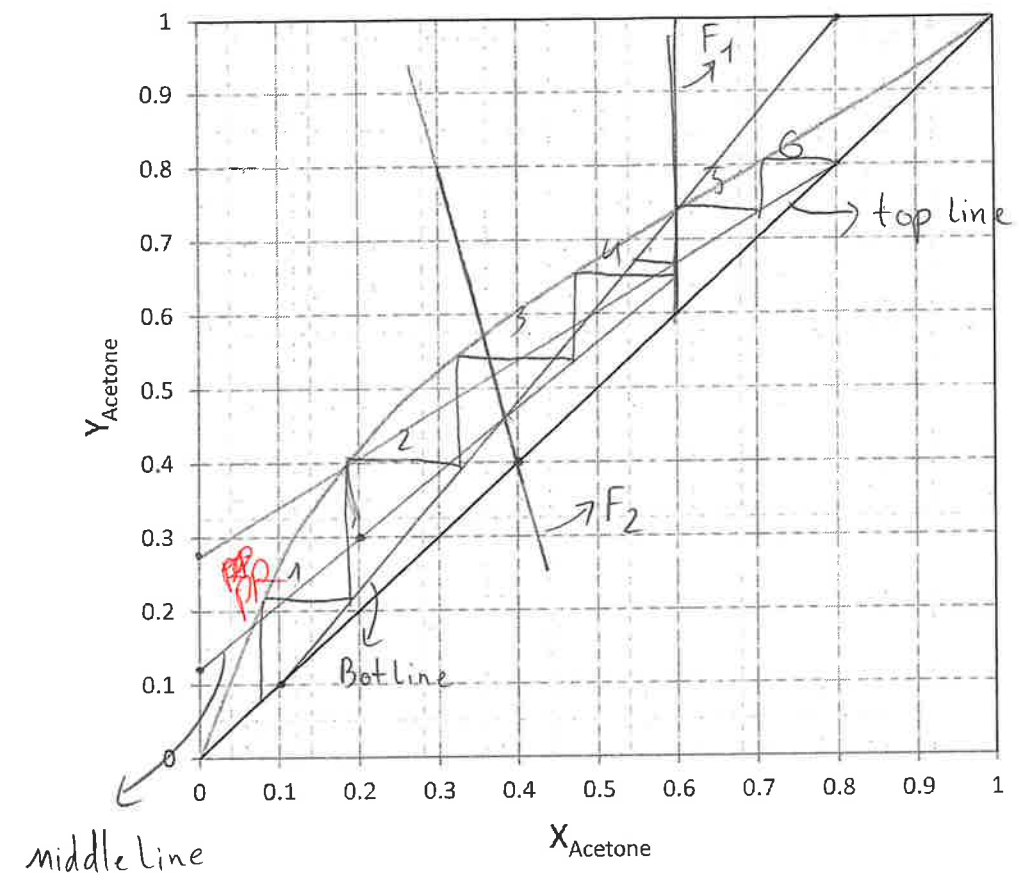
7) 10/10

8) 4/5

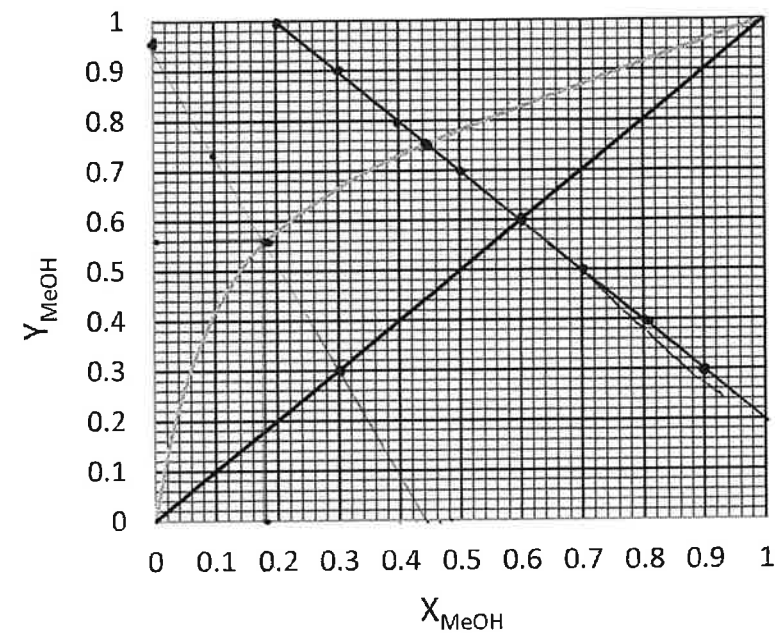
9) 3/3

10) 0/2

10) (0.49, 0.67)



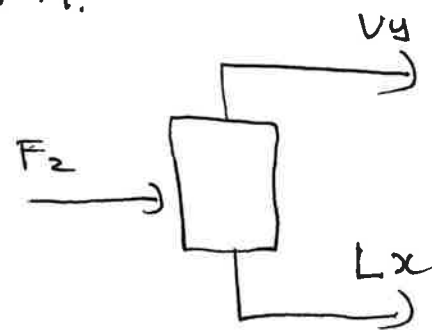
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Question 1.

①

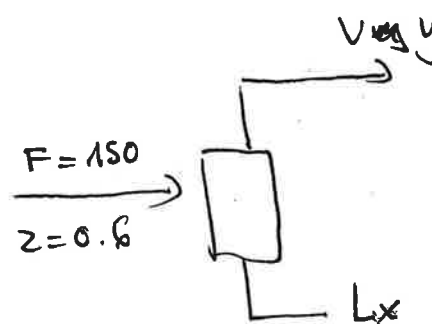


3/3

$$MB: F = V + L$$

$$CMB: Fz = Vy + Lx$$

②



$$\frac{V}{F} = f = 0.5$$

$$\Rightarrow V = 0.5F = 0.5 \times 150 = 75 \text{ kmol/h}$$

$$\Rightarrow L = 150 - 75 = 75 \text{ kmol/h}$$

$$MB: F = V + L$$

$$CMB: 150 \cdot 0.6 = Vy + Lx$$

$$\text{Op equation: } y = \frac{-L}{V}x + \frac{F}{V}z$$

$$= \frac{f-1}{f}x + \frac{z}{f}$$

$$= \frac{0.5-1}{0.5}x + \frac{0.6}{0.5}$$

$$= -x + 1.2$$

$$\Rightarrow x = 0.44$$

$$y = 0.74$$

5/5

6.

a) $q_1 = 1 \Rightarrow y_{z=1} = x_i = 0.6$ ✓
 slope unknown.

b) $q_2 = 0.8 \Rightarrow \frac{0.8}{0.8-1} = -4$ ✓

9/10

$$\frac{z_i}{1-q_i} = 0.4$$

$$\Rightarrow \frac{0.4}{1-0.8} = 2$$

$$\Rightarrow -4x + 2 = y$$

x-intercept: $x = 0.5$

7. Middle op

$F = V^2 = L^2 + D$ ✓

CMB

$F_2 + V^2 y = L^2 x + D \times D$

$\Rightarrow y = \frac{L^2 x + D \times D - F_2}{V^2}$ ✓

$\Rightarrow y = \frac{L^2}{V^2} x + \frac{D \times D - F_2}{V^2}$

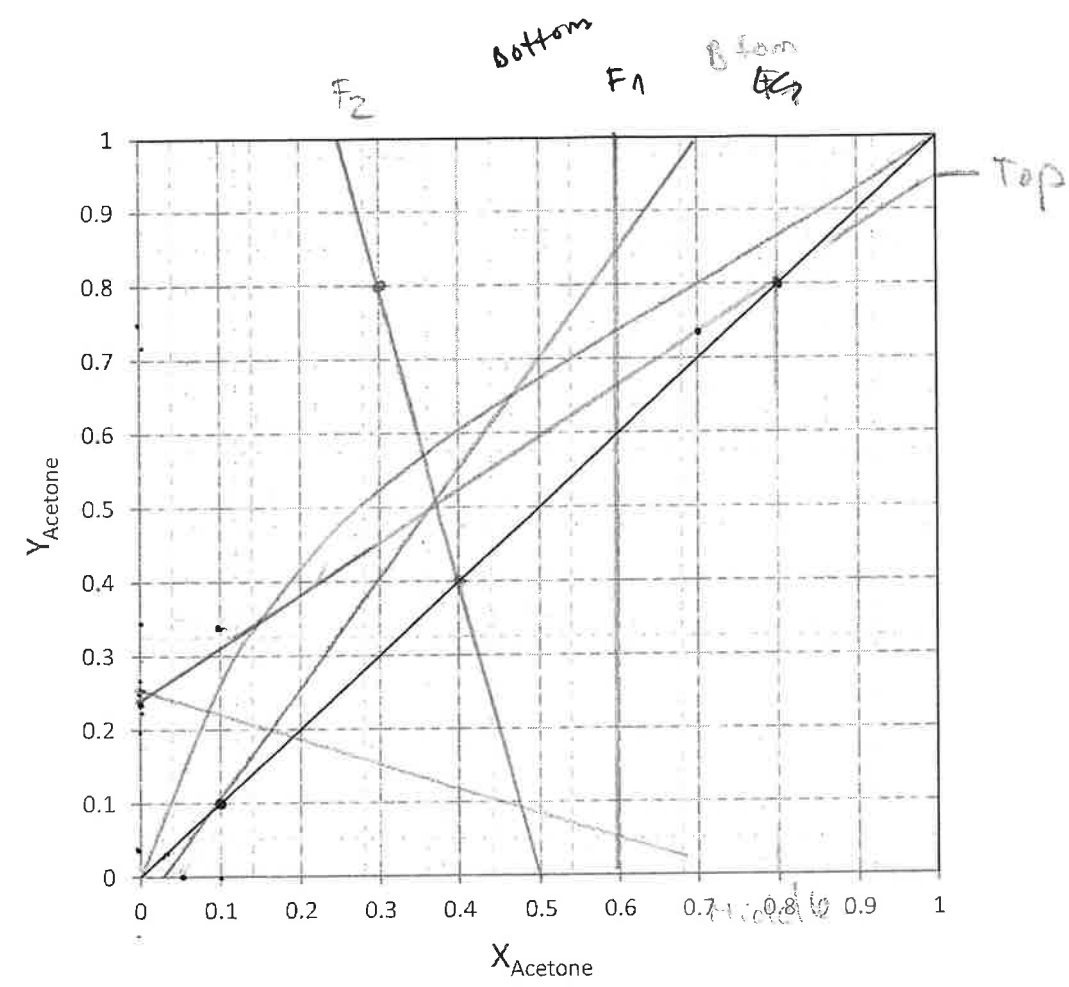
$\Rightarrow y = \frac{L^2}{V^2} x + \frac{97.144}{V^2}$

$= 0.8276x + 0.2345$

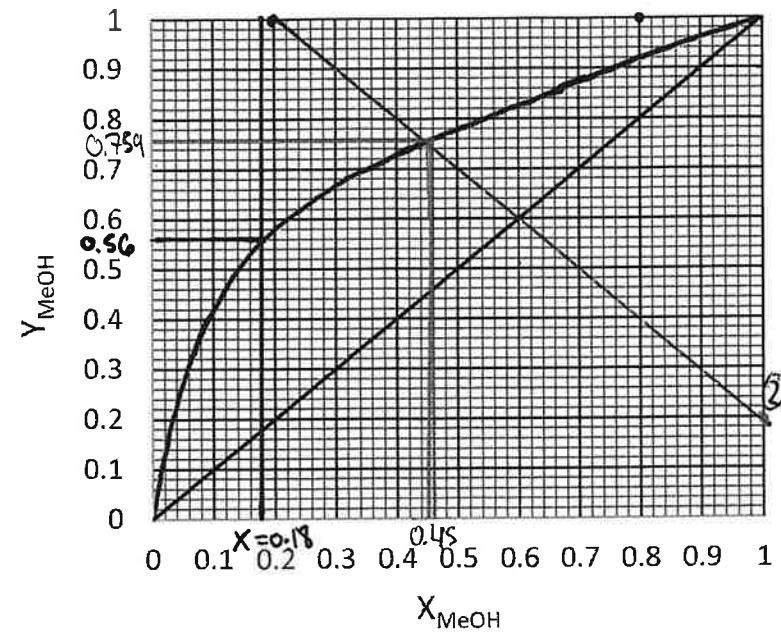
$\Rightarrow$

method is o/c!

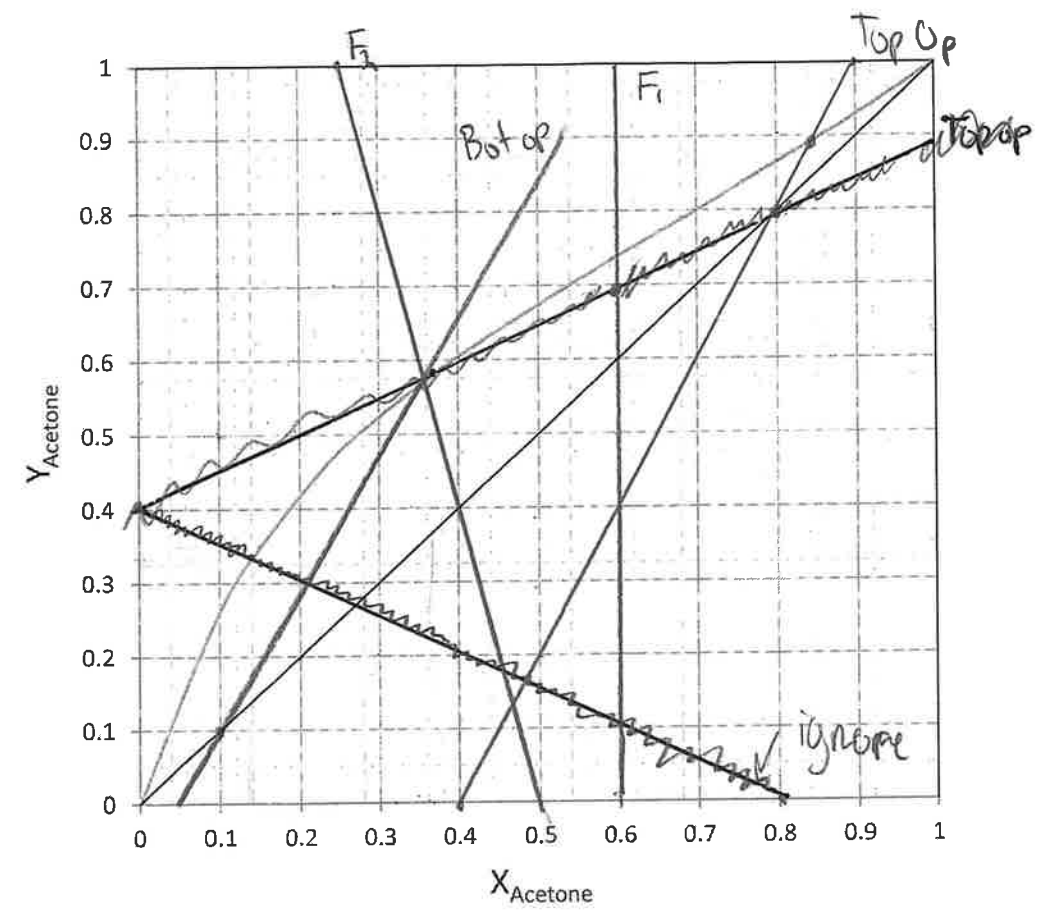
8/10



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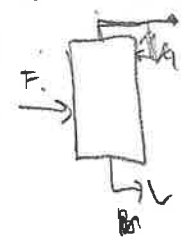


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Matthew Lobson

①



$$F = L + V$$

$$Fz = Lx + yV$$

$$y = \frac{F}{V}z - \frac{L}{V}x \Rightarrow y = -\frac{F-V}{V}x + \frac{F}{V}z \quad \left| \frac{V}{F} = f \right.$$

$$y = -\frac{1-f}{f}x + \frac{z}{f}$$

3/3

② $F = 150 \frac{\text{kmol}}{\text{hr}}$

$$\frac{V}{F} = f \quad V = fF = 0.50 \cdot 150 = 75 \frac{\text{kmol}}{\text{hr}} = V$$

$$75 \frac{\text{kmol}}{\text{hr}} = L$$

b) $z = 0.60$

$$f = 0.50 \quad y = -1x + 1.2$$

$$(0.2, 1)$$

$$x \approx 0.45$$

$$y \approx 0.759$$

5/5

③ $z = 0.30$

a) $x = 0.18$

$$0.56 = -\frac{1-f}{f} \cdot 0.18 + \frac{0.30}{f}$$

$$F = 150$$

$$0.56f = 0.18 - 0.18f + 0.30$$

$$0.75f = 0.48$$

$$f = 0.649 = \frac{V}{F}$$

$$V = 0.649(150) = 97.35 \frac{\text{kmol}}{\text{hr}}$$

$$L = F - V = 52.65 \frac{\text{kmol}}{\text{hr}}$$

3/10

$$(0.18, 0.56)$$

$$x = 0.18$$

$$y = 0.56$$

line

Find y-intercept: $0.9 = \frac{F}{V}z \rightarrow \frac{V}{F} = \frac{1}{3}, L = 100$

② Guess $T_D \xrightarrow{K_i = K_i(T)}$ find $K_i \xrightarrow{RR}$ find $\frac{V}{F} \xrightarrow{\text{Yes/No}}$ find comp enthalpies from (L, V, x_i, y_i)

$$E_{in} = E_{out}$$

Converge (T)
Newtonian method $(T_{j+1} = T_j + \frac{E(T)}{(\frac{\partial E}{\partial T})})$

New Temp

until

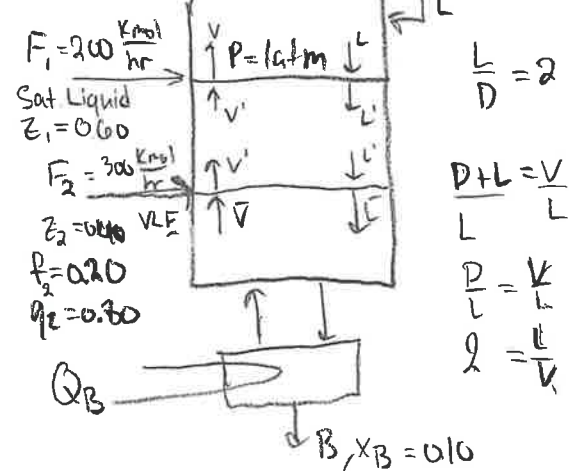
$$\epsilon = 0.005$$

$$T_{Guess+1} = T_D$$

9/10

Matthew Holstrom

3



CMO valid $V_j = V_{j+1}$, $L_j = L_{j+1}$

Overall MB
 $V' + L_x = V + L'$

$V = D + L$

$y = \text{und}$ (0.6, ∞)

$m = \frac{\#}{0} = \text{und}$

$q_1 = 1$

slope: ∞

(0.6, 0.6) on $q = X$

$q_2 = 0.80$

Overall MB
 $V + L' = V' + L$ $y = \frac{q_1}{q_1 - 1} x + \frac{z}{1 - q_1} = y = 4x + 2$ (0.75, 1)

Overall: $F_1 + F_2 = B + D$ $B = 500 - D$

$F_1 z_1 + F_2 z_2 = B x_B + D x_D$

$240 = B(0.10) + D(0.80) | B = 500 - D$

$140 = 0.7D$

$271.43 \frac{\text{kmol}}{\text{hr}} = D$

$228.57 \frac{\text{kmol}}{\text{hr}} = B$

$V + L = L' + V'$

enriching:

$D + L = V$ (0.8, 0.75) (on $y = x = x_D$)

$D + L = V$ $F_1 - D = L' - V'$ (0.8, 0.8)

$y = \frac{L}{V} x + (1 - \frac{L}{V}) x_D = 2x + 0.8$

stripping:

$F_2 = \frac{V'}{L'}$ $\frac{V'}{L'}$

$F = \bar{V} + B$ $f_2 = \frac{V_{F_2}}{F_2} = \frac{\bar{V} - V'}{F_2} \Rightarrow V' = \bar{V} - f_2 F_2 \Rightarrow \bar{V} = V' + f_2 F_2$ *envelope!*

$1 - f_2 = q_2 = \frac{L_{F_2}}{F_2} = \frac{L' - L}{F_2} = L' = L + q_2 F_2 \Rightarrow L = L' - q_2 F_2$

Middle Section

$V' = L' + D$

$V' \neq D + L'$

$V' = L' x + D x_D$

$D = V' - L'$

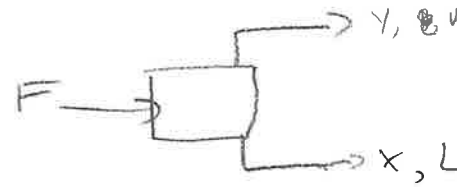
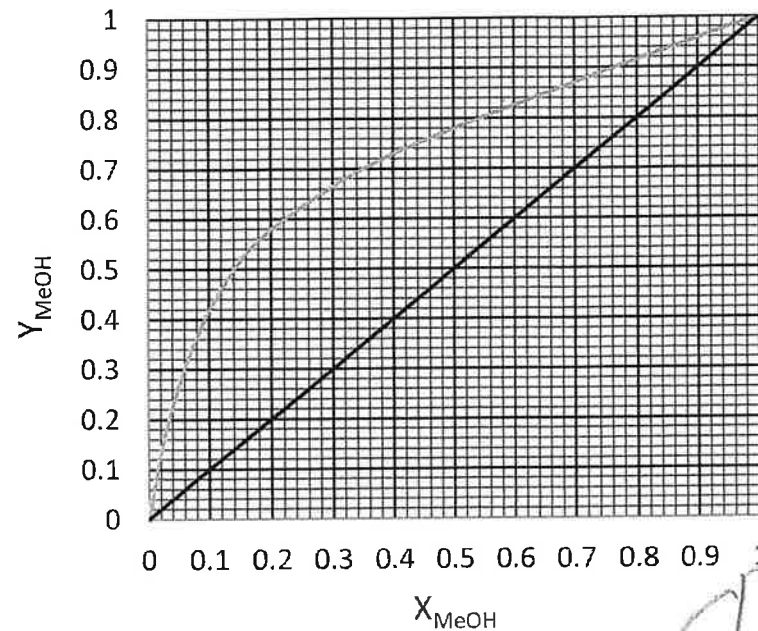
$y = \frac{L'}{V} x + (1 - \frac{L'}{V}) x_D$

2) 4/8

3) 5/15

4)

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



$$F = 150 \text{ kmol/h}$$

$$150 = 0.5V$$

$$150 = 0.5L$$

$$y = -\frac{L}{V}x - \frac{Fz}{V}$$

- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.

$$Fz = Vy + Lx$$

$$F = V + L$$

$$V = 75 \text{ kmol/h}$$

$$L = 75 \text{ kmol/h}$$

- 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.

$$V, L = 75 \text{ kmol/h}$$

$$150(0.6) = 75y + 75x$$

$$y = -x + 1.2$$

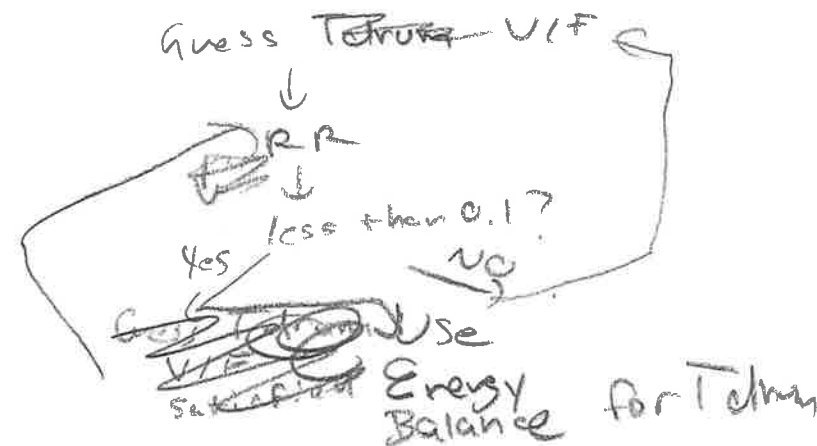
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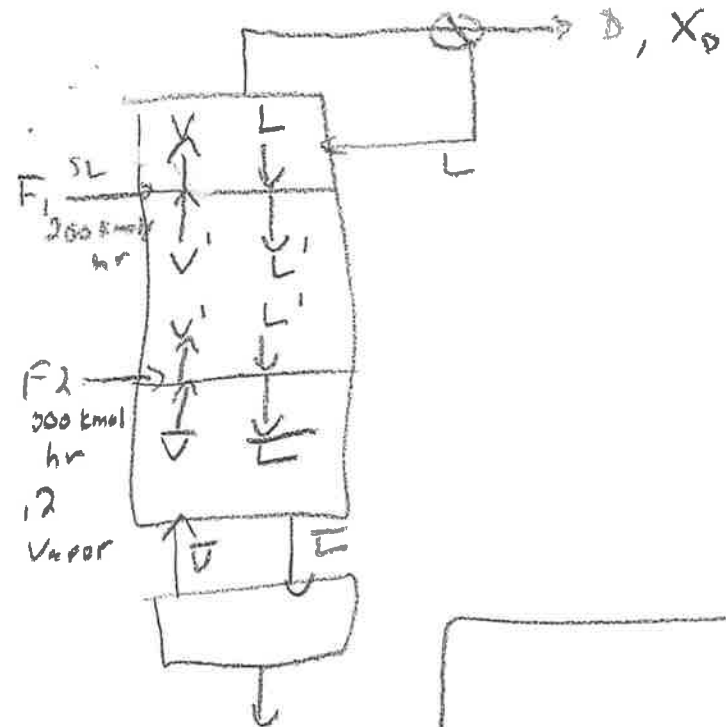
$$90 = 1.5V + 0.5L$$

$$150 = V + L$$

$$-75 = -0.5V + -0.5L$$

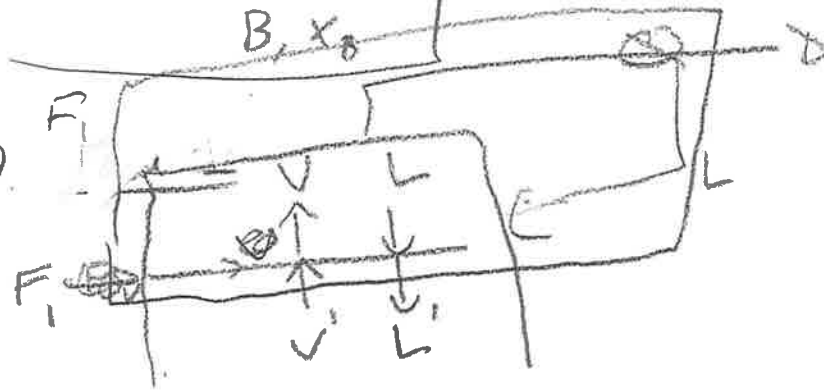


1)



Alex
luuu

2)



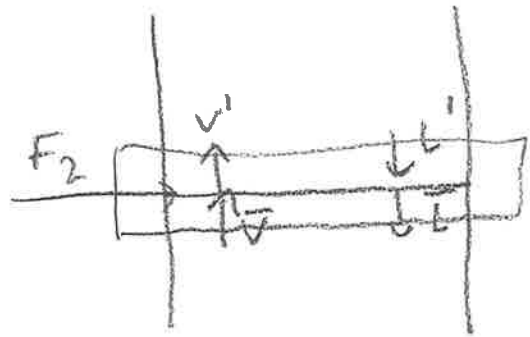
← Feed 1

$$200 + V' = L' + 271.43$$

$$F_1 + V' = L' + D$$

$$200 + V' = L' + D$$

← Feed 2



$$F_2 + L' + \bar{V} = V' + \bar{L}$$

$$300 + L' + \bar{V} = V' + \bar{L}$$

~~K₂~~, v', L', $\bar{L}$, $\bar{U}$

$$\frac{L}{D} = 2 \quad \frac{L}{271.429} = 2, \quad \boxed{L = 542.858 \text{ kmol/h}}$$

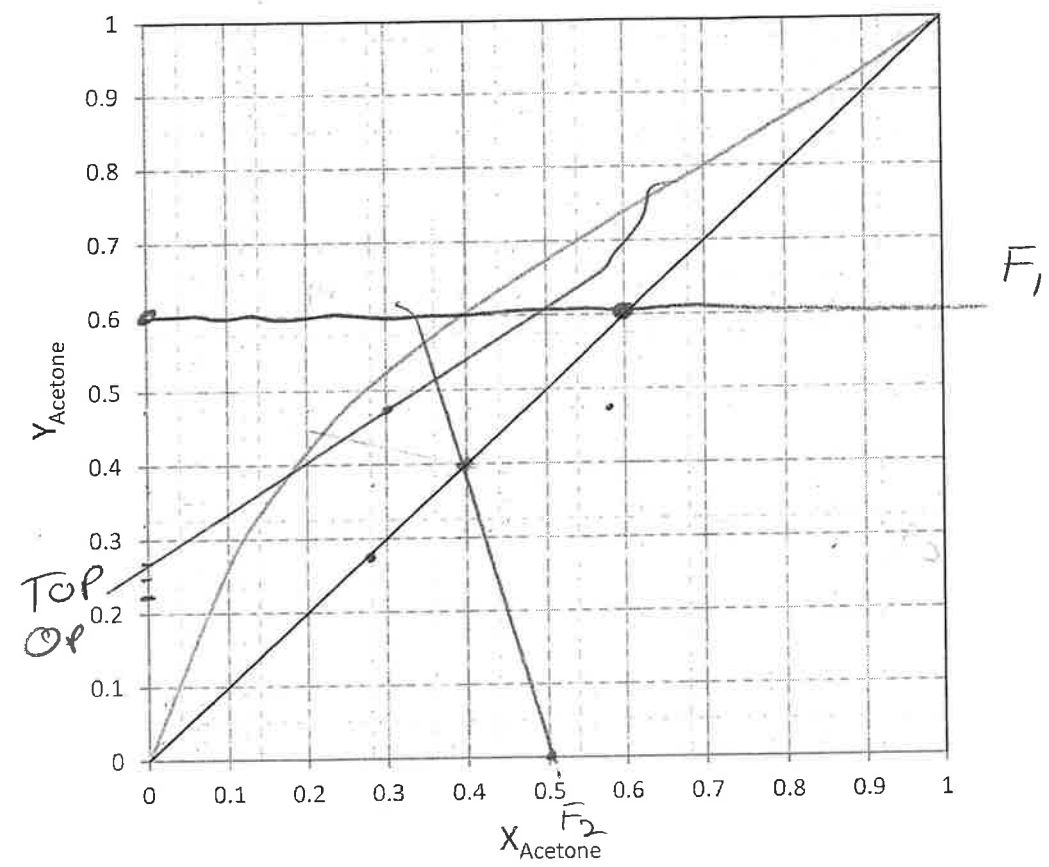
$$V = D + L$$
$$\boxed{V = 542.858 + 271.429 = 814.287 \text{ kmol/h}}$$

$$4.) Y = \left(\frac{L}{V}\right)X + \left(1 - \frac{L}{V}\right)X_D$$

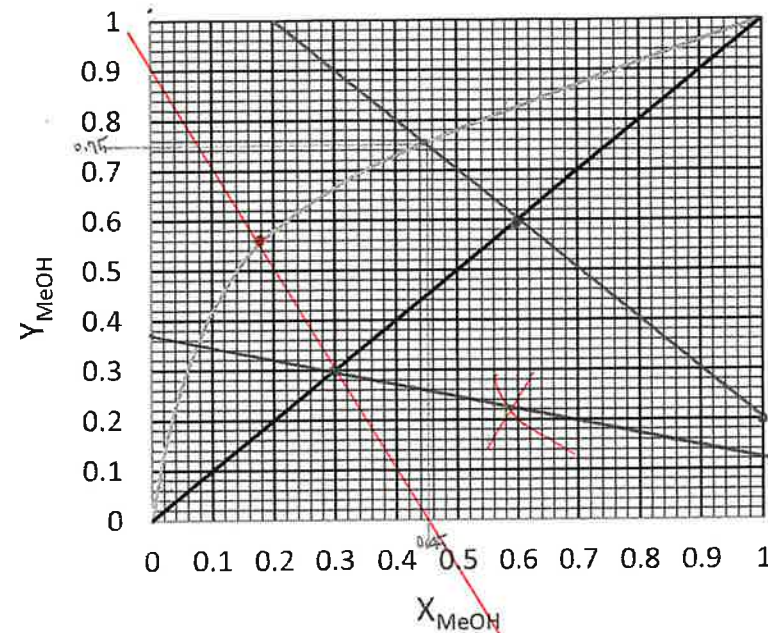
$$\frac{L}{V} = \frac{L}{L+D} = \frac{\frac{L}{D}}{\frac{L}{D} + 1} = \boxed{\frac{2}{3} \text{ Slope}}$$

$$\left(1 - \frac{L}{V}\right)X_D = \left(1 - \frac{2}{3}\right)(0.8) = \boxed{0.27 \text{ Intercept}}$$

$$5.) Y = \left(\frac{\bar{L}}{\bar{V}}\right)X - \left(\frac{\bar{L}}{\bar{V}} - 1\right)X_B$$



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find **graphically** V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1. 1)

∴ overall mass balance: $F = L + V$
 component mass balance: $Fz = Lx + Vy$

3

2) $F = 150 \text{ kmol/h}$ $z = 0.6$
 $V/F = 0.5 \rightarrow V = 0.5 F = 0.5 \times 150$
 $= 75 \text{ kmol/h}$

5

$F = L + V \rightarrow L = F - V = 150 - 75$
 $= 75 \text{ kmol/h}$

$Fz = Lx + Vy$

$Vy = Fz - Lx$

$y = -\frac{L}{V}x + \frac{Fz}{V}$

$y = -\frac{75}{75}x + \frac{150}{75} \cdot 0.6$

$y = -x + 1.2$

$(0, 1.2) \text{ \& } (0.6, 0.6) \text{ \& } (1, 0.2)$

$V = 75 \text{ kmol/h}$, $L = 75 \text{ kmol/h}$
 $x_{\text{MeOH}} \approx 0.45$, $y_{\text{MeOH}} \approx 0.75$

3) $F = 150 \text{ kmol/h}$ $z = 0.3$

2/10

liquid product = 0.18

vapor " = 0.82

$y = -\frac{L}{V}x + \frac{Fz}{V}$

$y = -\frac{21}{123}x + \frac{150}{123} \times 0.3$

$y = -0.22x + 0.37$

$(0.3, 0.3) \text{ \& } (0, 0.37)$

$(0.18, 0.55) \rightarrow y\text{-intercept} = 0.9$
 $\text{ \& } (0.3, 0.3)$

$Fz = 0.9$

$V/F = 0.82 \rightarrow V = 0.82 F = 0.82 \times 150 = 123 \text{ kmol/h}$
 $F = L + V \rightarrow L = F - V = 150 - 123 = 27 \text{ kmol/h}$

$\frac{V}{F} = \frac{1}{3}$

$V = 50 \text{ kmol/h}$
 $L = 100$

$V/F = 0.82$
 $y = -0.22x + 0.37$

2. ① Guess T

④ Energy difference = 0 $\rightarrow$ calculate T (guessing)

② At T, P $\rightarrow$ Find $K_i(T, P)$
 (in chart)

⑤ Calculate x_i, y_i

if not?

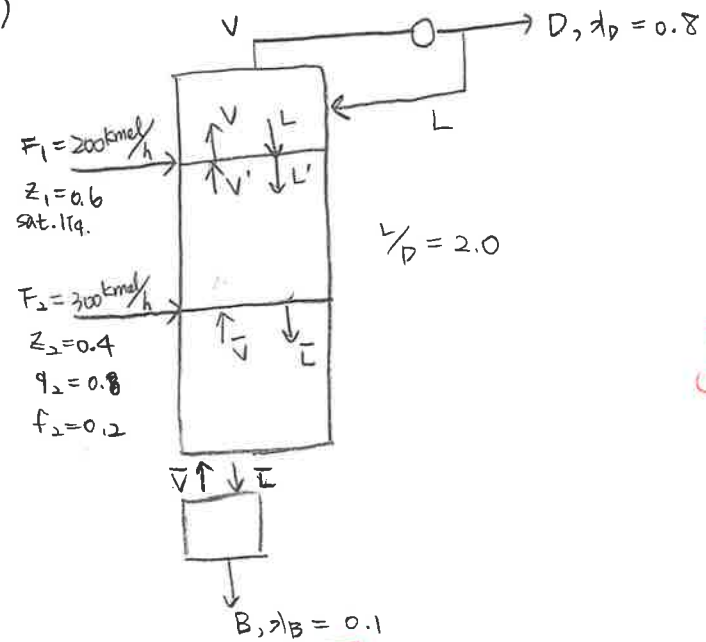
③ Using R-R equ., find V/F

⑥ Calculate L, V

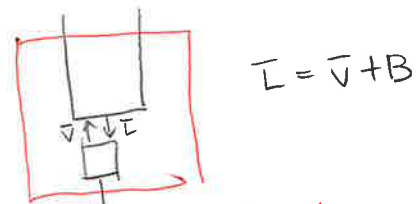
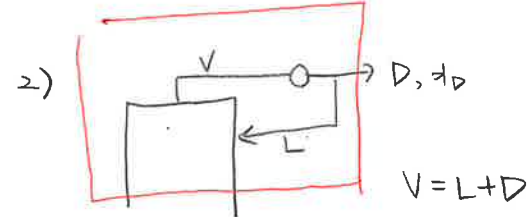
if $f(V/F) \neq 0$

5/10

3. 1)



3/3



$$F_1 + V' + L = V + L'$$

$$V = V_{F_1} + V'$$

$$L' = L_{F_1} + L$$

$$F_2 + \bar{V} + L = V + \bar{L}$$

$$V = V_{F_2} + \bar{V}$$

$$\bar{L} = L_{F_2} + L$$

$$\therefore V = L + D, \bar{L} = \bar{V} + B,$$

$$F_1 + V' + L = V + L', F_2 + \bar{V} + L = V + \bar{L}$$

3) overall M.B: $F_1 + F_2 = D + B$

comp. M.B: $F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$L/D = 2 \rightarrow L = 2D \quad V = L + D \rightarrow V = 2D + D = 3D$

$q_2 = \frac{\bar{L} - L}{F_2} = 0.8 \rightarrow \frac{\bar{L} - L}{300} = 0.8$

$\bar{L} = L' + 0.8 F_2 = 942.86 \text{ kmol/h}$

$\bar{V} = 754.29 \text{ kmol/h}$

$\bar{V} = \bar{L} - B = 942.86 - 188.57 = 754.29 \text{ kmol/h}$

$q_1 = \frac{L' - L}{F_1} \rightarrow \frac{L' - L}{200} = 1 \quad L' = 200 + 942.86 = 1142.86 \text{ kmol/h}$

$200 + 300 = D + B$

$120 + 120 = 0.8D + 0.1B \times 10$

$500 = D + B$

$2400 = 8D + B$

$D = 1900 \Rightarrow D = 271.43 \text{ kmol/h}$

$L = 2 \times 271.43 = 542.86 \text{ kmol/h}$

$V = 271.43 + 542.86 = 814.29 \text{ kmol/h}$

$B = 500 - 271.43 = 228.57 \text{ kmol/h}$

$\bar{V} = 754.29 \text{ kmol/h}$

$\bar{L} = 942.86 \text{ kmol/h}$

$V' = V + L' - F_1 - L = 814.29 + 1142.86 - 200 - 542.86 = 1194.29 \text{ kmol/h}$

3. 4) Top of line: $y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$

$$= \frac{542.86}{814.29}x + (1 - \frac{542.86}{814.29})0.8$$

$$y = 0.67x + 0.21$$

on $y=x$

$$(0.6, 0.6) \text{ \& } (0, 0.21)$$

$$\text{slope} = 0.67$$

2/2

5) Bottom of line: $y = (\frac{L}{V})x - (\frac{L}{V} - 1)x_B$

$$y = \frac{182.86}{554.29}x - (\frac{182.86}{554.29} - 1) \times 0.1$$

$$y = 1.41x - 0.041$$

$$(0.1, 0.1) \text{ \& } (0, -0.041)$$

$$\text{slope} = 1.41$$

$$1.41x = 0.041$$

x-intercept @ $y=1$

$$x = 0.79$$

6) (a)

$$q_1 = \frac{L' - L}{F_1} = 1$$

$$y = \frac{1}{1-1}x + \frac{0.6}{1-1}$$

$$\text{slope} = \infty \quad y\text{-intercept} = 0.6$$

$$(0.6, 0.6)$$

9/10

$$(b) \quad q_2 = \frac{\bar{L} - L}{F_2} = 0.8$$

$$y = \frac{0.8}{0.8-1}x + \frac{0.4}{1-0.8}$$

$$y = -4x + 2$$

$$\text{slope} = -4 \quad y\text{-intercept} = 2$$

$$(0.4, 0.4)$$

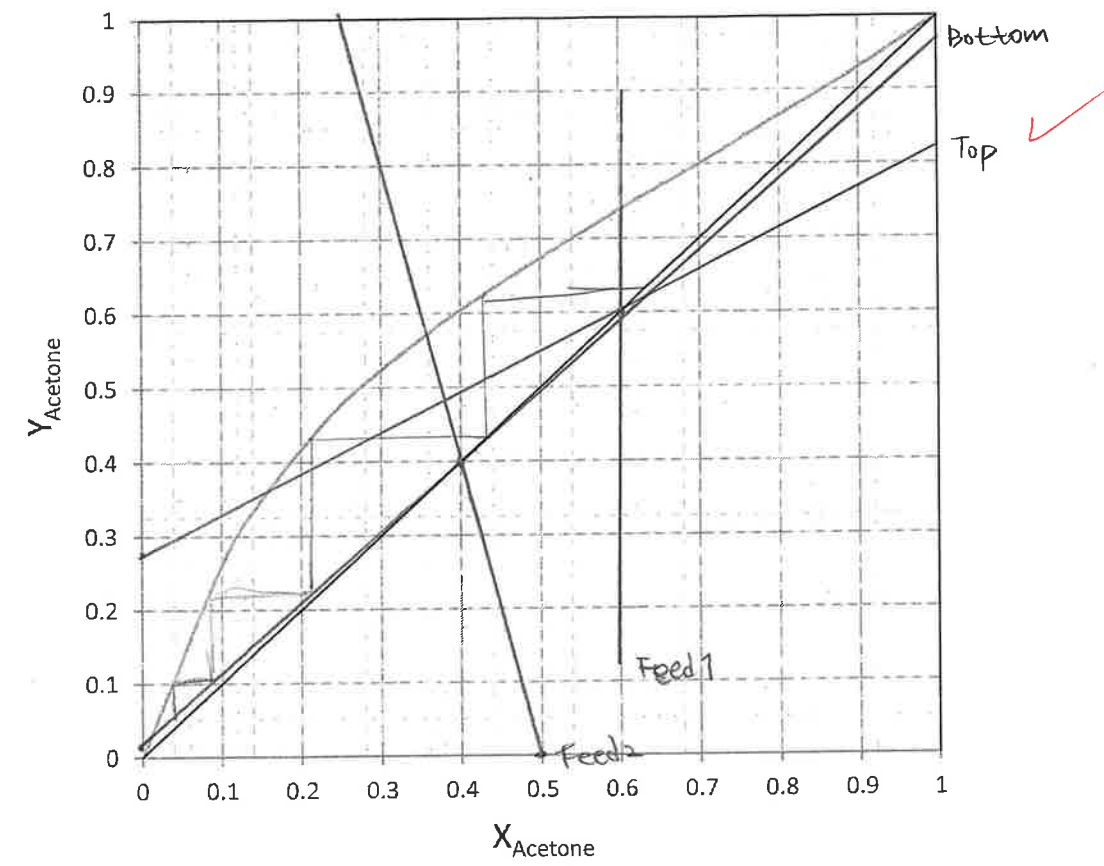
x-intercept: $x = 0.5$

8) 2/5

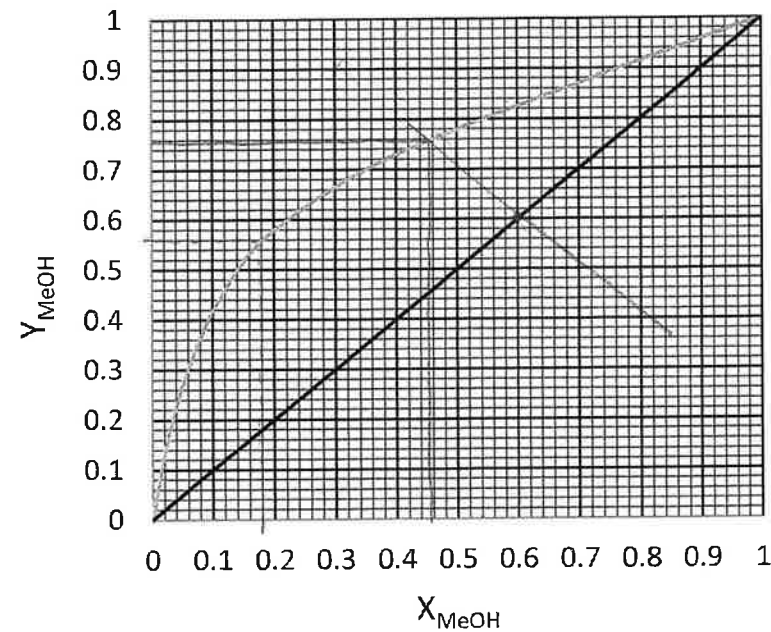
9) Feed 1 = 3

" 2 = 4

2/3



1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

1a.) M.B.) $F = L + V$ C.B.) $Fz_i = Lx_i + Vy_i$

or eq) $Fz_i = Lx_i = Vy_i \Rightarrow (L+V)z_i - Lx_i = Vy_i \Rightarrow y = \left(\frac{L+V}{V}\right)z - \left(\frac{L}{V}\right)x$

1b.) $Z = 150 \frac{\text{kmol}}{\text{hr}}$, $z_{\text{meOH}} = 0.6$, $\frac{V}{F} = 0.5 = \frac{V}{L+V}$

y_{int}) $\frac{1}{\frac{V}{L+V}} z = \frac{1}{0.5} (0.6) = 1.2$

$V = 150 \times 0.5 = 75 \frac{\text{kmol}}{\text{hr}}$
 $L = 75$

Slope) $\frac{F}{V} = \frac{L+V}{V} \Rightarrow \frac{F}{V} - 1 = \frac{L}{V} = 1 = m \Rightarrow y = -x + 1.2$

Point: $(0.6, 0.6)$ on $y = x$

$y_{\text{meOH}} = 0.76$ (graphically)
 $x_{\text{meOH}} = 0.46$

$Fz_i - (F-V)x_i = Vy_i \Rightarrow Fz_i - Fx_i = Vy_i - Vx_i$
 $\Rightarrow F(z_i - x_i) = V(y_i - x_i)$

$\Rightarrow \frac{150(0.6 - 0.46)}{0.76 - 0.46} = V = 70 \frac{\text{kmol}}{\text{hr}}$

$L = 150 - 70 = 80 \frac{\text{kmol}}{\text{hr}}$

1c.) $z = 0.30$, $x_{\text{meOH}} = 0.18$, $F = 150 \frac{\text{kmol}}{\text{hr}}$

$Fz = Lx + Vy \Rightarrow \frac{F}{F} z = \frac{L}{F} x + \frac{V}{F} y \Rightarrow \frac{V}{F} = z - \frac{L}{F} x$

$y = -(0.18) + 1.2 \approx 1.0$, $\frac{V}{F} = 0.30$

At $x = 0.18$, $y = 0.56$ (graphically)

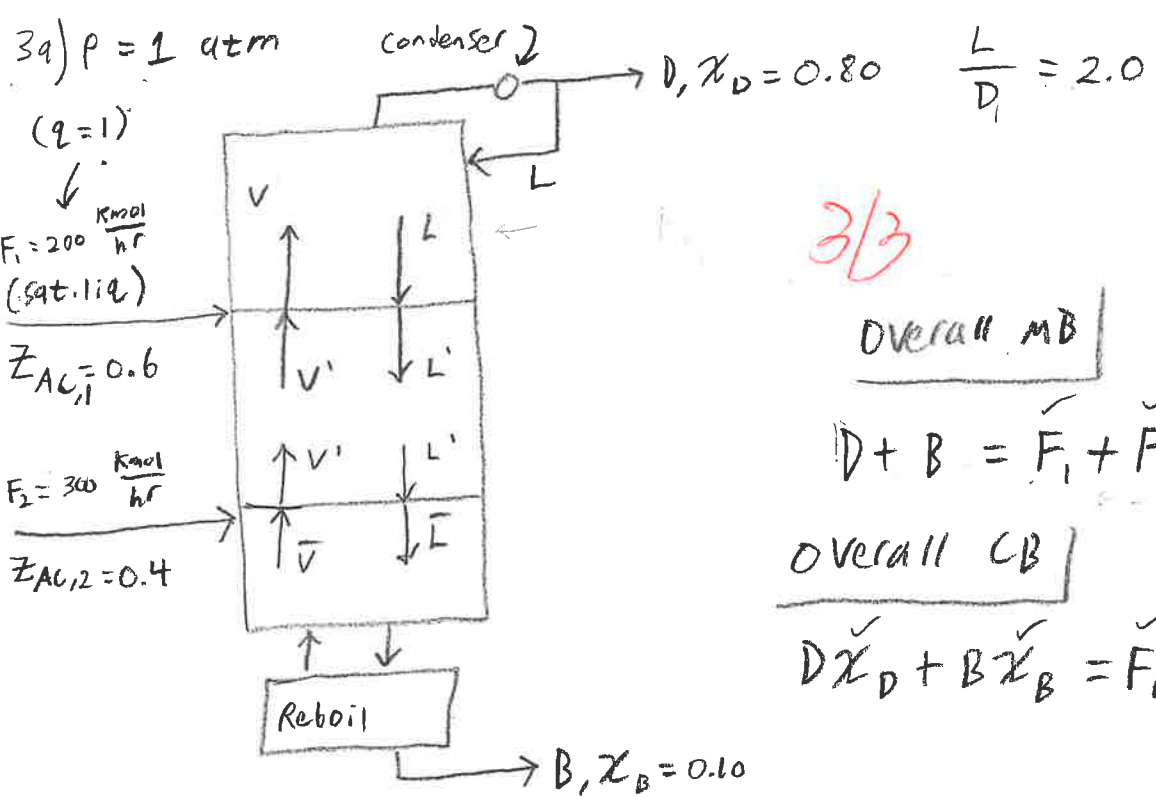
$\rightarrow z = 0.3 \Rightarrow (0.3, 0.3)$ on $y = x$

$L = F - V$, $Fz = Lx + Vy \Rightarrow Fz = (F-V)x + Vy \Rightarrow Fz = Fx - Vx + Vy$

$\Rightarrow F(z - x) = V(y - x) \Rightarrow V = \frac{F(z - x)}{(y - x)} = \frac{150(0.3 - 0.18)}{0.56 - 0.18} = 47.37 \frac{\text{kmol}}{\text{hr}}$

$\frac{V}{F} = \frac{47.36}{150} = 0.316$ OK!

$\frac{F}{V} z \approx 0.9$
 $V/F = 0.33$



3/3

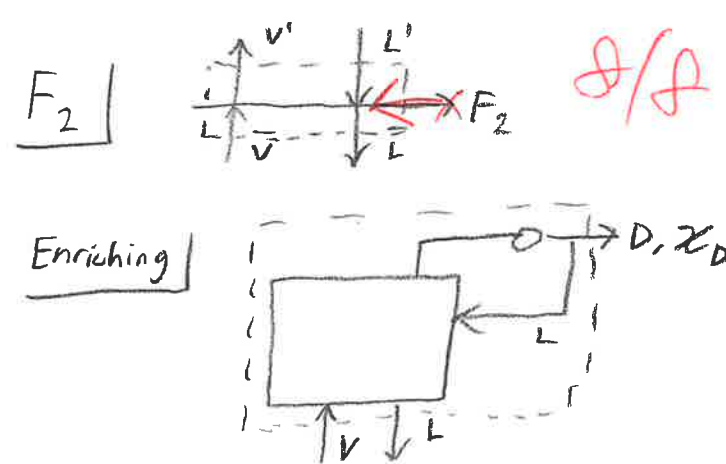
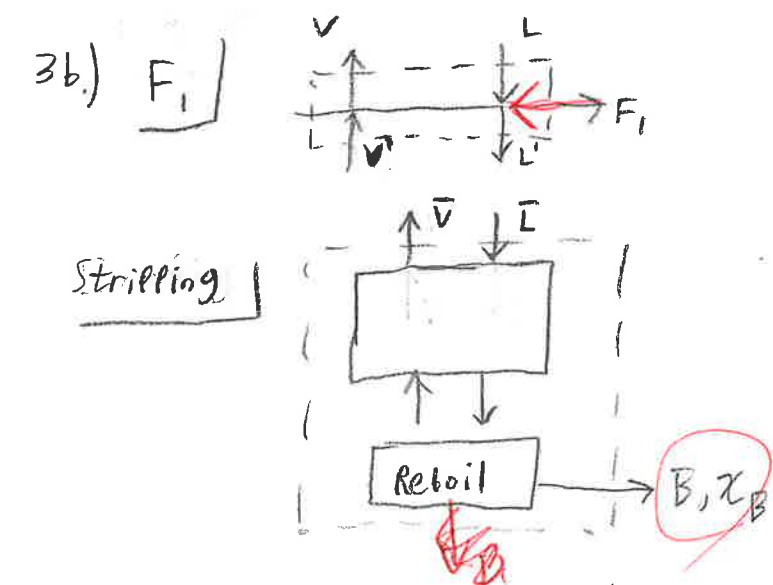
Overall MB

$$D = F_1 + F_2 - B$$

$$D + B = F_1 + F_2$$

Overall CB

$$D \check{x}_D + B \check{x}_B = F_1 \check{z}_{AC,1} + F_2 \check{z}_{AC,2}$$



3c.) F_1 MB) $V' + L + F_1 = V + L'$ CB) $F_1 z_{AC,1} +$

F_2 MB) $\bar{V} + L' + F_2 = V' + \bar{L}$ CB)

Enriching MB) $V = L + D$ CB) $V = L + D x_D$

Stripping MB) $\bar{V} = \bar{L} + B$ CB) $\bar{V} = \bar{L} + B x_B$

$F_1, F_2, z_{AC,1}, z_{AC,2}$
 Known x_D, x_B

3c: (continued.) $D = F_1 + F_2 - B$

$$D = \frac{F_1 z_1 + F_2 z_2 - B x_B}{x_D}$$

$$F_1 + F_2 - B = \frac{F_1 z_1 + F_2 z_2 - B x_B}{x_D}$$

5/15

$$F_1 x_D + F_2 x_D - B x_D = F_1 z_1 + F_2 z_2 - B x_B$$

$$\Rightarrow B(x_B - x_D) = F_1 z_1 + F_2 z_2 - F_1 x_D - F_2 x_D$$

$$\Rightarrow B = \frac{F_1(z_1 - x_D)}{(x_B - x_D)} + \frac{F_2(z_2 - x_D)}{(x_B - x_D)} = \frac{200(0.6 - 0.8)}{(0.1 - 0.8)} + \frac{300(0.4 - 0.8)}{(0.1 - 0.8)}$$

$$\Rightarrow B = 228.57 \frac{\text{kmol}}{\text{hr}}$$

$$D = F_1 + F_2 - B = 200 + 300 - 228.57 \Rightarrow D = 271.43 \frac{\text{kmol}}{\text{hr}}$$

$$V' = V + L' - L$$

$$V' = \bar{V} + L' + F_2 - \bar{L} \Rightarrow V + L' - L = \bar{V} + L' + F_2 - \bar{L} \Rightarrow V = \bar{V} - \bar{L} + L + F_2$$

$$V = L + D$$

$$L + D = \bar{V} - \bar{L} + L + F_2 \Rightarrow \bar{V} = D + \bar{L} - F_2$$

Given more time, I could solve this system.

Simply $V = V$
 $L = (\frac{L}{D})D$
 $L' = L + F_1$
 $\bar{V} = V' - 60$

3d./3e./3f./3g) Knowing the above information, These lines

could be graphed using a combination of points (depending on the type of line) For some, (z, z) and setting x or $y = 0$ gives the combination. For others, an intersection of known lines is used (where top, bottom, and feed intersect). slope should be discussed (4/6)

2/8

3h/3i/3j) I believe this is where $\frac{L}{D} = 2.0$ would

come in handy. An intersection between specified lines would also need to be used to draw it.

To find the number of stages, I would begin stepping

from x_B along the equilibrium curve and bottom operating

equation until reaching the 1st feed line, where I would

switch to stepping along the middle op. and equilibrium curve

until reaching the 2nd feed line. Finally, I would step along

the top operating eq and equilibrium line until reaching

x_D . To count the steps, I would include the reboiler as

a step, but not the total condenser since it has no equilibrium

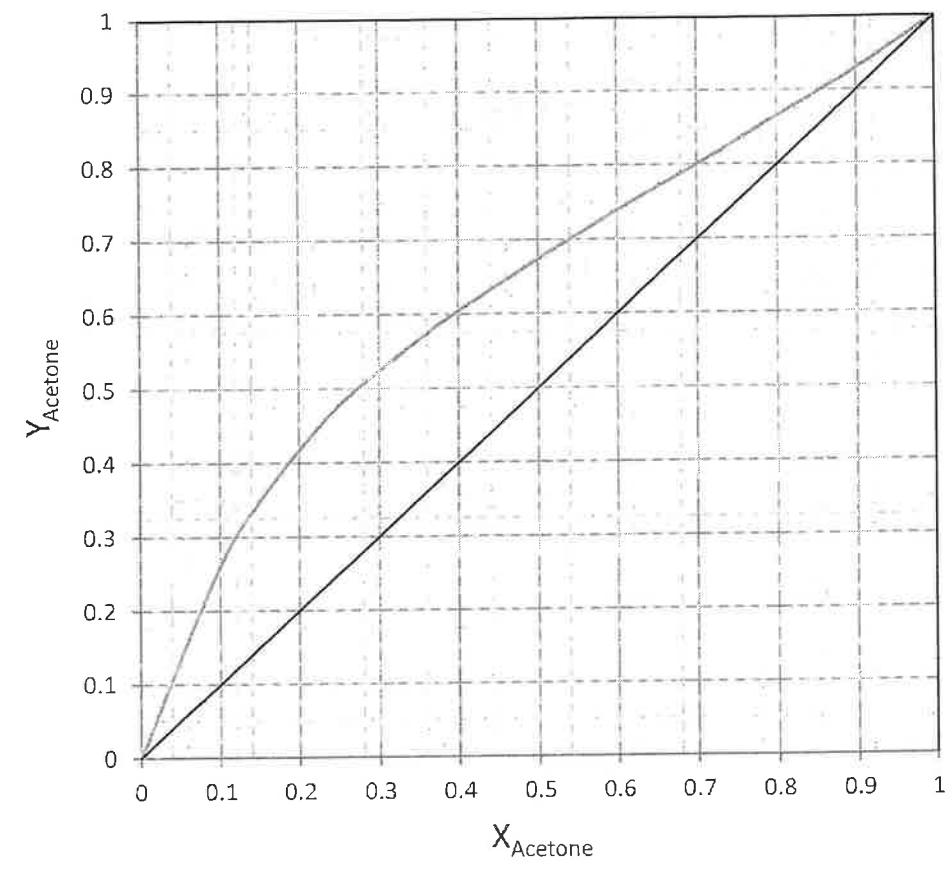
phase. The optimal feed stages would be those closest to either

feed line, and the fraction of vapor and liquid at stage 3

would be determined graphically from that stage

(using the eq. from class.)

8) $\frac{3}{5}$

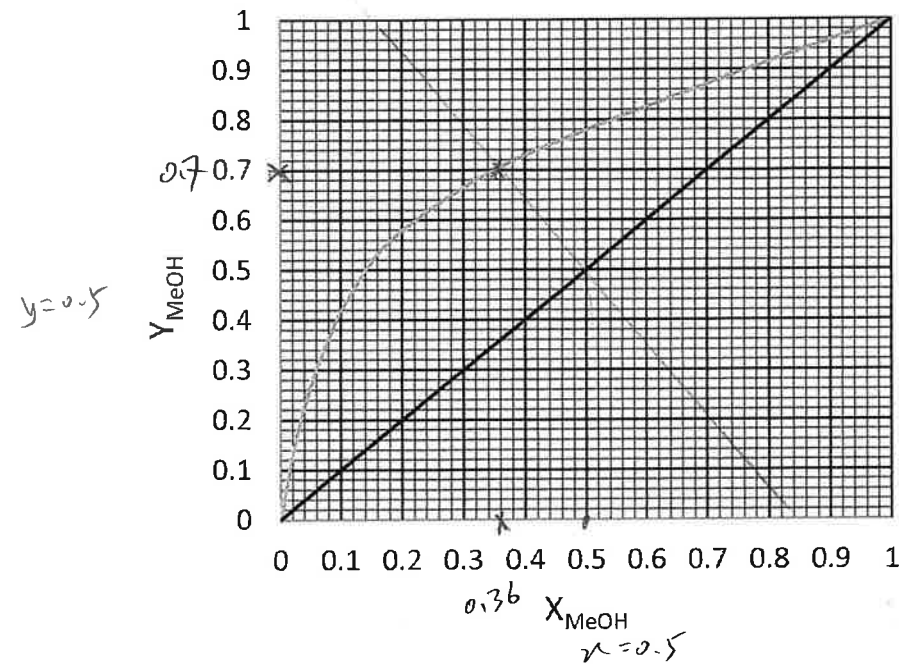


CHE 313 Transport Phenomena III

Exam #1 (2/24/26)

Name: Jeremy Wang

1. (18pts) We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. The equilibrium curve is given below.



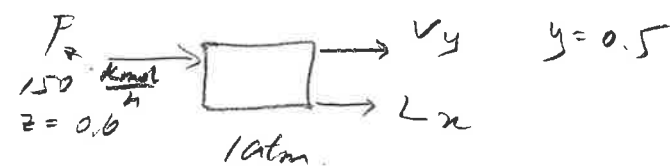
- 1) (3pts) Set up overall mass balance and component mass balance. Set up the operation equation.
 - 2) (5pts) Feed flow rate is 150 kmol/h. Feed is 60 mol% methanol, and 50% of the feed is vaporized. Find flow rates of vapor and liquid. After drawing the operation line on the VLE graph, find the vapor and liquid mole fraction.
 - 3) (10pts) When the feed is 30 mol% methanol and we desire 18 mol% methanol of liquid product, find graphically V/F by using the y-intercept, the flow rate of liquid and vapor, and vapor & liquid composition. Feed flow rate is 150 kmol/h.
2. (10pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make a flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V

$$\frac{(k_i - 1)z_i}{1 + (k_i - 1)(\frac{V}{F})}$$

$$1.0127 = 2.2$$

$$y = x$$

Q1



$$1) F = V + L$$

$$Fz = Vy + Lx$$

$$Vy = Lx \quad \text{if } Fz$$

$$y = \frac{L}{V} x + \frac{F}{V} z = \frac{L}{V} x - \frac{V+L}{V} z = \frac{L}{V} x - (1 + \frac{L}{V}) z$$

$$2) f = \frac{V}{F} \quad q = \frac{L}{F} \quad F = 150 \frac{\text{kmol}}{\text{h}}$$

$$V = 0.5 \times 150 \frac{\text{kmol}}{\text{h}} = 75 \frac{\text{kmol}}{\text{h}}, \quad L = F - V = 75 \frac{\text{kmol}}{\text{h}}$$

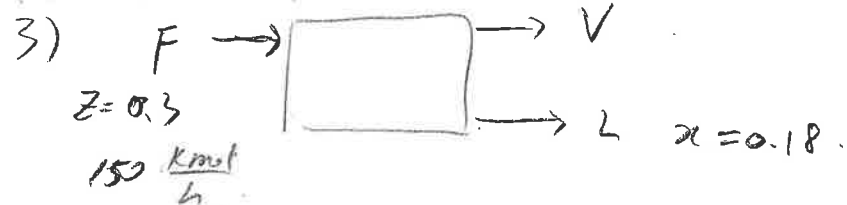
slope: -1
 Q → (0.45, 0.75)

$$\text{op line } y = \frac{75}{75} x - (1+1)(0.6) = -x - 1.2 \rightarrow f = \frac{75}{150} = 0.5 \quad q = \frac{75}{150} = 0.5$$

$$\text{feed line} = \frac{q}{q-1} x + \frac{1}{q-1} z = \frac{0.5}{0.5-1} x + \frac{1}{0.5-1} (0.6)$$

$$\text{feed} \rightarrow y = -x + (-1.2) = -x - 1.2$$

$$\text{By op line} \rightarrow x = 0.5, y = 0.5$$



$$F = V + L$$

$$Fz = Vy + Lx$$

$$45 =$$

0/10

Q 2

$$F = L + V$$

$$Fz_i = Lx_i + Vy_i$$

$$y_i = x_i K_i$$

$$\text{solve } x_i = \frac{Lx_i}{F} = \frac{Fz_i - Vy_i}{F}$$

$$x_i = \frac{Fz_i - Vy_i}{F}$$

$$x_i = \frac{z_i}{1 + (K_i - 1)(\frac{V}{F})} \rightarrow \sum x_i = \sum \frac{z_i}{1 + (K_i - 1)(\frac{V}{F})}$$

$$y_i = \frac{K_i z_i}{1 + (K_i - 1)(\frac{V}{F})} \rightarrow \sum y_i = \sum \frac{K_i z_i}{1 + (K_i - 1)(\frac{V}{F})}$$

$$\sum y_i - \sum x_i = \sum \frac{(K_i - 1) z_i}{1 + (K_i - 1)(\frac{V}{F})}, \quad \frac{df}{d(\frac{V}{F})} = - \sum \frac{(K_i - 1)^2 z_i}{(1 + (K_i - 1)(\frac{V}{F}))^2}$$

$$(\frac{V}{F})_{n+1} = (\frac{V}{F})_n - \frac{f(\frac{V}{F})}{\frac{df}{d(\frac{V}{F})}}$$

$$(\frac{V}{F})_{n+2} = (\frac{V}{F})_{n+1} - \frac{f(\frac{V}{F}_{n+1})}{\frac{df}{d(\frac{V}{F}_{n+1})}} \rightarrow \text{keep repeat Newton method process.}$$

until $f(\frac{V}{F}_{n+m}) < 0.005$ then stop.

then use the correct $\frac{V}{F}_{n+m} < 0.005$ value to plug in x_i and y_i

then plug in Component Balance equation solve L, V with overall

Mass Balance. and then find T_{drum} .

Read Question carefully!
& should have paid attention to instructions!
this is flowsheet making!

3/10

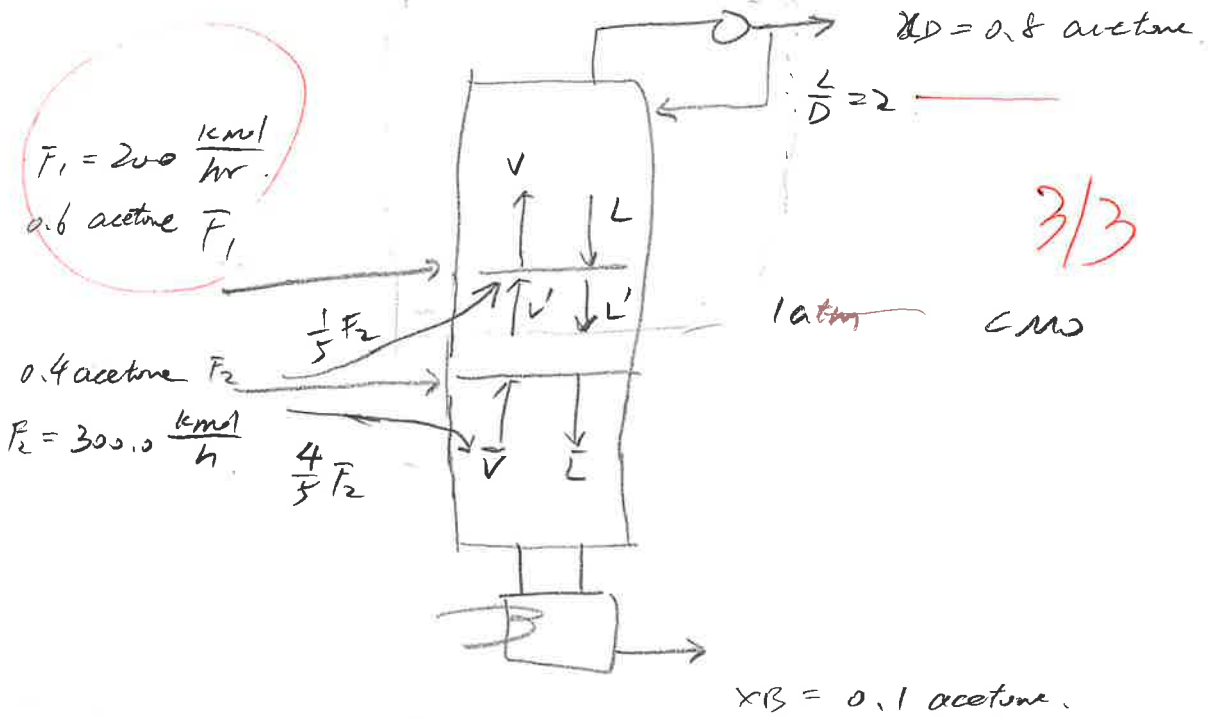
+3

Q3

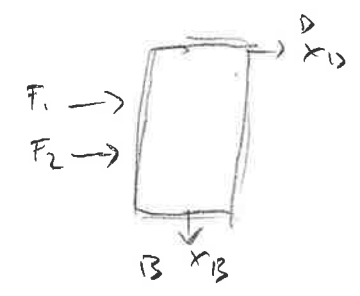
①

Sat liq

vapor liq

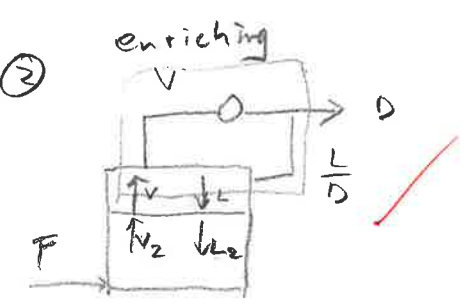


overall B =



$$F_1 + F_2 = B + D$$
$$F_1 z_1 + F_2 z_2 = B x_B + D x_D$$
$$Q + F_1 h_{F1} + F_2 h_{F2} = B h_B + D h_D$$

②



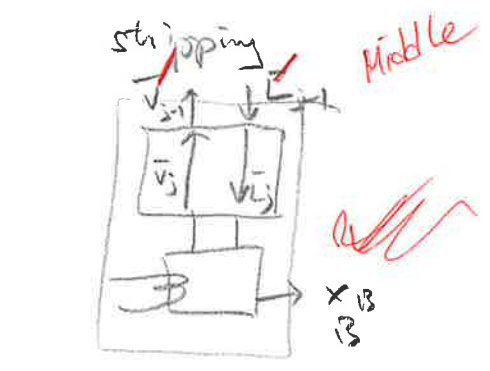
$$V = D + L$$
$$V y = D x_D + L x$$

Feed 2



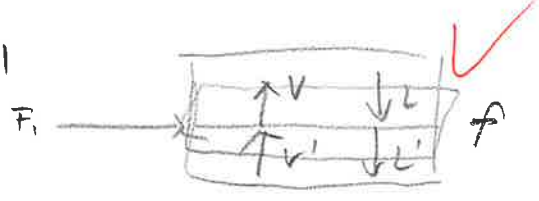
$$\frac{4}{5} F_2 + L' + \bar{V} = V' + \bar{L}$$
$$\frac{4}{5} F_2 z_2 + L' x + \bar{V} y + V' y + \bar{L} x$$

7/8



$$\bar{L} = \bar{V} + B$$
$$\bar{L} x = \bar{V} y + B x_B$$

feed 1



$$F_1 + V' + L = V + L'$$
$$L + F_1 + V' + \frac{1}{5} F_2 = V + L'$$
$$L x + F_1 z_1 + V' y + \frac{1}{5} F_2 z_2 = V y + L' x$$

Q3
③

$$D \cdot B = F_1 + F_2 = B + D$$

$$F_{12} + F_{22} = B \times B + D \times D$$

2/15

$$Top: \quad V_y = D \times D + L \times L \quad V = D + L \quad D = V - L$$

$$y = \frac{D}{V} x_D + \frac{L}{V} x_L$$

$$y = \frac{V-L}{V} x_D + \frac{L}{V} x_L$$

$$y = \frac{\frac{V-L}{V}}{\frac{L}{V}} x_D + \frac{L}{V} x_L \rightarrow = \frac{L}{V} x_L + (1 - \frac{L}{V}) x_D$$

$$F_{tot} = F_1 + F_2 = 200 + 300 = 500 \frac{1 \text{ mol}}{h}$$

$$500 = B + D$$

$$L = (\frac{L}{D}) D \quad a = \frac{V-L}{V}$$

$$L = 2D$$

$$V = D + 2D = V = 3D \Rightarrow \frac{V}{D} = 3 \Rightarrow \boxed{\frac{D}{V} = \frac{1}{3}} \quad D = \frac{V}{3}$$

$$\frac{L}{V} = \frac{2D}{V} = 2 \left(\frac{1}{3} \right) = \frac{2V}{3} \left(\frac{1}{V} \right) = \frac{2}{3} = \frac{L}{V}$$

$$y = \frac{D}{V} x_D + \frac{L}{V} x_L = \boxed{\frac{1}{3}(2) + \frac{2}{3} x_L = y} = \frac{2}{3} + \frac{2}{3} x_L = y$$

$$(0, \frac{2}{3}), ($$

$$b = \frac{L}{V} = \frac{2}{3} \rightarrow y = \frac{\frac{2}{3}}{\frac{2}{3}-1} x + \frac{1}{\frac{2}{3}-1} (0.6)$$

$$y = -\frac{3}{2} x + (-1.5)$$

